Explanation:
Cell fractionation is a technique used to separate different organelles depending on their size spinning in a centrifuge. The organelles will separate out in a particular order, according to their density and shape.

1) According to their density and shape, the nuclei settle out forming a pellet at the bottom.
2) The suspension containing can spun at higher speed the chloroplasts settle out.
3) The supernatant is spun at a higher speed still the mitochondria settle out.
4) At the last process the ribosomes separate out.

Therefore, the answer is – B – b/se the nuclei settle out first at the bottom of the mixture.

(25) If the size of a cell increases, which one of the following gets smaller?

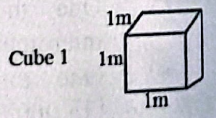
A. The volume of cell.
B. The surface area of the cell.
C. The surface area to volume ratio of the cell.
D. Volume to surface area ration of the cell.

Explanation:
When the size of a cell increases, it increases:

1) The volume of a cell.
2) The surface area of a cell.
3) The volume to surface area ratio of a cell but it decreases the surface area ratio of the cell.

Example – There are six sides to a cube.

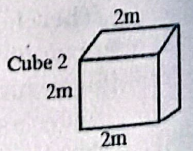
Surface area = $1m \times 1m = 1m \times 6 = 6m^2$
Volume = $1m \times 1m \times 1m = 1m^3$



Surface area to volume ration $= \frac{6}{1} = 6$

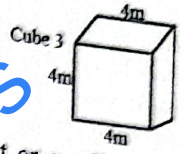
$A = 2m \times 2m \times 6 \text{ side} = 24\ m^2$
$V = 2m \times 2m \times 2m = 8m^3$

S. a. to volume ratio $= \frac{24}{8} = 3$



$A = 4m \times 4m \times 6 = 96m^2$
$V = 4 \times 4 \times 4 = 64m^2$

$S = \frac{96}{64} = 1.5$



Then when the size of an object or a cell increase, the surface area to volume ration gets smaller. Therefore, the answer is C.

(26) Which of the following is an important function of the Golgi apparatus?

A. Proteins synthesis
B. Packaging of protein for export out of the cell.
C. Removing of debris from the cell.
D. Storage of waste materials not needed by the cell.

Explanation:

- *The Golgi apparatus* consists of a number of flattened membrane bound sacs in which proteins are modified.
- For example, proteins may be converted into glycoproteins.
- Golgi apparatus acts as a cellular post office of a cell.
- Ribosomes are organelles used for protein synthesis vesicle is a small spherical organelle, bounded by a single membrane connected with Golgi apparatus and endoplasmic reticulum used to store and transport materials around the cell.
- The Golgi apparatus are involved processing, packaging and storing proteins, lipids and carbohydrates molecules with in a cell. **Therefore the answer is – B.**

(27) In which of the following features are eukaryotic cells distinguished from prokaryotic cells?

A. They have mitochondria.
B. Their nuclei lack membranes.
C. They have no DNA.
D. They have smaller ribosomes.

Explanation:
All cellular organism fall into two natural groups.

1) Prokaryotes and
2) Eukaryotes

These two groups are fundamentally different. The main difference b/n prokaryotes and eukaryote is the location of DNA.

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Both cells have DNA as agnatic material. In prokaryotes the DNA is not enclosed by nuclear membrane and lie freely in the cytoplasm. The cell lack true nuclei pro means before and karyon is nucleus. In eukaryotes contain true nuclei bounded by nuclear membrane. Both cells also have ribosomes. In prokaryotes there are 705 ribosomes smaller in size. In eukaryotes 805 (larger). **Therefore the answer is – A.** Because eukaryotic cell have mitochondria and prokaryotic cells have no mitochondria.

28. What will happen if human red blood cells are kept in a hypotonic solution?
   A. Lose water by osmosis by brust.
   B. Lose water by osmosis and shrink.
   C. Taking in water by osmosis, swell and brust.
   D. Take in water by osmosis, swell and remain turgid.

Explanation:

1) When red blood cells placed in hypertonic solution which means the solution has lower water potential than the cell contents, then water leaves the cell by osmosis and the cell shrinks.
2) In an isotonic solution, water potential of the cell equal that of the external solution and no net movement water occurs.
3) In hypotonic solution, the solution has a higher water potential than the content of the red blood cell, then water enters to the cell by osmosis and the cell brusts due to lack of cell wall like plant cells. **Therefore, answer is C.**

29. In which types of solution is the water potential more negative than in the cells.
   A. Hypotonic
   B. Hypertonic
   C. Isontonic
   D. Equal solute and solvent concentration.

Explanation:

1) Isotonic solution have the same water potential as the cell.
2) Hypertonic solution having a lower (more negative) water potential than the cell.
3) Hypotonic solution having a higher (less negative) water potential than the cell.

Therefore, the answer is – B – *hypertonic solution has more negative water potential to the cell.*

30. Which of the following paired organelles are membrane bounded?
   A. Ribosomes and peroxisomes.
   B. Chloroplasts and ribosomes.
   C. Mitochondria and ribosomes.
   D. Chloroplasts & mitochondria.

Explanation:

The main features of eukaryotic cells have membrane bounded organelles. Most organelles such as nucleus, mitochondria, Golgi apparatus, Endoplasmic reticulum chloroplasts and vacuole in plants have single and double membrane. Ribosomes and peroxisomes lack membrane bounded organelles. Chloroplasts and mitochondria are membrane bounded organelles. **Therefore, then answer is – D.**

31. Most cell membranes are primary composed of which compounds?
   A. Proteins and lipids
   B. DNA and ATP
   C. Chitin and starch
   D. Nucleotides and amino acids

Explanation:

Most cell membranes are composed of phospholipids, proteins, glycolipids and glycoproteins not DNA, ATP, Chitin, starch and nuclei lipids. **The answer is – A.**

32. Choose the one that is not a requirement of all living things.
   A. Ability to think
   B. Organization of parts
   C. Response to the stimuli
   D. Maintenance of internal constancy

Explanation:

All living organisms have common features to distinguishes from nonliving things. Some of them are – Reproduction, respiration, growth, nutrition, photosynthesis (not for all), movement, irritability (response to stimuli, homeostasis, maintenance of internal constancy, etc). Therefore, the answer is – A – b/se not all living things require the ability to think.

33. If the red blood cells shrink when placed in a certain solution, what is the strength of the solution relative to the strength of the protoplasm of the cells?
   A. Hypotonic
   B. Isotonic
   C. Hypertonic
   D. Isoosmosis

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Explanation:
Hypotonic solution has a lower solute concentration than the cell, then water enter to the cell. Hypertonic solution has a greater solute (less dilute) concentration than the cell, then water leaves from the cell and the cell become shrink. Isotonic solution has an equal solute concentration, then there is equal net movements of molecules into and out of the cell. **Therefore, the answer is C.**

34. Which of the following is the correct route that connects a stimulus and a response.

A. Receptor → Coordinator → Effector
B. Coordinator → Receptor → Effector
C. Receptor → Effector → Coordinator
D. Effector → Receptor → Coordinator

Explanation:
Receptors are sense organs which detect any stimulus to their surroundings.

- Coordinators are the central nervous system (the brain and spinal cord) which interpirate the stimulus.
- Effectors are organs (muscles and glands) which give response to the stimuli.

Therefore the correct pathway is receptor → coordinator → effector. **The answer is A.**

35. Which of the following cell types can be rich in lysosomes?

A. Red blood cells
B. Nerve cells
C. Phagocytic cells
D. Muscle cells

Explanation:
Lysosomes are small vesicles 0.2 – 0.5 µm in diameter. They contain a mixture of digestive enzymes called lytic enzymes. One of the functions of lysosomes is to digest material taken into the cell by the process of phagocytosis. White blood cell are known as a phagocyte is ingesting a bacterium. **Therefore, the answer is – C** b/se phagocytic cells are rich in lysosomes.

36. One of the following would be harder to see under the ordinary light microscope that is more likely to the available in school laboratories.

A. Nucleus
B. A bacterium
C. A mitochondria
D. Ribosome

Explanation:

- The nucleus is the largest and most prominent organelle in the cell. It has 10µm in size. It can be easily seen under light microscope than other organelles.

- A bacterium has different size and shape which is greater than ribosome and can be seen under high magnification of light microscope.
- Mitochondria are relatively large from 0.5 – 1.5 mm in diameter and 3 – 10 µm long and can be seen under higher magnification of light microscope.
- Ribosomes are small dense organelles about 20nm in diameter and it would be harder to see under light microscope.
- **Therefore the answer is – D.**

37. During chemiosmosis, what substance diffuses from one side of the other side of the membrane?

A. Water molecules
B. Protons
C. Electrons
D. ATP molecules

Explanation:
Chemiosmosis is the diffusion of hydrogen ions through the ATP synthase.

- Protons accumulate in the inner membrane space creating a higher concentration there than in the matrix (on the other side of the membrane).
- This proton gradient results in protons diffusion through ATP synthase.
- During chemiosmosis it diffuses protons through ATP synthase not water molecules, electrons and ATP molecules. **Therefore, the answer is – B – protons.**

38. Which of the following is not true about photo system II?

A. Its reaction center molecules P680.
B. It passes its excited electrons to photosystem I.
C. The energy lost from its excited electrons reduces NADP.
D. It replenishes its lost electrons from photolysis of water.

Explanation:
The chlorophyll and accessory pigment molecules are located in two types of photosystems called photosystem I (PSI) and photosystem II (PSII). In these photosystems light dependent reaction take place. Each photosystem has sensitive to light of different wave length.
In photosystem II (PSII):

- **The reaction center molecule of wave length is P680.**

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- The splitting of water molecules (hydrolysis can take place).
- It passes its excited electrons to photosystem I.
- Reduced NADP is not produced in the photosystem II.

Therefore, the answer is – C.

39. What is the importance of chemiosmosis is photosynthesis and cellular respiration?
A. Splitting of water molecules.
B. Operating the proton pump.
C. Combining hydrogen of carbon.
D. Synthesizing ATP.

Explanation:
The importance of chemiosmosis is the diffusion of hydrogen ions (protons) from the inner membrane space to the other side of the membrane through ATP synthase.

- Splitting of water molecule take place before chemiosmosis.
- Combining hydrogen and carbon take place after chemiosmosis.
- Synthesizing ATP also take place after chemiosmosis.
- During chemiosmosis operating the proton pump take place.

Therefore, the answer is – B.

40. Where does the light dependent reaction of photosynthesis occur in the chloroplast?
A. In the thylakoid membrane.
B. In the fluid of the stroma.
C. In all part of the chloroplast.
D. In the stomatal opening.

Explanation:
The process of photosynthesis is divided into two main stages. These are:
1) Light dependent reaction stage.
2) Light independent reaction.

Both stages are take place in the chloroplast of the legal. In the chloroplast there are many membrane systems which are the site of light dependent reaction.

- The membrane are covered with chlorophyll and other pigments, enzymes and electron carriers.

- The system consists of many flattened fluid filled sacs are called thylakoids. The stack of thylakoids are called grana where light dependent reaction take place.
- Light independent reaction take place in the stromal of the chloroplast.

Therefore the answer is – A.

41. For which of the following is the sugar produced by photosynthesis not used?
A. To produce biomass.
B. To make new DNA.
C. To produce ATP in espiration.
D. To produce enzymes.

Explanation:
During light independent reaction sugar is produced. The sugar produced by photosynthesis is used to produce ATP which involve in respiration.

- To produce enzymes together with proteins.
- To make new DNA molecule b/se DNA molecule consist of Deoxyribose sugar.
- Sugar is not used to produce biomass.

Therefore, the answer is – A.

42. In which process is ATP generated during short distance high speed running?
A. Aerobic respiration
B. Mitochondrial energy transformation
C. Anaerobic respiration
D. The Krebs cycle

Explanation:

- Long distance runners are slower than short distance runners.
- These long distance runners generate ATP by aerobic respiration, whereas short distance high speed runners generate ATP aerobically and to produce more energy they generate ATP also an aerobically.
- Aerobic respiration is producing energy with the presence of oxygen.
- Mitochondria energy transformation also we oxygen.
- The Krebs cycle also need oxygen.

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- An aerobic respiration is generating ATP with the absence of oxygen.

Therefore, the answer is – C – short distance high speed running generate additional ATP without oxygen or by using an aerobically respiration.

43. Which of the following kingdoms of life is consisting of prokaryotic organisms.
A. Fungi B. Monera C. Protista D. Plantae

Explanation:
Now a days there are five kingdom classification system. These are:
1) Kingdom plantae – all green plants.
2) Kingdom animalia – all animals which include vertebrates and invertaberates animals.
3) Kingdom fungi – including mucor (common nucleus) yeast, mush rooms etc.
4) Kingdom protistate all protozoans and algae. These from no. 1 – no. 4 kindoms are under eukaryotic organisms.
5) Kingdom monera – which include Archa bacteria, blue green algae (cyano bacteria) and lubacteria (true bacteria). This kingdom is under prokaryotic organism. **Therefore, the answer is – B** k. monera.

44. Which of the following human diseases can be prevented by taking proper diet?
A. Degenerative diseases C. Social diseases
B. Genetic diseases D. Deficiency diseases

Explanation:
The term health is define as eh state of complete physical metal and social wellbeing. Disease can be grouped into infectious (contagious) and noninfectious (noncontagious) diseases. Infectious disease are diseases which transmitted from an infected organism to infected organism by different factors. Non infections disease which do not transmitted.
Human disease can be grouped:
1) Human induced disease – caused by a person's life style of working condition.
2) Degenerative disease – the result of ageing.
3) Genetic diseases – transmitted through genes.

4) Social disease – social activities (addition).
5) Deficiency disease – lack of nutrition.

Therefore, the answer is – D – b/se deficiency disease are prevented by proper diet.

45. Which of the following particles does not normally transmit HIV?
A. Sexual intercourse through anus.
B. Sharing infection needles
C. Blood transfusion
D. Shaking hands

Explanation:
HIV/AIDS is an infectious disease which transmitted through.
- Sexual intercourse with many partners.
- Sharing infected any materials.
- Blood to blood contact and transfusion.
- From mother to her child during delivery.

But HIV/AIDS not transmitted during shaking hands, in swimming pools, toilet room etc. Therefore, the answer is – D.

46. To which of the following groups do those bacteriophages that integrate their DNA into the chromosome of their bacterial host belong?
A. Virulent viruses C. Lysogenic viruses
B. Lytic viruses D. Non parasitic viruses

Explanation:
Bacteriophages are viruses which attacks bacteria. These viruses have three reproductive life cycle.
1) Lytic cycle – It attach the host cell (the bacteria) and rupture (lysis) the host DNA and kill the bacteria. In this cycle it kills the host cell is called virulent viruses.
2) Chronic release cycle:
In this cycle the virus (bacleriophaphes) reproduce with in the host cell and release the viruses all at once. It release of new viruses is ongoing through the plasma membrane.
3) Lysofenic cycle
In this cycle the viruses DNA becomes in corporated or integrated with the host DNA. When the host cell's DNA with including the viral DNA.

Therefore, the answer is – C – b/se the bacteriophage or the viral DNA integrated with the bacterial DNA.

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47. Which group of microorganisms causes the disease known as athlete's foot?
A. Bacteria B. Fungi C. Protozoa D. Viruses

Explanation:
Different infectious disease can caused by different microorganism. For example: bacteria can caused diseases like pneumonia, cholera, typhoid, diphethec, tuberculosis etc.
Viruses – caused influenza, measles, AIDs, common cold, mumps, polio mellitus etc.
Protozoa – caused malaria, Bliharizia, sleeping sickness, etc.
Fungi – causes thrush, farmer's lung, Athlete's foot.
Therefore, the answer is – B.

48. Under which of the following groups can the fungi be more conveniently placed?
A. Autotrophs C. prokaryotes
B. Heterotrophs D. Plants

Explanation:
Autotrophs are those organisms which make their own food by the process of photosynthesis. Examples – all green plants and algae.
Heterotrophs are organisms which do not make their own food. They are directly or indirectly depending on all through. Examples – animals some prokaryotes and fungi. **Therefore, the answer is – B** under heterotrophs.

49. What is the advantage of using HAART (Highly Active Anti Retroviral Therapy) for the treatment of HIV?
A. It gives a lasting immunity to HIV.
B. It prevents mutation of HIV.
C. It prevents reinfection by HIV.
D. It helps to break the life cycle of HIV.

Explanation:
AIDS is cannot be treated and cured which caused by the virus called HIV.
- No vaccine to give immunity against infection.
- But there are a number of drugs called antiretroviral drugs. These drugs can be effective in slowing down the progression of AIDS.
- The advantage of these drugs work by blocking or breaking the reproduction of the virus in the CD4 lymphocytes because it can be taken in different way and different stages. These HAART cannot give a lasting immunity to HIV, cannot prevents mutation of HIV and cannot prevents reinfection by HIV. The main advantage of these HAART is it help to break the life cycle of HIV in different stages. Therefore the answer is D.

50. On which of the following principles aer most of the anti – HIV drugs currently in use working?
A. Inhibition of enzyme action
B. Degradation of viral DNA
C. Digesting of viral particles
D. Phagocytosis of the viruses

Explanation:
- Anti – HIV drugs currently is used to inhibition of enzyme action.
- These drugs are not degradation of the viral RNA and not digest the viral particles. Therefore, the answer is – A.

51. If a few anti – HIV drug is to be developed to prevent the virus from entering the host cell, which one of the following processes should be drug target?
A. Reverse transcriptase.
B. Binding of GP 120 and CD4.
C. Integration of viral DNA into host DNA.
D. Assembly of viral parts into a whole virus.

Explanation:
- Anti – HIV drugs are different drugs that act in different ways at different stages of the cycle of reproduction.
- It developed to prevent the virus from entering the host cell.
- It prevents the integration of viral DNA into host DNA.
- It prevents the assembly of viral particles.
- The anti – HIV drugs developed to prevent the entry to the host cell to do not binding of GP 120 and CD4. **Therefore, the answer is B.**

52. Which component of soil fertility is improved when farmers grow legumes in crop rotation?
A. Phosphorus B. Nitrogen C. Sulfur D. Carbon

Explanation:
- Crop rotation is very important to improve soil fertility.

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- It is used to add different nutrients to the soil.
- During crop rotation, when it is using by legumes they contain in bacteria which involved in the recycle of nitrogen. **Therefore, the answer is B.**

53. Which of the following biomes in Africa is supporting large wild mammals such as elephants, giraffes and lions.

A. The congo rain forest
B. The rain forest of western
C. Savanna grass land
D. The Sahara desert

Explanation:

There are different types of biomes in the world.

- **A biome** is a geographical or regional areas in the world.
- From different biomes which support large wild mammals in Savanna grass land. **Therefore, the answer is C.**

54. What is the important role player by microorganisms such as bacteria and fungi in the ecosystem?

A. Antibiotic production
B. Recycling of nutrients
C. Forming organic substances
D. Supplying energy to the ecosystem.

Explanation:

- Microorganisms such as bacteria and fungi are played different activities used for humans.
- Some of these activities are antibiotic production and production energy in the biogas process.
- These bacteria plays a great role in the decomposition of dead organisms in the recycling of nutrients in the ecosystem. Therefore, the answer is B.

55. In which one of the following aspects in the tropical rain forest biome poor?

A. Species diversity
B. Amount of sun light
C. Annual precipitation
D. Soil fertility

Explanation:

- The tropical rain forest biome is rich in species diversity, the amount of sun light and have high annual rain forest.
- This biome is poor in soil fertility b/se many of the minerals from the soil remain locked up in the huge trees.

Therefore, the answer is – D.

56. Which of the following is not usually true as an ecological succession progress to advanced seral stages?

A. More ecological niches are formed.
B. Species become more diverse.
C. The depth of the soil increases.
D. Less population are supported.

Explanation:

As an ecological succession developed into advanced seral stages:

- More ecological niches are formed.
- It increases the species diversity.
- The depth of the soil increases and
- More populations are supported and reach in to climax community.

Therefore, the answer is – D.

57. In which one of the four phases of population growth is the number of population the highest?

A. Lag phase
B. log phase
C. constant phase
D. decline phase

Explanation:

- In lag phase – popn is low and stable b/se they start to adapt a new surroundings.
- In log phase – the adapted popn increase rapidly b/se they have abundant resources.
- In stationary (constant) phase – number of popn are high and stable b/se the deaths are matched to birth.
- In decline phase – the numbers of popn decreases b/se of over use of resources and accumulation of toxic substances in the surrounding.

Therefore the answer is C.

58. What is the type of community called when it has reached the final and most complex stage of a succession?

A. Pioneer community
B. Several community
C. Climax community
D. Secondary community

Explanation:

A pioneer species (community) is species which is colonized a hostile area or base land. After several stages the most complex community. The development of different stage is called seres.

The answer is C.

59. Which of the following is an ecosystem?

A. A tropical rain forest.

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B. The African continent.
C. All the organisms in a given area.
D. The nonliving components of an environment.

Explanation:
Ecosystem is a system of interaction b/n living components to each other and a nonliving compounds. It contains biotic (living) and abiotic (nonliving) components. Therefore, the answer is D.

60. What is the main reason for high species richness of plants and mammals observed in Ethiopia?
A. Lack of predators.
B. Lack of disturbance.
C. Presence of several biomes with in the country.
D. Efficient management of the ecological resources.

Explanation:
Ethiopia is an important regional center of biodiversity. The country has a wide range of altitude and climate with the isolation of high lands. There are many different biomes present in the country.

61. Which one of the following concepts contains all the others?
A. Species B. Genus C. population D. Community

Explanation:
- Species is a group of related organisms that can reproduce to each other and give fertile off springs.
- Genus is a contains a group of species.
- Population is a group of organisms of one species occupying a defined area.
- Community is any group of organisms belonging to a number of different species that coexist in the same area.

Therefore, the answer is D – b/se community contains the concepts of all others.

62. If an age pyramid of a certain country is narrowing at the base, what does this tell about the trend of the popn size of the country?
The popn size is:
A. Declining C. Increasing slowly
B. Growing fast D. establishing

Explanation:
When an age pyramid of a certain country is broader at the base, it tell the rapid growth of the popn size. When an age pyramid is

narrowing at the base it tell about the trend of increasing slowly the popn size of a country. Therefore, the answer is C.

63. Before making crosses which part of the flower did meadel remove to avoid self-pollination?
A. Stigma B. Ovule C. Ovary D. Stamens

Explanation:
To make cross breeding to find out the breeding line and avoid self-pollination Mendel remove the statement of the flower. This process is called emasculation. Therefore, the answer is D.

64. Which of the following is the correct F_2 phenotypic ratio of a monohybrid cross?
A. 1: 2 B. 1: 1 C. 3: 1 D. 2: 2

Explanation:
Phenotype is an external appearance of an organism which determine by a particular genotype. Genotype is the genetic make up of traits/ organisms. The F_2 phenotypic ration of a monohybrid cross is 3: 1.

Example – P_1 (parental generation phenotype) Tall × Short
" " genotype TT × tt
" " gametes T × t
F_1 (First filial generation) phenotype all are tall
" " genotype – Tt
Cross b/n F_1 generation = $F_1 \times F_1$ = Tt × Tt
F_2 generation genotype 1 TT = 2Tt = 1 tt
Phenotype ration 3: 1
Genotype ratio 1: 2: 1
Therefore, the answer is C = 3: 1.

65. What are the most likely causes of variations within species?
A. Mitosis and a sexual reproduction.
B. Over population and over production.
C. Vegetable propagation and cloning.
D. Mutations and sexual reproduction.

Explanation:
Variations of species mostly caused by mutation and sexual reproduction which produce gametes by miosis and forming crossing over. Therefore, the answer is D.

66. One of the following is an important cause of gene mutation.
A. Old age C. smoking
B. Alcoholic drinks D. Radiation

Explanation:

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There different causes of gene mutation.

- The most causes of mutation electromagnetic radiation such as ultraviolet light, x – rays, gama rays.
- Other causes of mutations are called mutagenes, which include mustared gas, formal dehyded caffeine, to bacco constituents colichiline, some drugs and food preservers. Therefore, the answer is D.

67. How many chromosomes do humans inherit for each of their parents?
A. 23 chromosomes
B. 23 pairs of chromosomes
C. 46 chromosomes
D. 46 pairs of chromosomes

Explanation:
Each body cell has 23 pairs or 46 chromosomes. But sex cells half number chromosomes which passes to the off spring. Humans off spring inherited half number of 23 chromosomes from their parents through sex cells which produced during meiosis. Therefore, the answer is A – 23 chromosomes.

68. Which of the following is not true true about the gene called sex?
A. It found on the y – chromosome.
B. It determine maleness.
C. Tastes develop in its presence.
D. Females have two copies of this gene.

Explanation:
- SRY is an abbreviation of sex determining region of y chromosome.
- When the SRY gene is activated the bi potential gene develop in to a testes and the embry is male.
- When the SRY gene is absent (in activated) the bi potential gonads develop into ovary and the embryo in female. Then the SRY gene is found on Y chromosome that determine sex.

Answer: D

69. The sheep "dolly" is an example of which biotechnological manipulation of animals by human?
A. Transgenic animal
B. Genetically engineered animal
C. Cloned animal
D. Hybrid animals

Explanation:
To transfer gene in transgenic animal and genetically engineered animal it needs the vector cell and the host. But in cloned animal

it is in direct transfer of gene b/n animals. Dolly sheep is the first cloned animal by human. Therefore, the answer is C.

70. Choose the one that is different from all the others.
A. Genetically modified organisms.
B. Genetically engineered organisms.
C. Pathogenic organisms.
D. Transgenic organisms.

Explanation:
Pathogenic organisms are different from the others b/se these organisms are disease causing organisms. Therefore, the answer is C.

71. Which of the following crops is considered to be the best choice for a better of essential amino acids in a human diet?
A. Maize B. Quino C. Rice D. Wheat

Explanation:
There are two types of amino acids in our body. These are:
1) Nonessential amino acids which can made by our body by the process of transamination.
2) Essential amino acids which obtained from our diet. The best choice to balance these amino acid is taking non animals resource is guinoa. Therefore, the answer is B.

72. In which of the following are flowers and fruits found?
A. Ferns and relatives C. Mosses and confiers
B. Gymno spersm & ferns D. monocots and dicots

Explanation:
Ferns are reproduce by frons which consists of spores, mosses also bears archegonia (male) and antheriadia (female) organs.
- Gymnosperms plants are cone bearing plants and their seeds are found in cones.
- Angiosperms plants (monocots and dicots) are reproduce by flowers and producing fruits.

Answer: D

73. In a cross b/n heterozygotes what proportion is expected to be homozygotes recessive?
A. 25% B. 50% C. 75% D. 100%

Explanation:
When you cross b/n heterozygous the proportion of homozygous is 25%.
Given = Tt × Tt = both side heterozygous

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25% pure dominant
50% dominant
25% homozygous recessive
tt homozygous recessive

Answer: A

74. Gene silencing is the function of one of the following molecules:
A. Molecules
B. mRNA
C. SiRNA
D. ERNA

Explanation:

- ds RNA is double strand RNA produced in the nucleus with their range of gene.
- SiRNA short interfering RNA can act to express gene action and impotant in gene silencing.
- tRNA transfer RNA which transfer information from the cytoplasm to the nucleus.

Answer: C

75. Which process is held responsible for chronic myelogenous leukemia?
A. Translocation
B. Translation
C. Transcription
D. Duplication

Explanation:

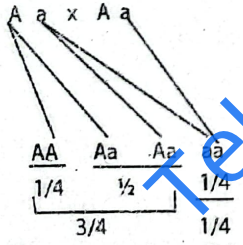
In chromosomal mutation translocation is a process that held responsible to form for chronic myelogenous leukemia.

Answer: A

76. Two parents of genotype Aa are corss bred. The alleles show complete dominance what proportion of the offspring wil phenotypically look like their parents?
A. 0 B. ¼ C. ½ D. ¾

Explanation:

The cross b/n the two heterozygous form complete dominance.



The proportion of their offspring which phenotypically similar to their parents are ¾.

Answer: D

77. Among the following mating, the ABO blood genotypes of the parents are shown, identifying the mating in which all the children will have the same blood type.
A. AO × BO B. AA × OO C. AB × BO D. BB × AO

Explanation:

The children which have the same blood type to their parents having AO × BO blood group.

Answer: A

78. If a new mutant allele a rises in a certain popn which of the following factors determines if the allele is going to be adaptive or non-adaptive?
A. The environment in which the populations lives.
B. The rate at which the gene mutates.
C. The popn in which the gene is found.
D. The use and disuse of the gene by the popn.

Explanation:

The factors that determine the allele is going to be adaptive or non-adaptive is the rate at which the gene mutates.

Answer: B

79. Suppose the amino acid coding region in a mRNA is 1200 nucleotides long, how long is the protein in terms of amino acid number?
A. 1200 amino acids
B. 600 amino acids
C. 400 amino acids
D. 300 amino acids

Explanation:

Genetic code is the sequence of bases in the nucleotides of the DNA that makes a gene that codes for the protein and that each amino acid in the protein is coded for by a tri (sequence of three) of bases.

- A gene is a sequence of base triplates in the DNA molecule that carries the code for a proton.
- A mRNA is 1200 nucleotides and divided by the triplet of protein codes and have 400 amino acids.

Answer: C

80. Which one of the following choices show the end products of a miltotic cell division?
A. Gametes and ordinary cells
B. Four genetically different cells
C. Two genetically different cells
D. Two genetically identical cells

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Explanation:
The three choices (Letter A, B and C) are the end products of meiosis (meiotic cell division). Choice teller D is the end products miotic cell division.
Answer: D

81. What exactly happens during a substitution mutation?
A. One base is replaced by different base.
B. An extra base is added during DNA replication.
C. A base is missed out during DNA replication.
D. Changes occur in the arrangement of structure of a chromosome.

Explanation:
- One base is replaced by different base substitution mutation.
- An extra base in adding during DNA replication is addition mutation.
- When a base is missed out it is deletion mutation.

When changes occur in the arrangement or structure of a chromosome is chromosome mutation.
Answer: A

82. In which homid species did scientists find the smallest brain size (cranial activity)?
A. Homo sapiens
B. Homo habilis
C. Homo erectus
D. Homo neander the lensis

Explanation:
Brain capacity of Homosapiens is 1400 cm^2 in size.
" " of Homo habilis is 700 cm^2 in size.
" " of Homo erectus 840 cm^2 in size.
" " of Homo neander the lensis 1200 cm^2
The smallest brain size observed in homo habilis.
Answer: B

83. What is the specialist in biology called if he/she studies?
A. Geneticist
B. Ecologist
C. Palenotolgist
D. Comparative Biochemist

Explanation:
- Geneticist study about gene and heredity.
- Ecologist study about ecology.
- Paleontologist study about chemical in living things.

Answer: C

84. When two species are compared which of the following sources of evidence is least informative about the degree of relationships b/n the species?
A. Nucleotide sequence of DNAs
B. Amino acid sequence of proteins
C. Glucose sequence of polysaccharides
D. DNA – DNA hybridization

Explanation:
Nucleotide sequence of DNAs, amino acid sequences of protein and DNA – DNA hybridization are the source of evidence of evolution. Glucose sequences of polysaccharides cannot uses as a source of evolution.
Answer: C

85. What is the reproductive isolating mechanism called if two species of frogs do not interbreed become they cannot understand the melting cells of one another?
A. One another
B. Seasonal isolation
C. Temporal isolation
D. Isolation by distance

Explanation:
- Seasonal isolation is members of two species (popn) reproduce at different times of the year.
- Temporal isolation is members of the two popns reproduce at different time of the day.
- Behavioral isolation is members of the two popns have different courtship patterns.
- Isolation by distance is geographical isolation. Therefore the answer is B. Because the two species of frogs have different courtship patterns.

Answer: B

86. Which of the following theories explains evolutionary changes of living things in terms of changes in their allele frequencies?
A. Darwin's natural selection.
B. Spontaneous generation.
C. Inheritance of acquired characteristics.
D. Neo – Darwinism.

Explanation:
Natural selection is related to members of species will be adapted to their environment better to survive and reproduce. The inheritance of acquired characteristic is that traits (alleles) changed (acquired) during the individuals life time.
Answer: C

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87. Among the following which one is the best criteria to show that the populations belong to the same species?
A. Morphological similarity
B. Physiological similarity
C. In habiting the same geographical area
D. Production of fertile offspring

Explanation:
Species is a group of closely related species (organisms) which interbreed to each other and produce fertile off springs.

Answer: D

88. In the process of the evolution of life on earth, which of the following four processes evolved last?
A. Photosynthesis
B. Aerobic respiration
C. Chemosynthesis
D. Photo anto trophism

Explanation:
Photosynthesis is a process in which plants produce their own food and oxygen by using of sun light. Plants are then they are photo autotrophs.

Answer: A

89. What is the reproductive isolating mechanism called it two species of frogs do not inter breed because they cannot understand the mating calls?
A. Seasonal isolation
B. Behavioral isolation
C. Temperature isolation
D. Isolation by distance

Explanation:
- Seasonal isolation is members of two species (popn) reproduce to different times of the year.
- Temporal isolation is members of the two popn reproduce at different time of the day.
- Behavioral isolation is members of the two popn have different courtship patters.
- Isolation by distance is geographical isolation.

Answer: B

90. Which of the following theories explains evolutionally changes of living things in terms of changes in their allele frequencies?
A. Darwin's natural selection
B. Spontaneous generation
C. In heritance of acquired characterstics
D. Neo Darwinism

Explanation:
Natural selection is related to members of species will be adopted to their environment better to survive and reproduce. the inheritance of acquired characteristic is that trait (alleles) changed (acquired) during the individuals life time.

Answer: C

91. Among the following which one is the best criterion to show that the populations belong to the same species?
A. Morphological similarity
B. Physiological similarity
C. In habiting the same geographical area
D. Production of fertile offspring.

Explanation:
Species is a group of closely related species (organisms) which interbreed to each other and produce fertile offsprings.

Answer: D

92. In the process of the evolution of life on earth, which of the following four process evolved last?
A. Photosynthesis
B. Aerobic respiration
C. Chemosynthesis
D. Photo auto tropism

Explanation:
Photosynthesis is a process in which plants produce their own food and oxygen by using of sun light plants are then they are photo autotrophs.
Aerobic respiration is evolved last than others b/se heterotrophic organisms can use oxygen and used to release energy than others.

Answer: B

93. Which of the following characteristics can show the evolutionary relationships among organism?
A. Structures having similar functions.
B. Structures having same size.
C. Structures having common origin.
D. Structures having different origins.

Explanation:
The evolutionary relationships among organisms can show those structures having common anscestors or common origin.

Answer: C

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94. Which one of the following is not important for evolutionary change of a population?
A. Over reproduction
B. Insufficient of natural resources
C. Existence of heritable variation
D. Survival of all that are born.

Explanation:
The theory of natural selection that was proposed by Darwin have main observations:
1) Over production (fecundity)
2) Variations among the offspring
3) The struggle for existence b/se resource are insufficient the survival of the filest.

Answer: D

95. In some human liver cells there are 92 chromosomes per cell what is the prodigy level of such cells?
A. Haploid C. Tetraploid
B. Diploid D. Hexaploid

Explanation:
Each gamete cells aer half chromosomes are called haploid (n). Body cells are diploid (2n), and some cells in humans like liver cells are tratrapliod (4n). For example, humans such sex cell is 23 chromosome. Each body cell (nerve cell) has 46 chromosomes. Some body cells (liver cell) has 46 chromosomes.

Answer: C

96. During the course of evolution, which of the following events came before all the others?
A. Origin of dinosaurs and other reptiles.
B. Availability of free oxygen in the atmosphere.
C. Origin of the oldest eukaryotic organisms.
D. Origin of the first multicellular animals and plates.

Explanation:
To be survive on the earth mostly higher and lower animals need oxygen. The availability of free atmospheric oxygen is produced by photosynthesis.

Answer: B

97. What kind of natural solution is at work in a situation where individuals at both extreme ends of the range get better survival advantages than those around the middle?
A. Stabilizing selection
B. Directional selection
C. Disruptive selection
D. Selection that removes the extremes.

Explanation:

- In stablizing selection, individuals at both endes of the range of valucs for a features are the least adapted and favor the middles.
- In directional selection, individuals at one extreme could have disadvantage and the other extreme have advantage.
- In disruptive selection, individuals at both extremes have some advantage over those displaying the mean value and the middle of the range have disadvantage.

Answer: C

98. In which of geologic period does the fossil record show more diverse and relatively higher forms of organism?
A. Devonian C. Jurassic
B. Cretaceous D. Permian

Explanation:
In cretaceous geological period there were observed more fossil records and more diverse higher forms of organisms tha others geological periods.

Answer: B

99. What is the most frequent cause tine agents of food poisoning?
A. Bacteria C. Viruses
B. Protozoa D. Worms

Explanation:
Bacteria is the most causative agents of food poisioning than other microorganisms.

Answer: A

100. Which one of the following do bees use to inform other bees about the location and distance of a new nector discover?
A. Pheromones C. Buzzing noise
B. Waggle dance D. Vibration of wings

Explanation:
Waggle dance is in which worker bees used to inform the location and distance of nectar where it is found for other bees.

Answer: B

101. Which of the following is not classified as a learned behavior?
A. Insight C. Latent
B. Innate D. Conditioned

Explanation:
Insight learning, latent behavior and conditioning are grouped in learned behavior but innate behavior is not under learned behavior.

Answer: B

102. What do we call the learned behavior is a mouse that had just escaped from the mouth of a cat Jumped violently at a slight touch by a trivial object?
A. Latent learning C. Conditioning
B. Sensitization D. Imprinting

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Explanation:

- Latent learning's laking about hidden learning orlearing that is not apparent as it takes place.
- Conditioning is a technique used to get an animal to lear a particular behavior.
- Sensitization is an increases in response to a harmless stimulus, when that stimulus occurs after a harmful stimulus.
- Imprinting is another kind of instinctive behavior in which the fixed action pattern in organized. For example a newly born (hatched) organism followed any movement.

Answer: B

103. The group of the Ethiopia wolf which does not contribute to territory marking with their urine containing pheromones is _____.

A. Adult males
B. Adult females
C. Sub – adult males
D. Sub – adult females

Explanation:

The Ethiopia wolf in a social animal. They maintain a group territory by making with urine containing pheromones. In this territory making behaviors, the adult males contribute to marking the territory but sub – adult females do not have contribution.

Answer: D

104. To what kind of animal behavior can the spinning of a web by a spider be classified?

A. Learned behavior
B. Experimental behavior
C. Instinctive behavior
D. Accidental behavior

Explanation:

There are two types of behavior. These are innate behavior include instinctive behavior. It involve a complex sequence of actions. A good example is spinning of a web by spider.

Answer: C

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1. While conducting research, which of the following should a biologist do first?

A) Generate hypothesis
B) Conduct experiments
C) communicate the results
D) Indentify a research question/problem

Explanation: To conduct scientific research biologists follow the scientific methods.

- The stage of scientific method starts from
→ observation
→ questioning and exploration about the problem
→ constructing experiments to test the hypotheses
→ conducting experiments to test the hypotheses
→ the development of theories and laws

Therefore the answer is D identify a research question

2. Suppose in an experiment conducted to test a new drug, 20 people were given the drug and 20 people a placebo, what is the control of the experiment?

A. people who received the placebo
B. people who received the new drug
C. the new drug itself
D. the new drug and the placebo.

Explanation: In a typical controlled experiment there are two groups

a) The first group is in which the variable is manipulated in a particular way and bring change during experiment is an experimental group
b) The another group is in which there is no manipulation or change is called the control group

The people those who are taking the new drug are experimental group and those people who received the placebo are the control group. Therefore the answer is A

3. Which molecule in the cell is the constituent of the gene?

A. Nucleic acid
B. protein
C. lipid
D. carbohydrate

Explanation: The answer is A. Because the genes are the constituent of Nucleic acid composed of DNA and RNA.

4. Which of the following is a distinguishing feature of amino acids that is not found in carbohydrates
A. carbon B. oxygen C. hydrogen D. Nitrogen
Explanation: Amino acid is the monomer of protein, which made up of carbon, hydrogen oxygen, nitrogen and sometimes sulphur, But carbohydrates are only made up of three element such as carbon, hydrogen and oxygen. **Therefore the answer is D**

5. Which one of the following molecules in living things is **not** an organic molecule?
A. sugar B. Nucleotide C. amino acid D. water
Explanation: Different organic molecules contain different combinations of carbon and hydrogen in addition to oxygen. Sugar (carbohydrate), lipids, proteins are organic molecules made up of carbon and hydrogen with other elements. Inorganic molecules may contain one or the other (carbon or hydrogen) but not both. Therefore the answer is D water contains hydrogen not carbon.

6. How many carbon and oxygen atoms are there in a molecule of maltose?
A. 6 carbon and 6 oxygen C. 11 carbon and 12 oxygen
B. 12 carbon and 11 oxygen D. 24 carbon and 12 oxygen
Explanation: Maltose is a double sugar (disaccharide) which is derived from two 2-glucose molecules and a molecule of water (H_2O) is formed by the process of condensation.
$C_6H_{12}O_6 + C_6H_{12}O_6 \rightarrow C_{12}H_{22}O_{11} + H_2O$ **The answer is B**

7. Which of the following is true about saturated fatty acids?
A. They have single bonds between carbon atoms
B. They are the same as polyunsaturated fatty acids
C. They are the same as monounsaturated fatty acids
D. They have double bonds between the carbon atoms
Explanation: The natural of the hydrocarbon chains in fatty acids can differ based on the bonds between the carbon atoms.
* When one of the carbon – carbon bonds is a double bond, then it is a mono unsaturated fatty acid.
* When more than one carbon – carbon bond is a double bond then the fatty acid is polyunsaturated fatty acids.
* When all the carbon – carbon bonds in the hydrocarbon chain are single bonds the fatty acid is a **saturated fatty acid.** ⇒ **Therefore the answer is A**

8. Which of the following is **true** about the substance represented by the chemical formula "$C_2H_5NO_2$"?
A. simple sugar C. fatty acids
B. inorganic substance D. amino acids
Explanation: The chemical formula of $C_2H_5NO_2$ is represented an amino acid because it contains nitrogen element.
Therefore the Answer D

9. Choose the one that represents a chemical formula of an organic compound?
A. $C_{12}H_{22}O_{11}$ C. NH_3
B. $Mg(OH)_2$ D. NaCl
Explanation: Organic compounds contain carbon and hydrogen in the addition to oxygen.
Then the answer is A because it contains carbon hydrogen and oxygen elements

10. How are unsaturated fats are distinguished?
A. They are made up of glucose and fructose
B. They are made up of amino acids and glycerol
C. They have double bonds in their carbon chains
D. They have only single bonds at the points of attachment
Explanation: Unsaturated fats are consists one double bond which is mono unsaturated fatty acids and more than one double bond with in the carbon chains which are polyunsaturated fatty acid. ⇒ **The answer is C**

11. Which of the following is produced when a lipid molecule is hydrolysed?
A. Amino acid and water C. Fatty acids and glycerol
B. Amino acid and glucose D. Glucose and glycerol
Explanation: Ehen lipid molecule is hydrolysed, it produce fatty acid glycerol. **Therefore the answer is C**

12. In the cause of enzymatic chemical reactions, what do you call the substance that is acted up on by on enzyme?
A. polypeptide B. Coenzyme C. vitamin D. substrate
Explanation: The substance that is acted up on by an enzyme is the substrate. **Therefore the answer is D**

13. Which of the following is the common characteristic of all enzymes?
A. They act inside cells only C. They require cofactors
B. They are proteins D. They operate at any pH

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Explanation: The answer is B because all enzymes are proteins.

14. Which group of organic compounds includes the enzymes?
A. proteins B. lipids C. starches D. carbohydrates
Explanation: Enzymes are globular proteins that have a tertiary structure that resembles a globule or ball.
Protein is one of the organic molecule that makes enzyme.
Therefore the answer is A

15. Which of the following pH values represents the strongest base?
A. 2 B. 7 C. 13 D. 15
Explanation: The pH scale is starting from zero (0 – 14) up to fourteen bellow 7 is acidic and 7 is neutral and from 7 up to 14 is basic condition. Then the strongest base is around 14
Therefore the answer is C

16. Which region of human digestive tract contains enzymes that perform well at low pH?
A. mouth B. stomach C. small intestine D. large intestine
Explanation: The pH of the intestinal tract of human changes from one region to the next.
⇒ The concentration of acid (proton donor or electron acceptor) is greatest when the pH number is lowest.
⇒ As the pH number increases the concentration of base (proton acceptor or electron donor) increases.
⇒ At a pH of 7.0 the concentration of H^+ and OH^- are equal
Therefore the answer is B because enzymes which are found in stomach works in acidic

17. For a biological detergent to effectively remove oily and greasy dirt, which of the following enzymes should it contain?
A. cellulose C. amylase
B. protease D. Lipase
Explanation: Oils and greasy dirt are lipid containing molecules Lipase is an enzyme used to hydrolys lipids
Therefore, the answer D

18. Which one of the following cellular forms did Robert Hooke observe under his crude microscope?
A. Bacteria B. protozoa C. yeast Dd. Empty cell wall
Explanation: Robert Hooke was the first person who use the term cell. He choose to study thin slices of cork from the bark of a cork oak tree.

He saw only the cell walls that surrounded the living portions of plant cell under his crude micro scope.Therefore the answer is D

19. One of the following organelles of the eukaryotic cells does protein synthesis take place?
A. The nucleus C. The chloroplast
B. The ribosome D. The mitochondria
Explanation: The ribosomes are the organelles of the eukarioplic cells in which protein synthesis can take place. The answer is B

20. Which of the following structures is not present in animal cells
A. cell wall B. nucleus C. protoplasm D. nucleic acid
Explanation: From the given structures cell wall is not present in animal cells. Then the answer is A.

21. What is the general term for the pat of the protoplasm that lies outside the nucleus?
A. cytosol C. central vacuole
B. cytoplasm D. plasma memberane
Explanation: Every Eukaryotic cells have two main parts. these parts are the cytoplasm and the nucleus. The general term of these parts is protoplasm. Then the out side portion of the nucleus is cytoplasm. **The answer is B**

22. Which of the following cellular structures is possessed by all cells?
A. cell membrane C. cell wall
B. nucleus D. Golgri apparatus
Explanation: From the given choice nucleus is not found in prokaryotic cells, the cell wall is **not** present in animal cells and Golgi apparatus also **not** present in prokaryotic cells. The cell membrane is the only cellular structures found in all cells.
Therefore the answer is A

23. Which of the following carbohydrates has structural function?
A. cellulose B. Glycogen C. starch D. sucrose
Explanation: cell wall is the structural function of plants which is made up of polysaccharide called cellulose. Therefore the answer is A

24. Which one of the following eukaryotic cell organelles was a free living cell before eukaryotic cells evolved?
A. nucleus B. Ribosome C. chloroplast D. Nucleolus

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Explanation: From the given choice Ribosomes are the only organelles possessed in both prokaryotic and eukaryotic cells. Then the answer is B

25. Which of the following mechanisms moves digested amino acids and glucose across the plasma membrane of the cells lining the wall of the small intestine?
A. osmosis C. facilitated diffusion
B. simple diffusion D. Active transport

Explanation: The digested amino acides and glucose are move across the plasma membrane of the cells lining the wall of the small intestine by facilitated diffusion. Because the plasma membrane has the carrier proteins or channel proteins with ion pores. Therefore the answer is C

26. Which one of the living characteristics UNNECESSARY for an object to be considered a living things?
A. Ability to respond to stimuli C. Ability to grow
B. Ability to reproduce D. Ability to move

Explanation: The answer is D because not all living things show the ability to move.

27. If a cell fails to clear its cellular debris, which one of its organelles is most likely not functioning?
A. Nucleus C. Endoplasmic relialum
B. mitochondria D. lysosomes

Explanation: Lysosomes are formed in the Golgi apparatus and contain digestive enzymes that break down cellular waste and debris. Then a cell fails to clear its cellular debris, when hysosomes are not function properly. Therefore the answer is D

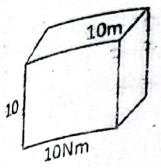
28. Among the following identify the organelle in which nucleic acid is not found?
A. mitochondria C. Ribosome
B. chloroplast D. Golgi apparatus

Explanation: From the given alternatives Golgi apparatus do not consists of nucleic acid (both DNA and RNA). Nucleus, mitochondira and chloroplast have their own DNA, but ribosome conteins RNA of nucleic acid. Then the answer is D

29. Suppose a hypothetical cube-shaped cell has sides of 10 micrometers, what is the surface area to volume ratio of this cell?
A. 6:10 B. 10:10 C. 3:6 D. 4:8

Explanation: Area of the cube is a = L × w

$10 \times 10 = 100 \times 6 = 600 Nm^2$



The volume of the cube (V) = L × w × h
V = 10Nm × 10Nm × 10m = 1000N

Then the surface area to volume ratio is = $\frac{A}{V}$

$= \frac{600}{1000} = 6:10$

Then the answer is A

30. In which of the following solution does an animal cell under go haemolysis?
A. In hypotonic solution
B. In hypertonic solution
C. In isotonic solution
D. In both hypotonic and hypertonic solution

Explanation: When an animal cell put in hypotonic solution, the cells gain water by osmosis and the hypotonic solution, the cells due to the weak plasma membrane. This is condition is called haemohysis. Therefore the answer is A

31. Which of the following will happen if a plant cell is kept in a solution that is stronger than its protoplasm?
A. The cell will become liquid
B. The central vacuole will expand
C. The protoplasm will get plasmolysed
D. The cell will swell and brust

Explanation: when the plant cell kept in a strong solution the cytoplasm of the cells loses its water by osmosis. Then the plant become shrink and the cell has been plasmolysed
Therefore the answer is C

32. How many net ATP molecules are generated through an aerobic respiration when as single glucose molecule is changed to pyruvale in the human body?
A. two B. three C. four D. six

Explanation: The first stage of the cellular respiration process take place in the cytosol of cytoplasm. This first step, known as glycolysis conists of the enzymatic break down of glucose without oxygen. Glycolysis is called an anaerobic process.
In glycolysis a single glucose molecule changed to pyruvate and generate four ATP molecules because to ATPs were utilized to

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start the process and a net two ATP molecule are stored in the process. Therefore the answer is A

33. During the krebs cycle, which of the following molecules temporarily stores most of the energy released from food molecule?
A. ADP B. ATP C. NADH D. FADH

Explanation: During the krebs cycle most of the energy, six $NADH^+$ released. From a glucose molecule stores temporarily. Only two $FADH^+$ and 2ATP molecule in krebscycle. Then the answer is C.

34. At which stage is most of the ATP generated in aerobic respiration?
A. Glycolysis C. krebs cycle
B. link reaction D. electron transport

Explanation: Most of the ATP produced by aerobic respiration comes from the electron transport system (ETS), because 10NADH and 2 FADH which are produced during glycolysis, link reaction and krebs cycle are deliver electrons to the enzymes responsible for the ETS and released out 34 ATPs during this stage. Therefore the answer is D

35. Which of the following substances is not necessary for photosynthesis to take place?
A. chlorophyle C. oxygen
B. carbon dioxide D. water

Explanation: Water and carbondioxide are the two raw materials of plants to photosynthesis take place. The necessary conditions of photosynthesis are chlorophyle and sunlight. Glucose is the main products of photosynthesis and oxygen is the byproduct of hotosynthesis of plants oxygen is the byproduct of photosynthesis of plants. Therefore the answer is C. oxygen is not necessary for photosynthesis to take place.

36. The molecule of which pigment is located at the reaction center of a photosystems?
A. chlorophyll a C. carotenid
B. chlorophyll b D. Accessary pigments

Explanation: The pigments in antenna complex contains chlorophyll a center of photosystem II (PSII) which PGSO and the reaction center of photosyptem I (PSI) P700 molecule is always chlorophyll a. Therefore the answer is A

37. What is the ultimate source of energy for all organisms living on the earth?
A. water B. oxygen C. Nitrogen D. the sun

Explanation: The sun light is the ultimate source of energy for all organisms living on the earth. The answer is D

38. Where exactly in the cell does the krebs cycle take place?
A. mitochondrial matrix C. cytoptosmic fluid
B. inner mitochondrial membrane D. outer mitochondrial membrane

Explanation: There are four stages of cellular respiration, they take place in different region of the cell. For example
1. glycolysis is take place in the cytosol of cytoplasm with the absence of oxygen (anaerobic)
2. The link reaction and the krebs cycle are take place in the fluid matrix of mitochondria with the presence of oxygen (aerobic)
3. The electron transport system (ETS) can take place in the inner membrane of the mitochondria, known as the cristae with the presence of oxygen (aerobic)

Therefore the answer is A. the krebs cycle take place in the mitochondrial matrix.

39. If present in the fermentation syptem, which one of the following would negatively affect alcohol production by yeast?
A. water B. oxygen C. Glucose D. Fermentation enzyme

Explanation: Fermentation system is anaerobic process which occur with out oxygen. If there is oxygen it affects the process.
The answer is B

40. Which term describes the process by which water is lost by evaporating through the stomata of leaves?
A. Transport C. Anchorage
B. Transpiration D. Adhesion

Explanation: The answer is B

41. Which of the following is an accurate representation of the relationship between the terms metabolism, anabolism and catabolism?
A. Anabolism = Catabolism
B. Metabolism = Catabolism
C. Anabolism = Catabolism + metabolism
D. Metabolism = catabolism + anabolism

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Explanation: Metabolism is the sum total of all the chemical changes that occur in the body of organisms. It consists of anabolism and catabolism. Anabolism is the energy requiring process by which small molecules are joined together (building up) to form larger molecules. Example . photosynthesis. Catabolism is the energy releasing process by which large molecules are broken down in the smaller ones. Example – respiration. Therefore the answer is D

42. In which of the following ways is the carbon dioxide of the atmosphere fixed in to the carbon found in organic molecules?
A. In the breathing processes of all animals
B. In the decomposition of organic molecules
C. In the process of photosynthesis by green plants
D. In all the cellular respiration processes of organisms

Explanation: The carbon atom which found in organic molecules such as in carbohydrates, lipids and proteins are processes of photosynthesis by algae and green plants. Therefore the answer is C.

43. Which of the following groups of plants carry out light dependent and light independent reactions of photosynthesis in separate cells?
A. C – 3 plants C. CAM plants
B. C – 4 plants D. plants without chlorophyll

Explanation: C – 3 plants are those plants living in temperate environments such as those found in Europe. The first compound formed in the light independent reactions of the calvin cycle is glycerate phosphate (Gp) which contains three carbon atoms.
- Both light dependent and light independent (calvin cycle) take place in the mesophyll cells, specially in the palisade cells.

C – 4 plants are mostly living in tropical environments including in Ethiopia.
The first compound formed in the light independent reactions is a C_4 compound (contains four carbon) is called oxalo acetate.
The light dependent and the light independent reactions take place in different (separate) cells. The light dependent reaction takesplace in the mesophyll cells, where as light independent reactions take place in the bundle sheath cell of vascular tissues. Therefore the answer is B

44. From which of the processes of cellular respiration is the majority of the ATP generated?
A. Anaerobic fermentation
B. Glycolysis and link reaction

C. Electron transport and chemiosmosis
D. krebs cycle and glycolysis

Explanation: The processes of cellular respiration the majority of ATP are generated in the electron transport chain and chemiosmosis. Because the molecule if reduced NAD and reduced FAD passing through ATP synthesis of electron transport chain.
Then the answer is C

45. One which of the following does the rate of algal photosynthesis in a Lake depend?
A. The oxygen content of the water
B. The nitrogen content of the water
C. The elevation where the lake is found
D. The amount of light that penetrates the take water

Explanation: The rate of algae photosynthesis in lakes depend on the amount of light that penetrates the lake water, because photosynthesis mostly take place in the presence of sun light. Therefore the answer is D.

46. During aerobic respiration, what is the route through which protons return from the mitochondria inter membrane space beck to its matrix?
A. proton pump C. ion channel
B. ATP synthase D. membrane lipid

Explanation: During the Electron transport chain of aerobic respiration, the protons accumulate in inter membrane space and creating a higher concentration there than the matrix. This proton gradient results in protons diffusing through the ATP synthase molecule which is called chemiosmosis. Therefore the Answer is B.

47. What does the absence of chloroplasts in a unicellular organism indicate?
A. Reproduction by sexual means
B. Autotrophic nutrition
C. Heterotrophic nutrition
D. complete lack of metabolic processes

Explanation: The absence of chloroplast in all living organisms including unicellular organisms are related with heterotropic nutrition. The presence of chloroplast is used to trap sun light and able to produce their own food by the process of photosynthesis. Therefore the answer is C

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48. Which of the following did the early heterotrophs do that probably assisted the origin of autotrophs?
A. They added oxygen and chlorophyle to the environment
B. They added carbon dioxide to the environment
C. They stored energy in the bonds of inorganic compounds
D. They manufactured food from carbondioxide and oxygen

Explanation: The early hetrotrops provide carbondioxide for the autotropic organisms. Auto trophs are able to make their own food by the process of photosynthesis. Then carbondioxide is one of the raw material for the process of photosynthesis. Therefore the answer is B

49. Which of the groups of bacteria consists of rod shaped cells?
A. Gram-positive
B. bacilli
C. cocci
D. spirilla

Explanation: Bacilli (singular bacillus) are rod shaped bacteria
The answer is B

50. Which of the followings do viruses share with cells?
A. cell wall
B. cytoplasm
C. Nucleic acid
D. protoplasm

Explanation: Viruses do not possess cell wall, cytoplasm and protoplasm. They only shared nucleic acid with other living cells.
Therefore the answer is C

51. Which of the following is true about protozoa?
A. They are a group of bacteria
B. They are one celled plants
C. They represent both plants and animals
D. They are unicellular animals

Explanation: The protozoans form a large group of kingdom protoctista, kingdom protoctista are grouped in to algae (plant like) and protozoa (animal like) eukaryotes. Protozoans are unicellular, animal like cells with heterophic nutrition. Therefore the answer is D

52. Choose the organisms that belongs to the eukaryotes?
A. Bacteria
B. Blue green
C. Amoeba
D. virus

Explanation: Bacteria and blue green algae found in kingdom monera and they are prokaryotic unicellular organisms. Viruses are neither prokaryotic nor eykaryotes. Amoeba are protozoans and they are eukaryotic organisms.
Therefore the answer is C

53. Choose the characteristics that viruses share with living things.
A. They are made up of many specialized cells
B. They contain genetic material
C. They reproduce by mitosis
D. They contain chlorophyll

Explanation: Viruses share genetic material which is nucleic acid such are DNA or RNA but not both. The answer is B

54. Which stage in the life cycle of HIV is inhibited by the antiretroviral drug known as integrase – inhibitor?
A. conversion of viral RNA to DNA
B. Replication of the viral genome
C. Insertion of the viral genome in to host genome
D. Release of the viral progeny from the cell

Explanation: There are several different antiretroviral drugs that act in different ways at different stages of the life cycle of reproduction
For example: 1) Reverse transcriptase inhibitors attach to HIV's enzyme preventing it from converting RNA to DNA.
2) protease inhibitors interfere with the action of HIV's protease enzyme preventing the assembly of new viruses particles.
3) Integrase inhibitor preventing the insertion of the viral genome in to host genome. Therefore the answer is B

55. From which organism is the first antibiotic used in medicine produced?
A. Gram negative bacteria
B. members of the fungi
C. Genetically modified protozoa
D. higher plants

Explanation: The first antibiotic which is called penicillin was discovered in 1928 by Dr. Alexander Fleming
Alexander Fleming was growing the mold penicillium notatum on some of the petridish. Penicillium notatum is a unicellular fungi.
Therefore the answer is B

56. What is the difference between HIV and AIDS?
A. AIDS is the virus and HIV is the disease
B. Both AIDS and refer to the disease
C. AIDS is the disease and HIV is the virus
D. Both AIDS and HIV refer to the virus

Explanation: AIDS (Acquired Immune Defiance Syndrome) in the disease and caused by HIV (Human immune Deficiency Virus)

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which attack our immunity system specially T-helper cells. Therefore the answer is C

57. Biotechnology can be applied in environmental remediation. To which of the major application areas of biotechnology is this most related?
A. Medicine and nutrition C. Biodiversity and industry
B. medicine and industry D. Agriculture and environment
Explanation: Medicine and industry are the major application areas of biotechnology. Biotechnology in medical products include, hormone (insulin), human growth hormone, drugs, antibiotics, monoclonal antibodies and vaccines. Industry products include, food industry, brewing industry in dairy farm industries etc. Therefore the answer is B

58. Which of the following parts of plant roots is harbouring nitrogen fixing bacteria?
A. Root nodule C. Root tip
B. Root hair D. Root cap
Explanation: Nitrogen fixing bacteria belonging to the genus Rhizobium are mostly found in nodules on the roots of legume plants. Then the answer is A

59. Which of the following statement is true about matter and energy in the ecosystem?
A. Both matter and energy are recycled
B. Matter is not recycled; energy is recycled
C. Matter is recycled; energy is not recycled
D. energy is recycled more than matter does
Explanation: Energy is continually lost as heat to the ecosystem and has to be replaced as light, where as nutrients or matter recycled in the ecosystem. Energy is not recycled. Energy flow in the ecosystem unidirectional and matter or nutrients recycled multidirectional. Therefore the answer is C

60. Which of the following forms of nitrogen is readily utilizable by green plants.
A. Atmospheric nitrogen C. ammonion ion
B. nitrite D. Nitrate
Explanation: Animals obtained nitrogen from the food which they consumed to build up their own proteins. Plants obtained nitrogen in the form of nitrate in the soil. Then the answer is D

61. In which one of the following ways the Tundrabiome differs from desert biome?
A. It is a treeless biome C. It receives little precipitation
B. It has little biodiversity D. It has permanently frozen subsoil
Explanation: Desert biome is divided in to hot desert and cold desert, when comparing with tundra regarding to the soil type thee tundra biome contains permafrost or frozen type than poor fertile soil desert biome. Therefore the answer is D

62. In Ethiopia, which one of the following vertebrate groups has the largest number of genera?
A. mammals B. Birds C. Reptiles D. Amphibians
Explanation: Ethiopia is an important regional center of biological diversity. This is because of having the wide range of altitude and climate. Ethiopia has many plant species and animal species. Regarding to the vertebrate biodiversity the country has a large number of genera of birds than other vertebrate.
Then the answer is B

63. Which of the following applies to the tropical rainforest?
A. Low rain fall and high temperature
B. Low rain full and low temperature
C. High rain fall and high temperature
D. High rain fall and low temperature
Explanation: In the tropical rain forest became has the characteristic of high rain fall, always warm temperature poor soil, many plant and animal species. Therefore the answer is C.

64. Which of the following resources is need by an ecosystem from out side in order to sustain itself?
A. produces C. micro nutrients
B. macro nutrients D. sun light
Explanation: To sustain life in an ecosystem the sun light is the main resources for all living things. Because, sun light is the primary condition for photosynthesis for produces to make their own food. The answer is D

65. Which of the following can create strong inter specific competition is shared by two or more species?
A. Habitat B. Ecosystem C. Niche D. Predator
Explanation: Habitat is an area where a population lives and finds the nutrients, water, living space and other essential needs to survive. Ecosystem is the interaction between the biotic (living) and a biotic

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(non living) components in the given area. In the given habitat organisms have their on function. The role of organism in the ecosystem is ecological niches. Which create strong inter specific competition. Therefore the answer is C.

66. Suppose sites A, B, C and D have Simpson's. Index of diversity (d) values of 20, 8, 16 and 3 respectively, which site is dominated by one or just a few species.

A. site A B. site B C. site D. site D

Explanation: Site D is dominated by on species. This is because a low value for the index of diversity suggests that an area dominated on or just a few species. Therefore the answer is D.

67. Among the following, which one has little or no contribution to the loss of biodiversity?

A. Overgrazing by domestic animals
B. Clearing forests to grow crops
C. Growing crops of improved varieties
D. growing crops land races

Explanation: From the given choice, overgrazing, clearing forests and growing crops of improved varieties have a great contribution to the loss of biodiversity. Growing crops landraces has a little contribution to the loss of biodiversity. Therefore the answer is D.

68. Which of the following is the largest ecological unit?

A. A community C. A population
B. A biome D. An ecosystem

Explanation: Ecosystem is the largest ecological of the plant, because it is made up of biotic (community and population) and a biotic (physical) component of the planet. Therefore the answer is D

69. What is the ecological succession in which plant communities establish after an area has been completely demolished by fire?

A. Tree succession C. primary succession
B. secondary succession D. Bare rock succession

Explanation: Succession is the process of changing from one types of community to another community. There are two types of successions. These are primary and secondary succession. Primary succession is in which a community of animals and plants develops on the bare land or rock. Secondary succession is in which community of organisms is disturbed by natural or human related events for example hurricane, volcano, fire, forest harvest etc, and

return to the previous stage in the succession. Therefore the answer is B.

70. Which of the following groups of animals have males with zz and females with zw sex chromosome constitution?

A. Grass hoppers B. Birds C. mammals D. honey bees

Explanation: There are reversal systems of sex determination in different organisms. For example in humans, the sex chromosomes of females are homogametic, which designated by letter xx and mates are hetero gametic which designated by xy. In birds the females are hetero gamatic represented by the letters of zw and the mates are homogametic which are zz. Therefore the answer is B.

71. In some crosses of maize, the progeny produces better yield than the parents. What is this phenomenon known as?

A. Dominance C. Hybrid vigour
B. Heterozygosity D. In breeding depression

Explanation: Hybrid vigour occurs when unrelated breeds or lines of the same species are cross bred. This hybrid vigour cases the offspring from this crosses are more productive than the average of their parent breeds. The answer is D.

72. In across between round grean pea of RRyy genotype and wrinkled yellow pea of rrYY genotype, what is the expected genotype of the F_1?

A. RRyy B. rrYY C. RRyy D. RrYy

Explanation: The inheritance of two genes at the same time is called dihybrid cross. Then when you cross between round green pea with the genotype of RRyy and wrinkled yellow pea of rrYY phenotype. The expected genotype of F_1 is RrYy. The answer is D

73. Which of the following is more true about the mall of the honey bee?

A. It has no father C. It is diploid
B. It is sterile D. It regulates the colony

Explanation: In honey bees, a queen bee mates only once and stores the sperm, she thin can control the release of the sperm.

⇒ If no sperm are released the eggs develop parthenogenetically in to drones, which are males.
⇒ If sperm are allowed to fertilize the eggs the fertile eggs develop in to other queens or workers bees which are females.
⇒ The queens and workers are diploid and drones (mates) are haploid.

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The queen and drones are fertile and the worker (females) are sterile. The male drones are produced by unfertilized eggs are called parthenogenesis. Therefore the answer is A

74. In to which of the following organic compounds can lipase, maltase and sucrose be grouped?
A. Hormones
B. carbohydrates
C. nucleic acid
D. enzymes

Explanation: These organic molecules are grouped in enzymes because all their suffix are ended by ase. Then the answer is D

75. In which of the activities of the honey bee colony are worker bees involved?
A. Laying eggs
B. serving as queens
C. Reproductive functions
D. Taking care of the carve

Explanation: The honey bees are social animals and have division of labor among themselves. For example, a few individuals of queens and drones (males) specialize in reproduction.

⇒ Whereas large numbers of sterile worker honey bees are involved in collecting food, defending the hive, and caring for the larve Therefore the answer is D.

76. Which one of the following is the mechanisms by which two genes located on the same chromosome are separated
A. Independent assortment
B. segregation
C. linkage
D. crossing over

Explanation: Mendel formulated several genetic laws to described how characteristic are passed from one generation to the next.

⇒ These mendel laws are

1. **Mendel's law of dominance**
   - When an organism has two different alleles for a given trait, the allele that is expressed, overshadowing the expression of the other allele is known as dominant.

The gene whose expression is overshadow wed is said to be recessive.

2. **Mendel's law of segregation:** When gametes are formed by a diploid organism, the alleles that control a trait separate from one another in to different gametes, retaining their individuality, because they are monohybrid crosses.
3. Mendel's law of independent assortment
   - members of one gene pair separate from each other independently of the members of other gene pairs. Here the

two genes located on the same chromosomes are separated and which is called dihybrid cross can take place. Therefore, the answer is A

77. During protein synthesis, where in the cell does transcription take place?
A. ribosome
B. Nucleus
C. Endoplasmic retialum
D. Golgi apparatus

Explanation: Transcription is the process in which the genetic information converts from a DNA code in to a mRNA code.

⇒ It take place in the nucleus.

Translation is the process in which the mRNA code is converted in a sequence of amino acids, it take place in the cytoplasm where ribosome are located. Therefore the answer is B.

78. Which of the following is not a term used to describe organisms that have had foreign genes added to them?
A. Genetically modified organisms
B. Transgenic organisms
C. Genetically engineered organisms
D. pathogenic organisms

Explanation:

⇒ From the given choice, genetically modified organisms, transgenic and genetically engineered organisms are the same terms (phases) which described, adding of foreign genes in organisms, such as in plants and animals to improve the desired characteristics.

⇒ Where as pathogenic organisms, those organisms which cause different diseases in plants, animals and including human beings. Therefore the answer is D

79. Which field studies the way in which individual traits of organisms are transmitted from one generation to the next?
A. Genetics
B. Ecology
C. Evolution
D. morphology

Explanation: Ecology is the branch of biology that studies about the relationships between organisms and their environment. Evolution is the changes in the frequency of genetically determined characteristics with in a population over time.
- morphology is the study to of the shape, size and the structure of living organisms.

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Genetics is also the branch of biology that studies of genes, how genes produce characteristics and how the characteristic are transmitted from one generation to the next. Then the answer is B.

80. Which molecules carry the instructions for protein synthesis?
A. carbohydrates and lipids
B. Amino acids
C. DNA-and RNA
D. Enzymes

Explanation: DNA and RNA are molecules that carry the instruction for protein synthesis. Therefore the answer is C

81. In protein synthesis, what is produced during transcription?
A. protein B. mRNA C. DNA D. polypeptide

Explanation: Transcription is the process that converts genetic information from a DNA code in to an mRNA code. In this process

$$DNA \xrightarrow{transcription} mRNA \xrightarrow{translation} proteins$$

⇒ Then the answer is B

82. What is a perfect flower?
A. A flower with petals
B. A flower with stames and sepals.
C. A flower with stamens and pistil
D. A flower with sepal and petals

Explanation: A flower which contains both male (stamens) and the female (pistil) part are called perfect flower. Where as a flower which contains male or a female part is called imperfect flower. So the answer is C

83. Which of the following types of chromosome mutations gives rise to an extra chromosome?

Explanation: Mutation is any change of nucleotide sequence of DNA in the genetic information. There are four types of chromosomal aberrations. These are:

1. Inversion = occurs when a chromosome is broken and a piece becomes reattached to it's the original chromosome.
2. Duplication occur when a portion of a chromosome is replaced and attached to the original section in sequence.
3. Deletion occurs when a broken piece becomes cost or destroyed before it can attached.
4. Translocation occurs when on broken segment of DNA becomes integrated in to a different chromosomes. From the given choice the answer is B.

84. Which of the following pairs of individuals have identical DNA finger prints?

A. Father and son
B. mother and daughter
C. Brother and sister
D. monozygotic twins

Explanation: An individual's finger prints are unique and do not change throughout life. An identical twins have identical DNA finger prints. Therefore the answer is D

85. Which of the following mating produces children all having the same ABO blood phenotypes?
A. AO × AB
B. BB × OO
C. AO × BO
D. AA × BO

Explanation: In mating process those children producing all having the same ABO blood phenotypes are blood group AO × BO. This mating type produce all blood types. By using pennet squr.

	B	O
A	AB	AO
O	BO	OO

Then it is possible for two parents with blood groups A and B to have four children, each with a different blood group. Their children may have the blood type AB, AO(A) and OO or AB, A, B and O blood. Therefore the answer is C

86. If a DNA contains, 10% thy mine what is the percentage of cytosine in the DNA?
A. 10% B. 30% C. 40% D. 90%

In the DNA molecules Adenine (A) is paired with thymine (T), then A = T and Guanine (G) is paired with cytosine (C) G = C
The percentage of thymine is equal to the percentage of adenine = A(10%) + T(10%) = 20%
The percentage of Guanine is equal to the percentage of cytosine
100% - 20% = 80% = Then percentage of cytosine is 40%. Therefore the answer is C

87. What is the branch of biology that studies about the origin and gradual changes of living things?
A. microbiology
B. Evolution
C. mutation
D. Reproduction

Explanation: Microbiology is a branch of biology that studies about microorganisms. Mutation is the change of the DNA sequence of organisms. Reproduction is a process in which the production of a new generation of individuals of the same species. Evolution is also the branch of biology that studies about the origin and gradual changes of living organisms. Therefore the answer is B

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88. Which of the following techniques is used to separate DNA fragments according to their size on a gel?
A. Southern blotting C. Radioactive labeling
B. Electrophoresis D. Digestion by restriction enzyme
Explanation: Restriction enzymes have the ability to cat DNA at places where specific sequences of nucleotides occur. The fragments are separated by gel electrophoresis. This is because the smaller fragments move further than larger molecules. Therefore the answer is B.

89. Which of the following can be taken as case for evidence of evolution from the field of plant and animal breeding?
A. Artificial breeding always creates new species
B. Artificial breeding only produces new organisms with higher yield
C. Artificial breading shows that natural selection can produce variations.
D. Artificial breeding can produce new variations in organisms
Explanation: Artificial breeding is used to produce the variation of domestic animals and plants with desirable characteristics such as improving the quality and quantities of yield, disease and drought resistance varieties and eliminating undesirable characteristics. Therefore the answer is D

90. If a radioactive substance that weighs one kilogram has a half life of 100 years what would be the percentage of the substance left after 300 years?
A. 50 B. 30 C. 25 D. 12.5
Explanation: When the radioactive of substance increase in duration or time it decrease it percentage and reduce its weight.
For example:- half timed ½ (50%) ½ is equal to 100 years
(¼) 25% is also 200 years the (1/8) = 12.5% is 300 years
Therefore the answer is D

91. According to lamark's theory of evolution, what is the mechanism by which evolving organisms acquire new structures?
A. mutation C. use-and-disuse of parts of the body
B. heredity variation D. Recombination of ancestral genes
Explanation: The French biologist Lamark proposed in 1809, a hypothesis to account for the mechanism of evolution based on two conditions. These conditions are
1. The use – and disuse of parts
2. The inheritance of acquired characteristics.

Therefore the answer is C

92. In which kingdom of life, is it most likely to find many examples of organisms that can undergo metamorphosis?
A. monera B. Fungi C. plantae D. Animalia
Explanation: Metamorphosis is derived from two words, meta and morphe meta means change and morphe is form. Then metamorphosis is a process in which a change of form or structure of the animal during the course of its life cycle. For example some insects can show complete metamorphosis. These insects have four stages of development, including egg, Larvae, maggot and imago (adult) eg. house fly, others can show incomplete metamorphosis, this include egg, nymph and adult eg. grass hopper and locust. Then the answer is D

93. What do you call structures that have the same evolutionary origin but now are different in structure or have different functions?
A. Endemic B. Analogous C. Homologous D. Indigenous
⇒ Organisms with similar anatomical features are closely related evolutionary because, they shared a common ancestor.
⇒ Some organisms have anatomical structures that are very similar inform and evolutionary origin, but have very different function. Such structures are called homologous structures.
⇒ Other organisms having the same function but different anatomy and different evolutionary origin. These structures are called analogous structures. Therefore the answer is C.

94. Which of the following is prevented from taking place if populations are separated by a geographic barrier?
A. mutation B. Evolution C. Gene flow D. Natural selection
Explanation:
- mutation is the change of the sequence of DNA
- Evolution is the gradual change of organisms through the over period of time.
- Natural selection is the theory that explain of species in terms of survival of those best adapted to a specific environment.
- Gene flow is the movement of genes from generation to the next or from one region to another.
- Then the answer is C

95. The half life of carbon -14 is about 5730 years. If a fossil is 17200 years old, about what percent of its original carbon – 14 is still present in the fossil?

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A. 75% B. 50% C. 25% D. 12%

Explanation: Half life ½ or 50% = 5730 years.

¼ or 25 , 11460 years or 5730 × 2

1/8 or 12.5% = 5730 × 3 = 17190 it is around 17200years, then the percentage is 12.5%. The answer is D

96. Which of the following pairs are analogous structures?

A. wing of a bird and wing of butter fly
B. front leg of a horse and a human arm
C. Awing of bird and a wing of bat
D. front leg of a frog and a wing of but

Explanation: based on the anatomical structure organisms are classified in to homologous and analogous structures.

⇒ Homologous structure are those structures having similar origin and the same form but have different functions.

For examples:- the human arm used for handling lools

- the front leg of a horse used to riding
- the front leg of a frog used for jumping
- the wings of birds and bal used for flying
- the flipper of wholes for swimming

These evidences can tell as these organisms can arose on the same ancestor.

⇒ Analogous structures are those structures have different origin but have the same function:

For example:- the wing of birds and the wing of butterfly. Therefore the answer is A.

97. Among the following fossil hominid species, which one is the oldest of all?

A. Homo erectus C. Australopithecus afarensis
B. Ardipithecus ramidus D. Australopthecus africanes

Explanation: There is a great deal of fossil evidence that several species of hominids of the genera ardipithecus, Australopithecus and genus homo. The earliest fossil record stats from ardipithecus ramidus (selam) found before 5 million years ago, australopithecs a farensis (Luci) around 3 million years ago

⇒ After Australopithecus afarensis it develop in to Australopithecus Africans.

Genus Australopithecus can developed in to genus homos such as homo habilis, homo erectus and then to homo sapiens.

Then from hominid species ardipithecus ramidus or selem is the oldest of all gnera of human ancentors. Therefore the answer is B

98. Which of the following conclusions can be drawn from the structural similarities observed between the flippers of wholes and the arms of humans?

A. The human species began life in the ocean
B. The human species and whales share a common ancestry
C. Whales are older than the human species
D. whales have evolved from early humans that went back to the ocean

Explanation: The flippers of whales and the arms of humans have similar anatomical strucutes and closely related evolutionary. This is because, they shared a common ancesters. These anatomical structures are called homologous structure. For example – the flippers of whales used for swimming where as the arms of humans for manipulation (handling) therefore, the answer is B

99. An experimental animal stopped responding to a stimulus that has been repeated to many times. What type of learned behavior is this?

A. Latent learning C. Imprinting
B. Habituation D. Operant conditioning

Explanation:

⇒ Latent learning is a hidden learning and it happens when the brain acquires knowledge at a certain time without reinforcement.

⇒ Imprinting is not learned behavior, it is innate behavior and instinctive behavior which inborn with an organism.

⇒ Operant conditioning is classical conditioning involves modifying an innate response by pairing it with a previously neutral stimulus.

⇒ Habituation is a type of learned behavior results in a decreased (stopped) response to a stimulus after repeated exposure to that stimulus over a period of time. Therefore the answer is B

100. Behaviour that is repeated on a daily basis is referred to as:

A. Lunar B. Circadian C. circannual D. seasonal

Explanation: Monthly cycle is lunar shythums, yearly rhythms is called cireannual rhythm. Daily rhythms are called circadian rhythms.

Answer: B

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DIRECTIONS: Each of the following questions is followed by four possible alternatives. Read each question carefully and **blacken** the letter of your choice on the separate answer sheet provided.

1. Which of the following step of the scientific method comes before all the rest?
   A. Hypothesis C. Conclusion
   B. Experiment D. Reporting the result

   Explanation: In scientific method the correct sequential order of the steps involved is:
   Hypothesis → Experiment → Reporting the result of the experiment → Conclusion **Answer: A**

2. What do we call a scientist who specializes in studying insects
   A. An ecologist C. An entomologist
   B. A paleontologist D. A microbiologist

   Explanation:

Scientist	Field of specialty
Ecologist	Environment study
Paleontologist	Studies about fossils
Entomologist	Studies about insects
Microbiologist	Studies about microorganisms

   Answer: C

3. To which one of the following organic molecules do enzymes belong
   A. Carbohydrates C. Proteins
   B. Amino acids D. Lipids

   Explanation: Enzymes are biological catalysts which are composed of globular proteins. **Answer: C**

4. Which of the following is a coenzyme?
   A. NAD C. Water
   B. Systematic errors D. Proteins

   Explanation: coenzymes are the non-proteins parts a cofactor which used to bind to an enzyme to give it its catalytic activity. The most commonly known examples of coenzyme are **NAD** (Nicotine amide adenine dinucleotide) and **FAD** (Flavin adenine dinucleatide)
   Answer: A

5. Which of the following units of measurement is more convenient to express the size of cellular organelles?
   A. Meter C. Millimeter
   B. Centimeter D. Micrometer

   Explanation: cells vary by their size some cells like egg cells of birds are large and can be easily seen by our naked eyes. However,

most cells are small and can only be seen by microscope. Such cells are measured by a unit called **micrometer (μm).**

6. Which tool of the biologist is more frequently used to study cell structure in the laboratory? **Answer: D**
   A. Dissecting kit C. Petri-dish
   B. Centrifuge D. Microscope

   Explanation: All the equipments listed are important in biological studies. However, microscope is the basic tool of a biologist which is most frequency used in biological laboratories. **Answer: D**

7. What colour do gram-positive bacteria stain with Gram's stain?
   A. Red C. Purple
   B. Pink D. White

   Explanation: the major difference between Grham positive and Grham negative bacteria is that, Grham positive bacteria have ability to easily interact and retain their primary stains. Hanu Christian Gram stained gram-positive bacterial purple.
   Answer: C

8. Which group of organisms in the ecosystem release nutrients locked up in dead bodies of organisms?
   A. Parasites C. Autotrophs
   B. Decomposers D. Carnivores

   Explanation: Decomposes (bacterial and Fungi) play a role in recycling activities by breaking down dead bodies of organisms, inorder to return nutrients locked in organisms body back to the nature. **Answer: B**

9. Which one of the following are the two major constituents of eukaryotic chromosomes?
   A. DNA and RNA
   B. DNA and protein
   C. DNA and carbohydrate
   D. RNA and lipid

   Explanation: the chromosome of eukaryotic cells contain DNA that is coined around proteins called histones. **Answer: B**

10. Which one of the following behavioral biologists is known for his study about imprinting behavior in animas?
    A. W.Kohler C. Ivan Pavlov
    B. B.F. Skinner D. Konrad Lorenz

    Explanation: Konrad Lorenz imprinted the goslings on himself during their sensitive phase. They formed along lasting bon to him and followed him readily.

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When an animal or person learns the characterstics of some stimulus, it is said to be "imprinted" on to the subject.

Answer: D

11. Which group of animals has the highest numbers of total and endemic species in Ethiopia?
A. Amphibians
B. Birds
C. Mammals
D. Reptiles

Explanation:

Animals endemic to Ethiopia	Number of species
Birds	27
Mammals	22
Amphibians	17
Fish	4
Reptiles	3

Answer: B

12. What is the molecule that supplies the quickest and suitable source of energy to cells?
A. Lactose
B. Sucrose
C. ATP
D. Lipid

Explanation: **ATP**(Adenosme triphosphate) is the immediate or the cash form of energy that can be directly used by our cells.

Answer: C

13. A carbohydrate compound is known to have 12 carbon atoms in the whole molecule. What could this compound be?
A. A polypeptide
B. A disaccharide
C. A monosaccharide
D. A polysaccharide

Explanation: Disaccharides are groups of carbohydrate that have two sugar units. One sugar unit contains 3 – 8 carbon atoms. The most commonly known examples of disaccharides are:

lactose
sucrose } $C_{12}H_{22}O_{11}$
maltose

Answer: B

14. Among the following organisms, which one belongs to the prokaryotes?
A. Paramecium
B. *Streptococcus*
C. Spirogyra
D. Tapeworm

Explanation: Prokaryotes are groups of organisms that have nuclear material lacking nuclear membrane, thus they freely suspend in the

cytoplasm. In addition, prokaryotes lack some organelles which are found in eukaryotic cell and they also lack double membrane bounded organelles. Prokaryotes belong to all organisms kingdom protista (bacteria and blue green).

Answer: B

15. Which one of the following crops has its centre of origin and diversity in Ethiopia has become a leading international commodity of commerce?
A. Teff
B. Enset
C. Coffee
D. Anchote

Explanation: All the plants listed are originated form Ethiopia. Above all, coffee (coffee Arabica) which is originated form Ethiopia has nowadays become the leading international commodity of commerce.

Answer: C

16. Baby ostriches tend to follow the first moving object that they see as they hatch out of the eggs. What kind of animal behavior does this demonstrate?
A. Positive taxis
B. Innate behaviour
C. Positive kinesis
D. Learned behaviour

Explanation: Innate behaviour is the behaviour that is genetically determined and the behaviour is fully functional at first attempt. Thus, the ability of baby ostriches to follow the first moving object that they see as they hatch out of eggs is an example of innate behavior.

Answer: A

17. Which of the following is true about mutations that occur in normal body cells?
A. They never lead to cancerous cells.
B. They never pass to the next generation.
C. They never damage the affected cells.
D. There is no way that they kill the affected cells.

Explanation: Mutation is a sudden change in the amount, arrangement or structure of the DNA of an organism. It may occur on mormal body cells (somatic cells) or sex cells (gametes). If mutations occur on somatic cells they do not pass to the next generation, but, mutation on sex cells are inheritable.

Answer: B

18. What is the process called when two bacteria directly contact cell to cell and exchange their genetic information?
A. Conjugation
B. Transformation
C. Co-transformation
D. Transduction

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Explanation: Bacterial conjugation is the transfer of genetic material between bacterial cells by direct cell to cell contact or by a bridge-like connection between two cells. **Answer: A**

19. Why is it that mutations are considered as one of the raw materials of evolution?
   A. They contribute to new variations in organisms.
   B. They are usually related to the environment in which they appear
   C. They are mostly beneficial to the environment in which they appear
   D. They usually become the causes for species extinction.

Explanation: If a mutated gene is advantageous to the organism, its frequency increases in successive generations. The number of organisms with the new gene will also increase as natural selection favours them. Thus, mutations are the raw materials for evolution because they contribute to variation. **Answer: A**

20. Which one of the following is consistent with understanding of human evolution?
   A. Bipedalism was never important in human evolution.
   B. Larger brain size had no contribution to the evolution of the human species
   C. Human ancestry had no relation whatsoever with that of the chimpanzees.
   D. Fossils of Lucy and Ardi provided evidence for human origin.

Explanation: Lucy and Ardi are both important explanation of modern man and chimpanzee evolution. Lucy was bipedal and the brain of the same size as chimpanzee. Ardi is bipedal and closer to the common ancestor of human and chimpanzees. **Answer: D**

21. If certain food stuff gave a positive result with Benedict's test only after it was hydrolyzed with hydrochloric acid, which of the following substances could it be?
   A. Non-reducing sugar C. Lipid
   B. Nucleic acid D. Protein

Explanation: Non reducing sugars such as are disaccharides such as sucrose. This, substances can get hydrolyzed when they react with hydrochloric acid. **Answer: A**

22. In scientific method what is the importance of a background research for a given problem
   A. To formulate a hypothesis
   B. To make a prediction
   C. To prepare report of the result
   D. To make a conclusion

Explanation: Inorder to formulate a hypothesis the researcher must have to do a back ground research for a collecting sufficient information about the problem under investigation. **Answer: A**

23. Which of the following is NOT true about mitochondria and chloroplasts?
   A. Both contain chlorophyll
   B. Both contain nucleic acid
   C. Both have double membrane
   D. Both transduct energy

Explanation:

Mitochondria	Chloroplast
✓ Have no chloroplast ✓ Have its own nucleic acid (DNA) ✓ Have double membrane energy transducting organelles.	✓ Contain chloroplast ✓ Have its own nucleic acid (DNA). ✓ Have double membrane ✓ Energy tranducting organelles.

Answer: A

24. Which of the following molecules is NOT a polymer? ?
   A. Protein C. RNA
   B. Lipid D. Starch

Explanation: Polymer is a molecule that is made from many smaller identical molecules called monomers.

Polymer	Monomer units
Protein	Aminoacids
RNA	nucleotides
Starch	Simple sugars (glucose

∴ Lipids are composed of two different types of molecules, fatty acid and glycerol. Thus, they are not ~~polysaccharides~~ polymers **Answer: B**

25. Which of the following is formulated as a possible outcome of an experiment?
   A. Hypothesis C. Data analysis
   B. Prediction D. Recording the data

Explanation: Hypothesis is a tentative explanation about an outcome of an experiments, which will be accepted or rejected as a result of the experiment. Hypothesis is sometimes called an educated guess: **Answer: B**

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26. Which one of the following classes of molecules CANNOT pass easily across the cell membrane by simple diffusion?
A. Small non polar molecules
B. Lipid soluble molecules
C. Non-polar molecules
D. Polar molecules

Explanation: Lipid soluble molecules are too large to pass easily across the plasma membrane by simple diffusion. **Answer: B**

27. Which one of the following substances has a shape which is similar to that of the substrate of an enzyme?
A. The reaction product
B. A competitive inhibitor
C. A cofactor
D. An allosteric inhibitor

Explanation: competitive inhibitors are molecules that have similar shape to that of the substrate. They reduce enzymic reaction by preventing substrate form binding to the active site taking its place.
Answer: B

28. At which level of structural organization do proteins have the alpha-helix shape?
A. Primary structure
B. Secondary structure
C. Tertiary structure
D. Quaternary structure

Explanation: In secondary structure of protein a peptide chain folds in to a α -helix plated sheet. Such as wool, hair, skin, nails etc.
Answer: B

29. Among the following scientists who contributed to the cell theory, identify the one who stated that '*a cell can arise only from another cell like it*'.
A. Robert Hooke
B. Theodor Schwann
C. Matthias Schleiden
D. Rudolf Virchow

Biologist	Contribution in cell biology
Robert Hooke	Discovered cells and coined the name cell
Theodor Schwann Matthias Schleiden	Formulated the cell theory
Rudolf Virchow	Developed on idea which says that' cells arise only from previously existing cells.

Answer: D

30. What does it mean when biologists express the cell membrane as a unit membrane?
A. A cell is covered by a single membrane.
B. A membrane is only one lipid layer thick.
C. All cells have essentially similar membrane.
D. A membrane is covered by a single layer of protein.

Answer: D

31. Which one of the following features makes human T-lymphocyte cells more vulnerable to HIV attack? Presence of;
A. Cell membrane.
B. DNA on the chromosomes.
C. CD4 on the membrane surface.
D. Pores in the cell membrane.

Explanation: HIV infects T-helper cells, specialized of the immune system. The virus attaches to protein receptors know as CO_4 on the host plasma membrane. **Answer: C**

32. Which one of the following terms refers to how fast an enzymes acts on its substrate?
A. Turn over number
B. Substrate number
C. Enzyme number
D. Product number

Explanation: Turn over rate is the number of enzyme that binds to substrate within a given period of time **Answer: A**

33. Which one of the following factors determines the rate at which organelles settle out of cell homogenate if spun in a centrifuge?
A. Mass of the organelle.
B. Function of the organelle in the cell.
C. Location of the organelle in the cell.
D. Thickness of the membrane covering the organelle.

Explanation: Centrifuge is a machine that spins at high speed to separate components of different molecular weights. **Answer: A**

34. Which of the following classes of organic molecules is the least important source of energy for cellular respiration?
A. Nucleic acids
B. Lipids
C. Carbohydrates
D. Proteins

Explanation: Carbohydrates and fats are the primary source of energy for the body. Nucleic acids (RNA and DNA) contain a five carbon sugar (carbohydrate). However, proteins are concerned with building and repairing the body, but are not source of energy.
Answer: D

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35. Which one of the following modes of transport is used by cells to move substances against their concentration gradients?
A. Osmosis
B. Simple diffusion
C. Facilitated diffusion
D. Active transport

Explanation: Osmosis and diffusion are passive transport systems that involve the movement of substances along the concentration gradient. However, active transport system occurs against the concentration gradient. **Answer: D**

36. Which of the following is **NOT** true about instinctive behavior?
A. It can be developed further through learning.
B. It is triggered by a key stimulus.
C. It has a fixed action pattern.
D. It is adaptive for the species.

Explanation: **Instnictive behavious** are generally pre programmed or pre-set, unlearned behaviours. They donot require learning or experience. **Answer: A**

37. In which of the following ways do retroviruses differ from other RNA viruses?
A. Their genetic material is DNA.
B. Their genetic material is RNA.
C. They copy RNA to DNA molecule.
D. They copy RNA from DNA molecule.

Explanation: Retro viruses have RNA as their genetic material in their host. They convert their RNA in to DNA by the process called reverse transcription. **Answer: C**

38. Which one of the following groups of micro-organisms does not contain parasitic members?
A. Bacteria
B. Algae
C. Fungi
D. Protozoa

Explanation: Algaes are not parasite, because all algaes have chloroplast and are photosynthetic. **Answer: B**

39. Which one of the following is NOT the correct characters of tropical rainforests?
A. Low biodiversity
B. High temperature
C. Heavy precipitation
D. Trees of different heights

Explanation: Tropical rain forest is a biome which gets the highest amount of rainfall than any other biomes. Thus, it contains variety of plant and animal species. Accordingly, tropical rain forest it the richest biome by its biodiversity. **Answer: A**

40. Which one of the following processes has a decreasing effect on the concentration of atmospheric carbon dioxide?
A. Cellular respiration
B. Combustion of fossil fuels
C. Decomposition of dead organisms
D. Photosynthesis

Explanation: the process of photosynthesis requires carbondixoide as one of the raw materials to synthesize sugar. Thus, the process of photosynthesis has declining effect on atmospheric CO_2. **Answer: D**

41. For what purpose do molecular biologists use the technology known as polymerase chain reaction or PCR?
A. To insert DNA into plasmids
B. To insert plasmid into bacteria
C. To multiply copies of DNA molecule.
D. To produce DNA form RNA.

Explanation: **Polymerase chain reaction** (PCR) is a procedure by which free nucleotides are assembled into a nucleic acid chain by enabling the activity of bacteria DNA polymerase to bind them together. **Answer:**

42. What is the long term primary effect of the current tree planting activities that Ethiopia is undertaking?
A. It will protect from harm of solar rays
B. It will increase the global temperature.
C. It will reduce the atmospheric CO_2.
D. Absence of natural selection

Explanation: Tree planting is vital in order to reduce the level of carbondixoide in the atmosphere which is causes global warming. **Answer: C**

43. Suppose when you first enter a room you notice an unpleasant smell which you eventually forget about its presence, what is this behavior called?
A. Latent learning
B. Insight learning
C. Habituation
D. Operant conditioning

Explanation: Habitation is a process which results in a decreased response to a stimulus after repeated exposure to that stimulus over a period of time. **Answer: C**

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44. Which one of the following is an evolutionary requirement for two sub-populations of a species to evolve into independent species?
A. Free exchange of genev
B. Geographic isolatoin
C. Free migration between populations
D. Absence of natural selection

Explanation: Geographic isolation is an example of allopatric speciation that results when a population from the existing species becomes geographically isolated and the isolated population develops into the new specie. **Answer: B**

45. Form evolutionary point of view, which of the following animals is expected to have hemoglobin proteins that are least similar to that of human?
A. Ape
B. Cow
C. Chicken
D. Frog

Explanation: Human beings have close evolutionary relationship with apes. Thus, there are various biochemical similarities among organisms from closer evolutionary tree. **Answer: A**

46. Which element is found in nucleic acids?
A. Calcium
B. Iron
C. Magnesium
D. Phosphorus

Explanation: Nucleic acids (DNA and RNA) are composed of chain of nucleotides which contain:
✓ Phosphate unit
✓ Nitrogenous bases
✓ Pentose sugar
Answer: D

47. From where do plants get most of their nutrients?
A. Chlorophyll
B. Soil
C. Light
D. Atmosphere

Explanation: plants absorb water and minerals that are used for synthesis of various components in plants. **Answer: D**

48. What do you call a group of genetically identical plants produced by vegetative reproduction?
A. Family
B. Clone
C. Hybrid
D. Genus

Explanation: Cloning is the production of identical organisms by asexual reproduction. **Answer: B**

49. What is the base found in RNA in place of thymine of DNA?
A. Cytosine
B. Guanine
C. Thymine
D. Uracil

Explanation: RNA and DNA are both nucleic acid, but they differ in various ways.

RNA	DNA
✓ Single stranded	✓ Double standard
✓ Contain base uricil instead of thymine	✓ Has thymine base but not uracil
✓ Cannot replicate	✓ It ruplicates
✓ Has ribose sugar	✓ Has deoxy ribose sugar

Answer: D

50. What does a restriction enzyme do?
A. Restricts transcription
B. Cuts DNA at specific sites
C. Prevents DNA from replicating
D. Hydrolyzes the DNA molecule

Explanation: Restriction enzymes are groups of enzymes that are capable of separating DNA at specific points along the chain. **Answer: B**

51. Which one of the following is the smallest of all?
A. A red blood cell
B. A virus
C. A bacterium
D. An amoeba

Explanation: Viruses are the smallest of all in size which ranges in between 0.01 – 0.1μm in diameter **Answer: B**

52. What is the ultimate source of all scientific knowledge?
A. Observation
B. Guessing
C. Trial and error
D. Observation and experimentation

Explanation: The primary source of scientific knowledge is observation, which will be later tested through experimentation. **Answer: D**

53. The optimum temperature of enzymes found in thermophilic bacteria is:
A. Lower than for enzymes in the human body
B. Higher than enzymes in the human body.
C. The same as enzymes found in human body.
D. Lower than enzymes found in warm blooded animals.

Explanation: Thermophilic bacterias are those bacterias that live at extreme temperatures which might be above 70°C. **Answer: B**

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54. A cow was found to yield much higher milk than any of the breeds of the parental cattle. What could be the most probable reason for this?
A. Dominant genes
B. Hybrid vigour
C. Recessive genes
D. Co-dominance genes

Explanation: Hybrid vigour is making two parents of different pure breeds in order to produce an organism that has an increase in such characterstics a rate, fertility and yield over those of its parents.

Answer: B

55. The most complex structure of proteins is called
A. Primary structure
B. Secondary structure
C. Tertiary structure
D. Quaternary structure

Explanation: Quaternary structure of proteins in the most complex one which is formed when two or more polypeptide chains become associated in the final structure of the protein. **Answer: D**

56. Which are the four most abundant elements in living cells?
A. Carbon, oxygen, sulfur, phosphorus.
B. Carbon, oxygen, hydrogen, nitrogen.
C. Carbon, oxygen, nitrogen, sulfur.
D. Carbon, oxygen, sulfur, magnesium.

Explanation: The four most abundant elements in living cells are **carbon, hydrogen, oxygen** and **nitrogen**. Because they are elements that make the organic substances (carbohydrate, lipid and protein).

Answer: B

57. When the F1 hybrid of a monohybrid cross is back crossed with the homozygous recessive parent, what percentage of the offspring would be homozygous recessive?
A. 0%
B. 25%
C. 50%
D. 75%

Explanation: for instance let us take Mendels hybrid tall pea plant when crossed with homozygous short pea plant as an example.

Hybrid tall pea plant = Tt
Homozygous short pea plant = tt

Tt × tt

	T	T	2Tt: 2tt → genotype
t	Tt	tt	50%hybrid tall : 50% short → phenotype
t	Tt	tt	

Answer: C

58. Which of the follwins is an inorganic molecule?
A. $CaCO_3$
B. CH_4
C. $C_6H_{22}O_{11}$
D. $C_{18}H_{36}O_2$

Explanation: All organic compounds necessarily contain carbon and hydrogen.

Answer: A

59. Which one of the following expression is more related to the phrase "survival of the fittest"?
A. Natural selection
B. Mendelian inheritance
C. Gene mutation
D. Inheritance of acquired characteristics

Explanation: The theory of natural selection which is stated by Charles Darwin has for major ideas:
✓ **Fecundity** – producing more offspring than that can possibly survive.
✓ **Variation** – differences among offspring.
✓ **Struggle for survival**
✓ **Survival of the fittest** – better adapted organism can survive among organisms of the same species.

Answer: A

60. In the formation of a macro molecules, what type of reaction should join two subunits together?
A. Hydrolysis reaction
B. Dehydration reaction
C. Denaturation reaction
D. Hydrophobic reaction

Explanation: Dehydration is the combination of two molecules which releases water as a byproduct. **Answer: B**

61. What did Francesco Redi prove through his scientific experiment?
A. Maggots appear spontaneously on foods placed anywhere.
B. Maggots do not appear in foods kept in jars that are protected with a cover.
C. Maggots do not appear in foods kept in open jars.
D. Flies appear spontaneously on rotting meat kept in closed or open jars alike.

Explanation: **Francesco Redi** was able o disprove the theory that maggots could be spontaneously generated from meat. What he did was, he took eight jars, placed meat in all the jars, but covered four of the jars. Maggots were developed in open jars, but did not develop in covered jars.

Answer: B

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62. Which one of the following shows the feeding method of decomposers?
A. Saprobiotic nutrition
B. Autotrophic nutrition
C. Parasitic nutrition
D. Intracellular digestion

Explanation: Decomposers are organisms such as bacteria and fungi which break down dead bodies of organisms. These groups of organisms are called **saprophytes.** **Answer: A**

63. What happens when the carrying capacity of an ecosystem is reached?
A. Excretory product accumulates and population numbers increase.
B. Population number decline rapidly.
C. Population number remain more or less constant.
D. Resources are plentiful and populations shoot up.

Explanation: **carrying capacity** is the condition where the number of organisms of a species which an environment can support at a particular time. At this stage population number stays constant. **Answer: C**

64. In which industrial products is pyruvate fermentation by yeast practically applied?
A. Brewing beer
B. Swiss cheese making
C. Production of vinegar
D. Yoghurt making

Explanation: At anaerobic situation yeast acts on sugar inorder to produce alcohol (ethanol). So, this process involves brewing of beer. **Answer: A**

65. To which group of organic compounds do the triglycerides and waxes belong?
A. Carbohydrates
B. Proteins
C. Lipids
D. Vitamins

Explanation: Triglycerides are neutral fats which are formed when glycerol chemically combines with three fatty acids.
Wax is a fatty acid which is combined with along chain of alcohol. **Answer: C**

66. What does an ethologist study?
A. Insect diets
B. Soil types
C. Fossil fuel
D. Animal behaviour

Explanation: **Ethologist** is a person who studies the behaviour of animals in natural settings. **Answer: D**

67. Which one of the following is a biotic factor operating within an ecosystem?
A. The amount of helium gas in the air
B. The rate of flow of water in a river
C. The carnivores that consume other animals
D. The type of climate in a given region

Explanation: In an ecosystem the biotic factors are living things that operate within an ecosystem. Thus, carnivores that consume other animals are the biotic factors. **Answer: C**

68. When athletes take part in short distance running, how do the cells generate most of the energy that is quickly needed?
A. Aerobic respiration in muscle cells.
B. Mitochondrial respiration in any cells.
C. Anaerobic respiration in muscle cells.
D. Yeast fermentation in the stomach.

Explanation: an aerobic respiration in humans takes place where people do not in hale sufficient amount of oxygen to break glucose as quickly as needed. **Answer: C**

69. What do you call the reasoning technique in science where general principles are used to analyze specific cases?
A. Induction
B. Deduction
C. Pseudo-deduction
D. Pseudo-induction

Explanation:
✓ **Inductive reasoning:** is a reasoning technique that process form specific to general.
✓ **Deductive reasoning:** a reasoning which proceeds from general to specific. **Answer: B**

70. If a substance that weighs 2,000 grams and has a half-life of 100 years is left with only 250 grams, for how long has the radioactive decaying activity been undergoing?
A. 200 Years
B. 250 Years
C. 300 Years
D. 500 Years

Explanation: **Half-life** is the time needed for half atoms of the radioactive element to decay. Thus, the substance having weight of 200 grams and half-life of 100 years decays into 250 grams within 300 years.

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2000 gram $\xrightarrow{100\text{ years}}$ 1000 gram $\xrightarrow{100\text{ years}}$

500 gram $\xrightarrow{100\text{ years}}$ 250 gram

Answer: C

71. Which one of the following will primarily happen if the enzymes in the isosomes of a cell are defective?
   A. Cellular debris will not be removed
   B. Chromosome replication will cease
   C. ATP production will stop
   D. Diffusion process will stop

Explanation: Lysosomes are cellular organelles that contain enzymes which are used:
- To remove cellular debris
- For intercellular digestion

Answer: A

72. Which of the following is true about all proteins?
   A. They are twenty amino acids long.
   B. They are globular in shape.
   C. They perform the same function.
   D. The monomers are held together by peptide bonds.

Explanation: substances which are produced form monomer units called **aminoacids.** The aminoacids that make protein molecule are joined together by a **peptide bond.**

Answer: D

73. Which of the properties of water allows mosquito larvae to protect themselves from sinking down into the water from the water surface?
   A. Its high specific heat capacity
   B. Its high latent heat of vaporization
   C. Its high surface tension
   D. Its low density in solid state.

Explanation: **High surface Tension** is the tension on the surface of a liquid resulting from un balanced forces acting on the molecules at the surface. This property allows small organisms such as insects to walk on the water without breaking its surface.

Answer: C

74. Which of the following properties of enzymes makes it possible that a single enzyme molecule can act on many substrate molecules?
   A. Enzymes are proteins
   B. Enzymes lower than energy of activation
   C. Enzymes are reused over and over again
   D. Enzymes are substrate specific

Explanation: Enzymes can be reused over and over again. Thus, because of this property of enzyme can react with many substrate molecules.

Answer: C

75. If the rate of the enzyme catalyzed reaction remains constant even when more substrate is added to the reaction, which of the following might be the reason?
   A. Saturation of the enzyme
   B. Inactivation of the enzyme
   C. Inhabitation of enzyme-substrate complex formation
   D. Loss of substrate specificity by the enzyme

Answer: D

76. Which stage in the life cycle of HIV is disrupted, if AIDS patients are treated with a drug that has a protease inhibiting activity?
   A. Entry of the virus into the host.
   B. Conversion of viral RNA to DNA
   C. Integration of viral DNA into host DNA.
   D. Assembly of viral parts into whole virus.

Answer: D

77. Of the following four cells whose surface area to volume ratio is given, which tell can more efficiently transport its needs of materials across the cell surface?
   A. 24 : 8 ratio
   B. 54 : 27 ratio
   C. 96 : 64 ratio
   D. 150 : 125 ratio

Explanation: An organism with large surface are to volume ratio can more efficiently transport its needs of materials across its cell surface.

Answer: A

78. Cells immediately use the energy that electrons lose as they pass along the chain of electron carriers to:
   A. Produce ATP
   B. Pump protons
   C. Spin the rotor of ATP synthase
   D. Reduce NAD

Explanation: In electron transport chain the electrons pass along a series of electron carriers that form a transport chain to lose energy as they pass from one carrier to the next. As electrons are transferred through the proton pumps, they lose energy which pumps the protons into the inner membrane space.

Answer: B

79. Which one of the following is a routine method used by clinics to test people for HIV infection?
   A. Microscopic examination for the virus

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B. Counting the number of white blood cells.
C. Testing for human anti-HIV antibody.
D. Measuring the amount of hemoglobin.

Explanation: the lymphocytes make antibodies (anti-HIV antibody) to fight free HIV in the blood when the virus is presenting the body. Thus, the presence of those antibodies is detected and the person is diagnosed being HIV-positive. **Answer: C**

80. The T-lymphocyte cells of AIDS patients are destroyed by:
A. Multiplication of HIV inside the cell
B. Infection of opportunistic organisms
C. The immune system of the host organism itself
D. The CD4 receptors on the cell surface.

Explanation: HIV infects the T-lymphocyte cells of AIDS patients by binding to the CD_4 receptors on the surface of T-helper cells. **Answer: D**

81. Suppose a fossil initially contains 100,000 atoms of a certain radioactive element whose half life is 10,000 years, after how many years would the number of the atoms be 12500?
A. Ten thousand years
B. Twenty thousand years
C. Thirty thousand years
D. Forty thousand years

Explanation: Initially the Fossil had 100,000 atom and if the half life is 10,000 years, the number of atoms be 12500 after thirty thousand years.

$$100,000 \xrightarrow{10,000\ years} 50,000\ atom \xrightarrow{10,000\ years} 25,000\ atom \xrightarrow{10,000\ years} 12,500\ years$$

Answer: C

82. The following are Simpson's Index of species diversity calculated for four areas containing the same types of species. Which index value is from the area dominated relatively by a fewer number of species?
A. 8.00 C. 3.5
B. 6.00 D. 2.5

Explanation Low value of diversity index shows that an area is dominated by one or just few species or organisms, while an area is dominated by more number of successful species, the diversity index will be higher. **Answer: D**

83. Which one of the following happens in both cyclic and non-cyclic photophosphorylation?
A. ATP is formed
B. Oxygen is generated
C. NADP is reduced
D. Water molecule splits

Explanation: The products of noncyclic phosphorylation are ATP, $NADPH^+$ and oxygen. While, the product of cyclic photophosphorylation is ATP only. **Answer: A**

84. Which of the following is NOT a true characteristic of the populations of most of the developing countries of the world?
A. High fertility rate
B. Increasing population size
C. More number of old people than young people
D. Birth rate greater than mortality rate

Explanation: In developing countries the population number is increasing at an alarming rate. Thus, there are more children than older people, because the fertility rate is greater than death rate. **Answer: C**

85. Which of the following is NOT true about the nature of the first form of organism on earth? They were:
A. prokaryotic C. aerobic
B. unicellular D. anaerobic

Explanation: The first form of organisms on the earth are:
- Prokaryotic
- Unicellular
- Anaerobic

Answer: C

86. Choose the step that comes at the last step in the scientific method
A. proposing the research hypothesis
B. formulation of a scientific theory
C. conducting experiments
D. making predictions

Explanation: The correct sequence of steps in scientific method is:
D → A → C → B **Answer: B**

87. Which of the following is NOT true about C4 plants such as tef (*Eragrostic tef*)?
A. CO_2 is harvested during the night time.
B. The bundle sheath cells contain chloroplasts.
C. Light-dependent reaction occurs in mesophyll cells.
D. Chloroplasts of bundle sheath cell lack thylakoids.

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Explanation: In C_4 plants and all other plants CO_2 is fixed in to mesophyll cells during the day time. **Answer: A**

88. What is the reason that plants do not use nitrogen directly from the atmosphere?
A. Nitrogen concentration is low in the atmosphere.
B. The molecular size of nitrogen is too large to pass through the stomata.
C. Nitrogen can enter plants only through the root hairs.
D. Plants lack the necessary process to use elementary nitrogen.

Explanation: Plants cannot use nitrogen directly from the air. Plants can absorb nitrogen only by their roots from the soil in the form of nitrates (NO_3^-). **Answer: C**

89. What does the fermentation of glucose by yeast normally yield?
A. Lactic acid, CO_2, and ATP
B. CO_2, H2O, and 35 ATP
C. Alcohol, CO_2, and 2ATP
D. Alcohol, CO_2, and 36 ATP

Explanation:

$$C_6H_{12}O_6 \xrightarrow{\text{yeast}} C_6H_6OH + 2CO_2 + 2ATP$$ **Answer: C**

90. Which of the following came first in the course of organic evolution?
A. Photosynthetic organisms
B. Free oxygen in the atmosphere
C. Land plants
D. Multicellular organisms

Explanation: The correct sequential order is:

$A \rightarrow B \rightarrow C \rightarrow D$ **Answer: A**

91. Which tow nitrogenous bases belong to the purines?
A. Adenine and thymine
B. Adenine and guanine
C. Guanine and cytosine
D. Thymine and uracil

Explanation: Adenine and guanine are purines. They are double ringed. Uracil, cytosine and thymine are pyramines and are single ringed. **Answer: B**

92. Which one of the following is true?
A. Recessive alleles are only expressed in the homogygote.
B. Dominant alleles are expressed only in the heterozygote.
C. Recessive alleles are expressed in the heterozygote.
D. Genetically modified organisms are never used to manufacture vaccines.

Explanation: Recessive alleles are expressed only when both alles are homozygous for the recessive trait. **Answer: A**

93. Which of the following is a functional group of a fatty acid?
A. A ketone group
B. An aldehyde group
C. An amino group
D. A carboxyl group

Explanation: Fatty acids are made up of a long chain carbon skeleton that has a carboxylic acid functional group. **Answer: D**

94. Which of the following is an example of an orientation innate behaviour?
A. Kinesis in woodlice
B. Blinking of the eyes
C. Sudden withdrawal of limbs from hot object
D. Nest building by weaver birds. **Answer: A**

95. Which hormone promotes human sleepfulness in darkness and controls the sleep-wakecycle?
A. Insulin
B. Adrenaline
C. Melatonin
D. Thyroxine

Explanation: Melatonin is a hormone that is used as a sleep aid in the treatment of some sleep disorders. **Answer: C**

96. In Pavlov's experiment on the classical conditioning of dogs, what does the reaction of the dogs to the sound of the bell represent?
A. The conditioned stimulus
B. The conditioned response
C. The unconditioned stimulus
D. The unconditioned response

Explanation: In Ivan Pavolv's experiment
i) Before conditioning
   - ✓ Food was unconditioned stimulus
   - ✓ Salivation was unconditioned response

ii) After conditioning
   - ✓ Bell was conditioned stimulus.
   - ✓ Salivation was conditioned response.

Answer: B

97. What is the purpose of the infoldings of the inner-membrane of the mitochondrion?
A. Increasing the photosynthetic capacity of the cell
B. Speeding up the loss of CO_2 during fermentation.
C. Spending up the process of glycolysis.
D. Increasing the surface area for ATP production.

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Explanation: The inner folds of mitochondoria (cristae) are used to increase the surface area for cellular respiration, which is used in the production of ATP.

Answer: D

98. Which of the following is a worthwhile biological problem for scientific investigation?
   A. Estimation of the amount of carbon in a forest ecosystem.
   B. How a crying dog can tell who among elders of the village is going to die next.
   C. How witchcrafts inflict bad health by cursing a healthy person
   D. How the traditional medicine men/women cure a possessed person.

Answer: D

99. Which one of the following biomes of the Earth has the greatest diversity of species?
   A. Deciduous forest
   B. Tropical rain forest
   C. Desert
   D. Tundra

Explanation: Tropical rain forest is the biome that gets the highest amount of annual rainfall. It is the richest biome because it contains diversity of plant and animal species.

Answer: B

100. One of the following is true about bacteriophages that have lysogenic life cycle.
   A. They are RNA viruses
   B. They integrate their nucleic acid into that of the host
   C. They multiply in the host immediately after infection
   D. Progeny viruses are released by chronic release method

Explanation: In lysogenic cycle the assembled viruses leave the cells (are released) either by causing cellysis (splitting) or by chronic (on going) exocytosis form the plasma membrane.

Answer: D

Biology (EUEE) 2008

DIRECTIONS: Each of the following questions is followed by four possible alternatives. Read each question carefully and blacken the letter of your choice on the separate answer sheet provided.

1. What is the molecule in plant cells that first captures the radiant energy from sunlight?
   A. ATP B. DNA C. Chlorophyll D. Carbon dioxide

Explanation: Chlorophyll is one of the requirement for the process of photosynthesis which used for absorbing sunlight.

Answer: C

2. Of the following functions, which one do triglycerides accomplish in cells?
   A. increase density of tissues and cells
   B. increase thermal insulation of cells
   C. yield limited amount of ATP for cells
   D. facilitate entry of excess water into cells.

Answer: B

3. Which of the following laboratory glassware is used for culturing bacteria?
   A. Petri dish
   B. Test tube
   C. Filter paper
   D. Measuring cylinder

Answer: A

4. Of the following, which one is the main source from which plants get the nutrients necessary for their growth and development?
   A. Light B. Chlorophyll C. Atmosphere D. Soil

Explanation: Plants get nutrients necessary for their growth and development from the soil by active transport system.

Answer: D

5. Which of the following is the correct sequence of the steps in scientific research?
   A. Hypothesis formulation → conclusion → experiment → question
   B. Question → hypothesis formulation → experiment → conclusion
   C. Hypothesis formulation → question→ experiment → conclusion
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step 1 : Question
step 2 : Hypothesis formulation
step 3 : Experiment
step 4 : Conclusion

Answer: B

6. Which of the following types of cancer is known to mostly develop in AIDS patients?
   A. Cervical cancer
   B. Stomach cancer
   C. Kaposi's sarcoma
   D. Breast cancer

Explanation: Kaposi's sarcoma is signal of the final infection of HIV.

Answer: C

7. How many fatty acids would a cell need to form a molecule of triglyceride?
   A. Two B. Four C. Five D. Three

Explanation: A cell need three fatty acid to form a molecule of triglyceride.

Answer: D

8. Which part of the human alimentary canal contains digestive enzymes that function at acidic pH?
   A. Stomach
   B. Mouth
   C. Esophagus
   D. Small intestine

Explanation: Stomach is one part of human alimentary canal which contains different enzymes like pepsin, rennin that function at acidic pH due to the presence of HCl.

Answer: A

9. Which of the following food types would most likely give a negative result upon addition of iodine solution?
   A. Bread B. Butter C. Biscuit D. Potato

Explanation: When iodine solution added to bread, biscuit and potato they give us positive result because of carbohydrate they contain.

Answer: B

10. Which process produces mRNA during protein synthesis?
    A. Translation
    B. Replication
    C. Mutation
    D. Transcription

Answer: D

11. What is the best term that expresses the movement of substances in cells against their concentration gradients?
    A. Active transport
    B. Passive transport
    C. Osmosis
    D. Diffusion

Explanation: Active transport involves the movement of substances from the region of lower concentration to the higher concentration which is against the concentration gradient.

Answer: A

12. In cell division, what is the phase that comes following the metaphase called?
    A. Extraphase B. Prophase C. Anaphase D. Telophase

Explanation: The correct stages in mitosis cell division is prophase → metaphase → anaphase → telophase

Answer: C

13. Who was the person that first observed living cells moving around when he examined drops of water under the microscope?
    A. Robert Brown
    B. Robert Hooke
    C. Anton van Leeuwenhoek
    D. Theodor Schwann

Answer: C

14. Which of the following is true about gene mutation?
    A. Altering the DNA sequence of a gene
    B. Change in the position of a block of genes
    C. Addition of genes to a chromosome
    D. Loss of genes from a chromosome

Answer: A

15. What is the circumstance that causes the health condition known as sickle-cell anaemia?
    A. DNA denaturation
    B. Haemoglobin mutation
    C. RNA mutation and decay
    D. Phosphate mutagenesis

Answer: B

16. Based on similarity in number of amino acids found in haemoglobin, which one of the following animals has closer phylogenetic relations to human beings?
    A. Chicken B. Horse C. Frog D. Gibbon

Answer: D

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17. In which of the following groups of living organisms do the cells lack organized nuclei?
A. Fungi B. Protozoa C. Bacteria D. Algae
Explanation: Bacterias are prokaryotic organism that have no true nucleus and membrane bound organelles.
Answer: C

18. Which of the following is the correct constitution of the sex chromosome of a normal woman?
A. Non-disjunction C. Deletion
B. Replication D. Double inversion
Answer: A

19. Which of the following is the correct constitution of the sex chromosome of a normal woman?
A. XY B. XX C. XO D. XXY
Explanation: Every one inherits *x* chromosome from their mother. If this join with sperm carrying another *x* chromosome you will be a girl, but if it is fertilized by sperm carrying *y* chromosome you will be a boy.
Answer: B

20. From which of the following does the O_2 released during the process of photosynthesis originate?
A. Pyruvic acid B. CO_2 C. Sugar D. Water
Explanation: Water is the main source of hydrogen in the glucose and oxygen released to atmosphere during the process of photosynthesis.
Answer: D

21. What do you call structures that have the same evolutionary origin even though they may now have different structural make up or functions?
A. Endemic C. Homologous
B. Analogous D. Homologous
Explanation: Analogous structures are structures that have similar function but dlt morphology and anatomy but homologous structure are structure that have the same basic anatomy and common evolutionary origin but different function.
Answer: C

22. For what purpose do biologists use the GPS receiver?
A. To produce area maps C. To estimate tree ages
B. to measure tree heights D. To count tree rings
Answer: A

23. Which of the following organisms usually forms the pioneer community in a primary biological succession?

A. Annual herbs C. Trees
B. Lichens D. Ferns
Explanation: Primary succession occurs on bare lifeless substrate such as rock. Different organisms occupy pioneer community in primary succession such as lichen Algae, and Cyano bacteria.
Answer: B

24. In an experiment designed to study the effect of temperature on the rate of seed germination, which of the following should the experiment vary?
A. seed number C. Temperature
B. water quality D. Seed size
Answer: C

25. In the carbon cycle, which of the following processes removes carbon dioxide from the atmosphere?
A. Respiration C. Combustion
B. Decomposition D. Photosynthesis
Explanation: Photosynthesis is the process of making food by green plants. Plants remove carbondioxide for this process from the atmosphere.
Answer: D

26. Which of the following sequences represents the correct change in number of chromosomes during fertilization?
A. $n + n \rightarrow 2n$ C. $n \rightarrow n$
B. $2n \rightarrow 2n$ D. $2n \rightarrow n + n$
Answer: A

27. "Rats can be produced by keeping rags and grains at a corner of a room". Which of the following line of thinking supports this statement?
A. Darwinian evolution C. Alternation of generation
B. Spontaneous generation D. Sexual reproduction
Explanation: Spontaneous generation (a biogenesis) an idea that suggests that life can be obtained from non-living objects "spontaneously".
Answer: B

28. What is the specific name of biological scientists who do research that tries to find evidence of life on other planets in the Solar System?
A. Neurobiologists C. Astrobiologists
B. Paleontologists D. Biogeographers

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Explanation:
- **Astrobiologists**: name of biological scientists that studies about life in the planets.
- **Neurobiologists**: Biologists that study about nerve and brain function in people and animals.
- **Paleontologists**: biological scientists that study the forms of life existing in prehistoric or geologic times.
- **Biogeographer**: a biologists engaged in biogeography.

Answer: C

29. If two species are known to belong to the same order, they must also belong to which taxonomic category?
A. Class B. Family C. Species D. Genus

Explanation: The correct stage of taxonomic categories are kingdom species→ genus → family → order → class → phylum → kingdom

Answer: A

30. What happens in the first reaction of the Krebs cycle during energy transformation?
A. A 2-C compound is produced.
B. A 6-C compound is produced.
C. A 4-C compound is produced.
D. A 5-C compound is produced.

Answer: B

31. Which one of the following events happened before all the others?
A. The cell theory was proposed.
B. The protozoa were discovered.
C. The compound microscope was invented.
D. The structure of DNA was described.

Answer: A

32. Under what conditions do C4 plants have more photosynthetic efficiency than C3 plants?
A. Low water supply C. Low light intensity
B. Low temperature D. Low CO_2 concentration

Answer: D

33. Which of the following is **NOT** a universal property of all living things?
A. Heritable characters C. Photosynthesis
B. Reproduction D. Growth and development

Explanation: Growth and development, reproduction and heritable characters are universal property for all living things but photosynthesis takes palace only by green plants, some bacteria and algae.

Answer: C

34. In which area of enzyme application is invertase injected to sucrose paste in order to produce liquid chocolates?
A. Detergent making industry C. Pharmaceutical industry
B. Pulp and paper industry D. Food processing industry

Answer: D

35. Which one of the following functions best at higher optimum pH than all the rest?
A. Pepsin C. Trypsin
B. Salivary amylase D. Enzymes in stomach

Explanation: The enzyme which are found in the stomach function best at higher pH due to HCl.

Answer: D

36. Which class of enzymes joins two molecules together by formation of new bonds?
A. Ligase B. Isomerase C. Lyase D. Hydrolase

Answer: A

37. Which of the following has a bigger size than all the others?
A. a ribosome taken from an animal cell
B. a mitochondrion taken from a plant cell
C. a nerve cell taken from a human brain
D. a glucose molecule taken from a plant cell

Answer: C

38. How does a simple microscope differ from a compound microscope?
A. A simple microscope has no lens.
B. A simple microscope has got two lenses.
C. A simple microscope uses mirror as lens.
D. A simple microscope has only one lens.

Answer: D

39. Which of the following industries can reduce more CO_2 emission by shifting to the use of enzymes in the manufacturing process?
A. Bread making C. Leather making
B. Cheese making D. Manufacturing cosmetics

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Answer: A

40. Which of the following is made of globular proteins?
A. Enzyme B. Keratin C. Collagen D. Glycogen

Answer: A

41. Which one of the following should be done in order to remove an enzyme inhibition caused by a competitive inhibitor?
A. Remove affected enzyme molecules
B. Add more substrate to the system
C. Remove the end product of the reaction
D. Add more inhibitor to the system

Answer: B

42. Which of the following parts of the plant cell is **NOT** a living component of the cell?
A. Cell membrane C. Cytoplasm
B. Cell wall D. Nucleus

Answer: B

43. What are the possible consequences of deforestation of the tropical rainforest?
A. an increase in existing ecological niches
B. reduction in species diversity of an area
C. increased removal of CO_2 from the atmosphere
D. an increase in the amount of nitrogen in the soil

Answer: B

44. Which of the following processes releases CO_2 into the atmosphere?
A. Respiration C. Feeding
B. Assimilation D. Photosynthesis

Explanation: Respiration is the process of getting energy from the food. This process releases CO_2 in to atmosphere but during photosynthesis CO_2 is removed from the atmosphere.

Answer: A

45. If an area is dominated by just one species having very many individuals, what would be its index of diversity?
A. Fluctuating C. Low
B. high D. Unpredictable

Answer: C

46. Which of the following is true about sex determination in birds?
A. They have the heterozygotic X and Y chromosomes.
B. Males have heterozygotic W and Z chromosomes.
C. Females have the homozygotic WW chromosomes.
D. They have the heterozygotic W and Z chromosomes.

Answer: D

47. How many amino acids are there in all known proteins?
A. About 10 B. About 35 C. About 20 D. About 46

Answer: C

48. Which of the following is the major mechanism by which AIDS is transmitted?
A. Mother to fetal transmission
B. heterosexual intercourse
C. Homosexual intercourse
D. contaminated blood transfusion

Answer: B

49. Which of the following molecules can serve as a raw material for industries that produce glucose?
A. Nucleic acids C. Lipids
B. Proteins D. Starch

Answer: D

50. In enzymes that contain non-proteins organic molecules, in addition to the protein component called?
A. Apoenzyme C. Holoenzyme
B. Coenzyme D. Cofactor

Answer: A

51. To which one of the following classes of stimuli do pheromones belong?
A. Auditory B. Smell C. Visual D. Touch

Answer: B

52. Which of the following types of microscopes is most suitable for a detailed study of the surface structure of an object?
A. Field microscope
B. Optical microscope
C. Scanning electron microscope
D. Dissecting microscope

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Answer: C

53. According to the fluid-mosaic modal of the plasma membrane, what does the word 'mosaic' refer to?
A. The hydrophobic property of fatty acids.
B. The bilayer nature of the membrane
C. The arrangement of the proteins
D. The movement of the phospholipids

Answer: B

54. Which of the following diseases is correctly matched with its causative agent?
A. Malaria – fungus
B. AIDS – virus
C. Ringworm – protozoa
D. Syphilis – worm

Answer: B

55. Which of the following organic compounds would release both nitrogen and sulfur to the ecosystem when decomposed?
A. Polysaccharides
B. Sucrose
C. Proteins
D. Lipids

Answer: C

56. Among the vertebrates found in Ethiopia, which class has the highest percentage of endemic species?
A. Amphibians B. Reptiles C. Birds D. Mammals

Answer: C

57. Which of the following terms refers to the movement of individuals out of a population?
A. Mortality
B. Immigration
C. Emigration
D. Natality

Explanation: - **Mortality** – death rate
- **Natality** – birth rate
- **Immigration**- movement of organism into population
- **Emmigration** – movement of organisms out of population

Answer: C

58. Which of the following crosses will produce progeny with phenotypic ratio of 3:1?
A. Crossing the F1 to the dominant parent
B. Crossing two homozygote individuals

C. Crossing the F1 to the recessive parent
D. Crossing two heterozygote individuals

Answer: D

59. What percentage of the F2 progeny of a monohybrid cross is expected to have the recessive phenotype?
A. 100% B. 75% C. 25% D. 50%

Answer: C

60. Which of the following is NOT true about innate behaviors?
A. Can be improved by trial and error
B. Common to all members of the species
C. Present at birth or no hatching
D. DO not have to be learned

Explanation: Innate behavirour is occurred at birth or hatching

Answer: C

61. In Ethiopian animal diversity, which group is represented by the highest number of orders, families, genera and species?
A. Birds B. Amphibians C. Fish D. Mammals

Answer: A

62. Why are mutations considered important in evolution?
A. They are usually related to the environment.
B. They contribute to new variations in organisms.
C. They are always beneficial to the organism.
D. They become causes for species migrations.

Answer: B

63. Which of the following processes involved in the water cycle is carried out by green plants?
A. Evaporation
B. Precipitation
C. Condensation
D. Transpiration

Answer: D

64. Deficiency of which of the following nutrients in human diet is likely to result in a deficiency of some co-enzymes like FAD?
A. Essential amino acids
B. Vitamins
C. Carbonhydrates
D. Saturated fatty acids

Answer: A

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65. If someone suddenly removes his/her hand from a very hot object, which of the following types of behavior is manifested?
A. Reflex action
B. Imprinting
C. Learned behavior
D. Sensitization

Explanation: Reflex action is sudden automatic response to stimuli and it's the simplest form of innate behaviours.

Answer: A

66. Which step in the HIV life cycle is disrupted by an anti-retroviral drug that competitively inhibits the reverse transcriptase enzyme?
A. Entry into the host cell
B. Assembly of viral parts into a virus
C. Formation of DNA from RNA
D. Integration of viral DNA into host DNA

Answer: C

67. During which of the following processes in cellular respiration are most of the ATPs formed?
A. Glycolysis
B. Chemiosmosis
C. Link reaction
D. Krebs cycle

Answer: B

68. Which of the following pairs are **ANALOGOUS** structures?
A. The human arm and the front leg of a mule.
B. The front leg of a frog and the wing of a bat.
C. The wing of a bird and the wing of a butterfly.
D. The wind of a bat and the wing of a bird.

Explanation: Analogous structure are structures that have the same function but different morphology and anatomy.

Answer: C

69. Which of the following changes that happened during human evolution had the most contribution to the evolutionary success of *Homo sapiens*?
A. Proportionately big brain size to body mass
B. Big body parts and big overall body mass
C. long legs, arms and more upright body posture
D. fast running ability and overall physical strength

Answer: C

70. Woodlice are observed avoiding light and heat by quickly moving to moist and darker areas. Which behavior of these animals helps them to

detect differences in light intensity and move to the darker and moist part of the habitat?
A. Instinctive learning
B. Negative phototaxis
C. Positive phototaxis
D. Inslight learning

Answer: B

71. If a codon on a messenger RNA is UUU, what is the complementary anticodon on the transfer RNA?
A. UUU B. GGG C. CCC D. AAA

Answer: D

72. If a clone is produced by transferring a nucleus of animal A to an enucleated egg of animal B and the egg is then implanted in the uterus of animal C, which animal would the clone resemble most?
A. Animal C B. Animal B C. Animal A D. Other animals

Answer: A

73. Which of the following classes of fatty acids is without carbon-carbon double bond?
A. Monounsaturated fatty acids
B. Unsaturated fatty acids
C. Saturated fatty acids
D. Polyunsaturated fatty acids

Explanation: Saturated fatty acids contain single bond between each carbon atom.

Answer: C

74. Choose the one that had the **LEAST** contribution to human evolution?
A. Development of bipedalism
B. Adaptation to flight
C. Attaining opposable thumb
D. Increasing brain size

Answer: B

75. In which biome are epiphytes typically present as a characteristic element?
A. Tropical mountain forests
B. Cold desert woodlands
C. The Tundra environment
D. Boreal deciduous forests

Answer: A

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76. If it is known that the total amount of DNA in a cell is 300 units and that adenine alone contributes 70 of these units, how many units go to cytosine?

A. 40 units
B. 70 units
C. 80 units
D. 35 units

Explanation: A DNA molecule contain four bases such as a denine, cytocine, guanine and thymine. The number of adenine equals with the number of thymine and the number of cytocine equals with the number of guanine.

Answer: C

77. What is the substance that helps to keep the biological membrane in a fluid state?

A. Waxes B. Water C. Cholesterol D. Phospolipids

Answer: D

78. Which of the following substances is **NOT** formed when glucose is fermented by yeasts?

A. Alcohol B. ATP C. Lactic acid D. Carbon dioxide

Answer: C

79. Which of the following involves trial and error learning?

A. operant conditioning
B. Habituation
C. Sensitization
D. Classical conditioning

Answer: A

80. Which of the following is the adaptation by C4 plants that helps them to avoid photorespiration?

A. harvesting of carbon dioxide at night
B. using separate cells for light and dark reactions
C. storing carbon dioxide in the vacuole
D. keeping the stomata closed during the day.

Answer: C

81. How many years have passed since Drawin's book on the theory of evolution was published?

A. About 50 years
B. About 160 years
C. About 120 years
D. About 100 years

Answer: B

82. Which of the following can be understood about living things from the study of how breeders improve domesticated plants and animals?

A. Living things tend to over-reproduce
B. Natural resources are of limited supply
C. Living things can be improved through selection
D. Individuals compete for resources

Answer: C

83. Which of the following is responsible for the bending of a young plant towards a unidirectional sources of light?

A. Reduced photosynthesis on dark side
B. Faster growth rate on the dark side
C. Reduced auxin concentration on dark side
D. Increased rate of cell division on the light side

Answer: D

84. In what way would AIDS patients benefit form treatment with anti-retroviral drugs?

A. Reduction of HIV replication
B. Provision of cure for AIDS
C. Immunizing against HIV
D. killing of opportunistic infectious agents

Answer: A

85. If a country has a large number of young people relative to the number of old people, to which category of countries does it belongs?

A. Industrial
B. Hunter-gatherer
C. Post-industrial
D. Developing

Answer: A

86. Which of the following is the best collective name for all bacteria with spherical shapes?

A. Spirochaetes
B. Bacilli
C. Cocci
D. Streptococci

Explanation: Bacteria cells have different shapes and arrangements this include cocci which contain bacteria with spherical shape, bacillus which contain bacteria with rod shaped bacteria and spirals bacteria with spiral, cork-screw or comma shaped.

Answer: C

87. Suppose two heterozygous round yellow (RrYyx RrYy) pea plants were crossed and 128 seeds were produced, how many of the seeds are expected to the heterozygous round yellow?

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A. Spirochaetes
B. Bacilli
C. Cocci
D. Streptococci

Explanation:

	RY	ry	Ry	rY
RY	RRYY	RrYy	RRYy	RrYY
ry	RrYy	rryy	Rryy	rrYy
Ry	RRYy	Rryy	RRyy	RrYy
rY	RrYY	rryy	RrYy	rrYY

Heterozygous round yellow (RrYy) = $\frac{4}{16} \times 128 = 32$ seeds

Answer: B

88. Which of the following is an important way by which green plants mitigate the greenhouse effect?
A. Use of fire wood to replace coal
B. Removing CO_2 from the atmosphere
C. Releasing water to the atmosphere
D. Releasing oxygen to the atmosphere

Answer: B

89. Which phosphate bond of the ATP is broken when the energy it contains is needed for cellular activity?
A. The first bond
B. The C – C bonds
C. The second bond
D. The third bond

Answer: D

90. What is the advantage that a photosystem containing molecules of different types of light sensitive pigments have?
A. To absorb light of different wavelengths
B. To increase the size of the photosystem
C. To increase the complexity of the pthososystem
D. To increase the surface area for light absorption

Explanation: Photosystem containing different pigment molecules in the antenna complex helps it to absorb different wave lengths of light.

Answer: A

91. **Why does sucrose give a negative result when mixed and heated with Benedict's solution?**
A. Because it is not a sugar molecule.

B. Because it is a reducing sugar.
C. Because it is a non-reducing sugar.
D. Because it is not a monosaccharide.

Answer: C

92. What does an evolutionary selective pressure that acts around the mean do?
A. It stabilizes
B. It terminates
C. It converges
D. It disrupts

Explanation: Stabilizing selection operates around the mean and works against the extreme features.

Answer: A

93. Which of the following is consistent with science?
A. Proof by investigation of the causes of a phenomenon
B. Proof by reference to a respected leader
C. Proof by reference to an unknown object
D. Proof based on lessons form a person one believes.

Explanation: Science requires proof based on investigation of the causes of a phenomenon.

Answer: A

94. In his theory of evolution, the cause of which of the following concepts was **MISSING** in Darwin's explanation?
A. Over-reproduction
B. Struggle for survival
C. Hereditary variation
D. Survival of the fittest

Explanation: Hereditary variation is incorporated by Neo-Darwinians.

Answer: C

95. Which of the following is a learned behavior?
A. Suckling of the newborn at mother's breasts
B. Salivation by conditioned dogs at the sound of a bell
C. Withdrawal of hands suddenly from hot objects
D. Blinking the eyes when something gets into them

Explanation: Salvation by conditioned dogs at the sound of belt is a demonstrative experiment of learned behaviour (conditioned replex) by Ivan Pavlov.

Answer: B

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96. Which of the following diseases is transmitted by mosquitoes?
A. Diabetes mellitus
B. *Kaposi's cancer*
C. Rabies
D. Malaria

Answer: D

97. In which of the following are all the tools mainly used in the laboratory rather than in field situation?
A. Centrifuges, microscopes, measuring cylinders, petri dishes
B. Flow meters, centrifuges, theodolites, microscopes
C. Centrifuges, plant presses, flow meters, microscopes
D. Theodolites, petri dishes, GPS receivers, measuring cylinders

Answer: A

98. Which aspect of biotechnology is considered strictly genetic engineering?
A. Providing gene therapy
B. Production of new types of plants
C. Monoclonal antibodies
D. Mapping of the human genome

Answer: B

99. What causes tomato fruits to ripen much more slowly when kept in a refrigerator than if left on a table at room temperature?
A. Low temperature slows the normal action of ripening enzymes.
B. Enzymes produced by bacteria normally inhibit ripening.
C. Humidity accelerates enzyme activity and ripening process.
D. Normal temperature arrests the action of ripening enzymes.

Answer: A

100. Which characteristic of RNA makes it suitable for moving out of the nucleus?
A. Inability to replicate
B. Absence of thymine
C. Its unstable nature
D. Smallness of its size

Answer: D

1. The word science comes from the Latin word 'Scientia' meaning what?
A. Study
B. Experiment
C. Life
D. Knowledge

Answer: D

2. Which of the following substances can be used over and over again in an enzyme catalyzed reaction?
A. Enzyme – substrate complex
B. Substrate
C. Enzyme
D. Reaction product

Answer: C

3. What is the intermediate compound that is formed during an enzyme catalyzed reaction?
A. Reactant
B. Substrate
C. Product
D. Enzyme – substrate complex

Explanation:
Enzyme + Substrate ⟶ Enzyme- Substrate Complex

Answer: D

4. Which of the following microorganisms are prokaryotic?
A. Bacteria
B. Protozoa
C. Yeast
D. Algae

Explanation: Bacteria are prokaryotes; while, protozoa, yeast and algae are prokaryotes.

Answer: A

5. Which of the following laboratory tools is **LEAST** accurate to measure 1 ml of liquid?
A. Pipette
B. Measuring cylinder
C. Burette
D. Beaker

Explanation:

6. From which of the following diseases can one be protected by sleeping under the cover of a mosquito net?
A. Sleeping sickness
B. AIDs
C. Malaria
D. Athlete's foot

Answer: C

7. Which of the following parts of the flower did Mendel remove from young flowers to prevent self-pollination?
A. Stamens
B. Sepals

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C. Petals D. Ovaries

Explanation: Stamens are the male reproductive parts of the flower that Mendel removed from a garden pea plant. This helped him to prevent self-pollination.

Answer: A

8. What do we call the place where organisms live in their ecosystem?
   A. Biotic C. Habitat
   B. Biotic D. Niche

Explanation: habitat is a place where an organisms lives.

Answer: C

9. Which of the following is the sex chromosome constitution of human males?
   A. XX B. XY C. ZZ D. ZW

Explanation: The chromosomes of human male are XY; while, the female sex chromosomes are XX.

Answer: B

10. Which of the following plants uses CAM photosynthesis?
    A. Sugarcane C. Maize
    B. Sorghum D. Cactus

Answer: D

11. Which of the following bases is **NOT** found in RNA?
    A. Adenine C. Cytosine
    B. Guanine D. Thymine

Explanation: RNA contains a nitrogenous base Uracil instead of thymine.

Answer: D

12. In a report of a scientific experiment, description of which of the following comes before all the others?
    A. Procedure
    B. Results
    C. Hypothesis
    D. Maltose

Explanation: The correct sequential order of the scientific methods mentioned is: C ⟶ A ⟶ B ⟶ D

Answer: C

13. Which one of the following I different from the other three?
    A. Starch C. Cellulose
    B. Glycogen D. Maltose

Explanation: Starch, glycogen and cellulose are polysaccharides; while, maltose is a disaccharide.

Answer: D

14. What happens when a plant cells absorbs in more water?
    A. It shrinks
    B. It becomes turgid
    C. It becomes wilted
    D. It swells and bursts

Explanation: When a plant cell absorbs water it becomes turgid because it is filled with water. In other ways, if a plant cell looses water it shrinks.

Answer: B

15. Which level of protein structural organization involves two or more polypeptide?
    A. Primary structure
    B. Secondary structure
    C. Tertiary structure
    D. Quaternary structure

Answer: D

16. Which of the following do biologists consider ancestral to the higher organisms of today?
    A. Plantae
    B. Animalia
    C. Fungi
    D. Protisa

Answer: D

17. Which type of fatty acid is healthier when used as a human food source?
    A. Saturated fatty acid
    B. Monounsaturated fatty acid
    C. Polyunsaturated fatty acid
    D. Fatty acids in animal products

Explanation: Poly unsaturated fatty acids prevent the accumulation of cholesterol in lining of arteries and other lining of the internal organs. Thus, they are healthier diets.

Answer: C

18. Which of the following is **NOT** recycled in the ecosystem?
    A. Sulphur C. Carbon
    B. Energy D. Phosphorus

Explanation: The flow of energy in an ecosystem is unidirectional. Thus, energy cannot be recycled in an ecosystem.

Answer: B

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19. What kind of cross is performed to determine whether the parent was homozygous or heterozygous?
A. Monohybrid cross
B. Dihybrid cross
C. Back cross
D. Double cross
Explanation: Backcrossing is a crossing of a hybrid with one of its parents or an individual genetically similar to its parent, in order to achieve offspring with a genetic identity which is closer to that of the parent.
Answer: C

20. Which type of animal behavior happens without learning?
A. Innate
B. Latent
C. Insight
D. Conditioned
Explanation: Innate behavior is unlearned behavior which develops without any kind of past experience.
Answer: A

21. When a drop of iodine solution is added to a solution of food stuff and the food stuff turns dark blue, what does the food stuff contain?
A. Reducing sugar
B. Disaccharide
C. Glucose
D. Starch
Explanation: Addition of iodine solution on starch forms a blue black color. This shows the presence of starch.
Answer: D

22. Which of the following has the highest risk of HIV transmission?
A. Hand shaking
B. Needle sharing
C. Towel sharing
D. Toilet sharing
Explanation: Sharing sharp objects and unprotected sexual intercourse have the highest risk of HIV transmission.
Answer: B

23. Which of the following elements is the most abundant in living things?
A. Nitrogen
B. Carbon
C. Hydrogen
D. Oxygen
Explanation:

Element	*% composition in the body*
Hydrogen	59
Oxygen	24
Carbon	11
Nitrogen	4
Others	2

Answer: C

24. Which of the following organelles is common to both plant and animal cells?
A. Cell wall
B. Chloroplast
C. Mitochondria
D. Central vacuole
Explanation: Cell wall, chloroplast and central vacuole are found in plant cells but not in animal cell. Mitochondria are found in both plant and animal cells.
Answer: C

25. Among the following organic molecules, identify the one that does not contain nitrogen.
A. Starch
B. Protein
C. DNA
D. RNA
Explanation: starch is a carbohydrate that does not contain nitrogen.
Answer: A

26. If a cell fails to form lysosomes, which of the following cellular functions will get disrupted first?
A. Protein synthesis
B. Chromosome replication
C. Removal of cellular debris
D. Transport across cell membrane
Explanation: Lysosome is known as a suicide bag of a cell; because it helps to remove the cellular debris or garbage.
Answer: C

27. Most co-enzymes are the derivatives of which the following nutrients?
A. Proteins
B. Vitamins
C. Carbohydrates
D. Fats
Answer: B

28. To produce glucose from starch, which of the following enzymes would one use?
A. Isomerases
B. Lyases
C. Hydorlases
D. Ligases
Explanation:

Enzyme	*Function*
Isomerase	Catalyse internal structure of substrate molecule.
Lyases	Hydrolytic removal of a group or addition of a group to substrate.
Hydrolases	Catalyze hydrolysis reaction.
Ligases	Catalyze joining of two molecules.

Answer: C

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29. Which class of enzyme inhibitors has similar shape as that of the normal substrate?
A. Allosteric
B. Competitive
C. Denaturing
D. Irreversible

Answer: B

30. For a substance to easily pass across the cell membrane by simple diffusion which of the following properties should it possess?
A. Large size
B. Ability to dissolve in lipid
C. Positive charge
D. Negative charge

31. Among the following factors that affect enzyme activity, which one does **NOT** cause enzyme denaturation?
A. High pH
B. Low pH
C. High temperature
D. Competitive inhibitor

Answer: D

32. Which of the following is a type of ecological succession that starts from a cleared forest area?
A. Hydorsere
B. Primary
C. Secondary
D. Pioneer

Answer: C

33. Which of the following states of photosynthesis can take place in the dark?
A. Photolysis of water
B. Photo system I
C. Photosystem II
D. Calvin cycle

Answer: C

34. Which of the following is the major role played by bacteria and fungi in the ecosystem?
A. Causing diseases
B. Nutrient recycling
C. Fixing carbon dioxide
D. Producing antibiotics

Explanation: Fungus and bacteria are decomposers that help to recycle substances in the ecosystem.

Answer: B

35. If heterozygous tall pea plant (Tt) is crossed with a short pea plant (tt), what percentage of the progeny is expected to be short?
A. 100% B. 75% C. 50% D. 25%

Explanation:
- Heterozygous tall (Tt)
- Short pea plant (tt)
- Tt X tt

	T	t
t	Tt	tt
t	Tt	tt

50% Tt (tall) :50% tt(short)

Answer: C

36. Which of the human cell types is attacked by HIV?
A. Red blood cells
B. White blood cells
C. T-lymphocytes
D. Leukocytes

Answer: C

37. Which of the following processes is most important to release nutrients from dead organic matter into the soil?
A. Fixation
B. Decomposition
C. Excretion
D. Respiration

Answer: B

38. Which of the following biomes supports the highest diversity of plant and animal life?
A. Tropical rainforest
B. Deciduous forest
C. Grassland
D. African savannah

Explanation: Tropical rain forest is the biome that gets the highest amount of rainfall; thus it has the largest species of plant, animals and microorganisms as inhabitants.

Answer: A

39. Which of the following demographic factors can increase the size of the world population?
A. Natality
B. Migration
C. Emigration
D. Immigration

Explanation:
Increase in birth rate (natality) results in increase in human population on the world.

Answer: A

40. When Mendel crossed a tall pea plant with a short plant, the F_1 progeny were all tall. What F_1 did he obtain when he made the reciprocal cross?
A. Short plants
B. Tall plants

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C. Extra – tall plants
D. Plants of average height

Explanation: **Reciprocal cross** in genetics, one of a pair of crosses in which the two opposite mating types (or sexes) are each coupled with each of two different genotypes and mated with the reciprocal combination.

Answer: B

41. Among the following ABO blood group genotypes, which one produces two types of antigens?
A. I^AI^o B. I^BI^o C. I^AI^B D. I^BI^B

Explanation: If you belong to the **blood group** B, you have B **antigens** on the surface of your red **blood** cells and A antibodies in your **blood** plasma. If you belong to the **blood group** AB, you have both A and B **antigens** on the surface of your red **blood** cells and no A or B antibodies at all in your **blood** plasma.

Answer: C

42. Which of the following methods do animal breeders use to produce domestic animals with hybrid vigor?
A. Feeding with nutritive feed.
B. Giving proper veterinary care.
C. Mating together related animals.
D. Crossbreeding their animals.

43. Which of the following groups are believed to be the first photosynthetic organisms to evolve on earth?
A. Green plants
B. Green algae
C. Blue green algae
D. Lichens

Answer: D

44. Which of the following factors brings about changes both during evolution and breeding of plants and animals?
A. Limited resources
B. Production of excess progeny
C. Selection
D. Competition

Answer: C

45. Among the following, which one is held most responsible for the present fast depletion of the world biodiversity?
A. Herbivores
B. Carnivores
C. Grazers
D. Humans

Answer: D

46. Which of the following reproductive isolating mechanisms keeps the horse and donkeys as two independent species?
A. Hybrid inviability
B. Hybrid infertility
C. Ecological isolation
D. Habitat isolation

Explanation:
Mule, the hybrid of donkey and horse is not fertile.

Answer: B

47. Which of the following activities of an organism do we call a behavior?
A. Reception of external stimulus.
B. Reception of internal stimulus.
C. Responding to stimulus.
D. Lack of receptors to receive stimulus.

Answer: A

48. A reflex action that involves internal organs such as the heart is referred to as _____?
A. Somatic reflex
B. External reflex
C. Spinal cord reflex
D. Autonomic reflex

Explanation: There are two **types of reflex** arcs: the autonomic **reflex arc**, affecting inner organs, and the somatic **reflex arc**, affecting muscles.

Answer: D

49. At which of the following generations of crosses between dominant and recessive homozygote parents are all the progeny heterozygous?
A. P_1 generation
B. F_1 generation
C. F_2 generation
D. F_3 generation

Answer: B

50. Which of the following organs serves as a coordinating system in a reflex action?
A. Spinal cord
B. Sense organs
C. Nerve cells
D. Muscle

Answer: A

51. Which type of enzymes catalyzes the joining of two molecules by the formation of new bonds?
A. Decarboxylase
B. Dehydrogenase
C. Ligase
D. Oxidoreductase

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Explanation:

Enzyme	*Function*
Dehydrogenase	Transfer of hydrogen molecule to coenzymes such as NAD+
Ligase	Joining of two molecules by the formation of new bonds.
Oxidoreductase	Transfer of hydrogen and oxygen atoms or electrons from one substance to another.

Answer: C

52. Which of the following statements about enzymes is **NOT** correct?
A. Enzymes speed up a chemical reaction.
B. Enzymes are very specific in their action.
C. The presence of an enzyme will change the outcomes of a reaction.
D. Enzymes lower the energy of activation for a reaction to occur.

Explanation: Enzymes;
- Speed up a chemical reaction.
- Very specific in their action.
- The presences of enzymes have no impact on the outcome of the reaction.
- Lower the activation energy required for a reaction to occur.

Answer: C

53. Glycolysis occurs in the _____?
A. Cytoplasm
B. Mitochondria
C. Nucleus
D. Chloroplast

Explanation: Glycolysis is the first step of respiration which occurs in the cytoplasm, while the other stages such as krebs cycle, link reaction and electron transport chain occur in the mitochondria. **Answer: A**

54. Anaerobic metabolism refers to the generation of ATP:
A. Without the involvement of ADP
B. Without the use of glycogen
C. Without the use of oxygen
D. By the conversion of pyruvate to lactate

Explanation: Anaerobic metabolism is the generation of ATP without utilization of oxygen. In such a case, only 2ATP is produced from 1 glucose molecule. **Answer: C**

55. The conversion of one molecule of glucose to two molecules of pyruvate results in the next formation of:
A. Two molecules of ATP
B. Six molecules of water
C. Three molecules of ATP
D. Thirty - eight molecules of ATP

Explanation: During glycolysis 4 ATP's are produced from the aerobic break down of glucose molecule. However, since the cell spends 2ATP's to break down the glucose molecule, the net of 2ATP's are produced. **Answer: A**

56. During alcohol fermentation, pyruvic acid in the presence of pyruvate decarboxylase is first converted to _____.
A. Glucose
B. Lactose
C. Lactic acid
D. Acetaldehyde

Answer: D

57. The total amount of living tissue within a given tropic level is called
A. Organic mass
B. Energy mass
C. Trophic mass
D. Biomass

Explanation:
The total amount of organic matter or living tissue within a given trophic level is called biomass. **Answer: D**

58. Biomes where trees predominate are called _____.
A. Polar biomes
B. Desert biomes
C. Forest biomes
D. Grassland biomes

Answer: C

59. The goals of biodiversity conservation include all of the following **EXCEPT**
A. Protecting individual species
B. Introducing exotic species into new environments
C. Preserving habitats and ecosystems
D. Making sure local people benefit from conservation efforts

Explanation: The goal of biodiversity conservation include:
- Protection of species.
- Preserving indigenous species of the environment.

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- Preserving habitats and ecosystems.
- Making sure local people benefit from conservation.

60. The maximum population growth rate characteristic of a species is called _____.
A. Limiting factor
B. Biotic potential
C. Carrying capacity
D. Exponential growth pattern

Answer: D

61. From among the following, choose the field which is a branch or a part of biotechnology.
A. Parasitology
B. Immunology
C. Phycology
D. Genetic engineering

Explanation: Genetic engineering is a field of biotechnology which includes altering an organism's genotype by inserting another organism's gene in to its DNA.

Answer: D

62. Which of the following is publicly and **UNDESIRED** research activity?
A. Breeding a new crop variety.
B. Synthesis of a new vaccine.
C. Manufacture of a new effective medicine.
D. Breeding a new pathogenic microorganism.

Explanation: Breeding a new pathogenic is undesirable research activity.

Answer: D

63. Which of the following alternatives contains **ONLY** organic molecules?
A. Calcium carbonate, glucose, methane
B. Carobon dioxide, sodium chloride, lysine
C. Linoleic acid, glycine, methane
D. Water, carbon dioxide, calcium carbonate

Explanation: Linoleic acid, glycine and methane contain carbon and hydrogen. Thus they are organic molecules.

Answer: C

64. Which test is suitable for checking the presences of lipids in food stuffs?
A. Biuret test
B. Benedict's test
C. Emulsion test
D. Iodine test

Explanation:

Test	Food stuff
Biuret test	Protein
Benedict's reagent	Carbohydrate
Emulsion test	Lipid
Iodine test	Starch (Carbohydrate)

Answer: C

65. How do heavy metal ions reduce the reaction rate of enzymes?
A. By destroying the 3 – dimensional shape of the enzyme's active site.
B. By raising the temperature above the optimum level for the enzyme.
C. By dropping the pH well below the optimum level for the enzyme.
D. By inducing mutations that disrupt the functioning of the enzyme.

66. What happens when enzymes become denatured?
A. They can normally catalyze the reactions.
B. Their active sites are changed.
C. Rate of the reaction increases.
D. Bonds holding amino acids are unaffected.

Answer: B

67. What happens when human red blood cells are kept in a hypertonic solution?
A. The same net gain and loss of water.
B. More water gets into the cell than leaving it.
C. More water goes out of the cell than getting in.
D. The cells prevent water from getting in or leaving out.

Explanation: when a human red blood cell is placed in a hypertonic solution, the water moves out of the cell and the cell shrinks.

Answer: C

68. Among the following which one best describes the cells of bacteria?
A. Eukaryotic
B. Prokaryotic
C. Both prokaryotic and eukaryotic.
D. Neither prokaryotic nor eukaryotic.

Answer: B

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69. Which of the following nitrogenous bases of nucleic acids is a purine base?
A. Adenine
B. Cytosine
C. Thymine
D. Uracil

Explanation: **Purines** and **Pyrimidines** are nitrogenous bases that make up the two different kinds of nucleotide bases in DNA and RNA. The two-carbon nitrogen ring bases (adenine and guanine) are **purines**, while the one-carbon nitrogen ring bases (thymine and cytosine) are **pyrimidines.**

Answer: A

70. Which of the plants with the following genotypes is heterozygous?
A. BBYY
B. BbYy
C. bbyy
D. YYBB

Answer: B

71. Which of the following groups of viruses copy their RNA to DNA?
A. Bacteriophages
B. DNA viruses
C. RNA viruses
D. Retroviruses

Explanation: A **retrovirus** is a single-stranded positive-sense RNA virus with a DNA intermediate and, as an obligate parasite, targets a host cell. Once inside the host cell cytoplasm, the virus uses its own reverse transcriptase enzyme to produce DNA from its RNA genome.

Answer: D

72. Which of the following can reverse an enzymes inhibition that is caused by an allosteric inhibitor?
A. Adding more substrate
B. Adding more end product
C. Depleting the end product
D. Depleting the substrate

Explanation: Non-competitive Inhibitors bind to a site **other** than the Active Site, called an **Allosteric Site**. Doing so **distorts** the **3D Tertiary structure** of the enzyme, such that it can no longer **catalyse** a reaction.

Answer: A

73. During the Krebs cycle, to which of the following molecules is most of the energy released from food transferred?
A. ATP B. NAD C. ADP D. AMP

Answer: B

74. If in a DNA molecule consisting of 1000 base pairs, there are 300 adenine bases, how many guanine bases will be present?
A. 200 B. 400 C. 600 D. 700

Explanation:
- The number of Adenine is equal to Thymine.
- The number of Guanine is equal to Cytosine.

Answer: A

75. During which stage of aerobic respiration does oxidative phosphorylation occur?
A. Glycolysis
B. Krebs cycle
C. Link reaction
D. Chemiosmosis

Explanation: Electron transport chain and chemiosmosis make up oxidative phosphorylation.

Answer: D

76. What mode of feeding do soil organisms that release nutrients from dead organic matter into the soil have?
A. Photoautotrophic
B. Saprophytic
C. Chemoautotrophic
D. Parasitic

Answer: B

77. When farmers are using fewer and fewer number of crop varieties, what would happen to the genetic diversity of our crop plants?
A. It would get eroded.
B. It would be more conserved.
C. It would be more diverse.
D. Nothing would happen to it.

Explanation:
If a farmer uses fewer number of crop varieties the nutrients in the soil get degraded. This results in the diversity of crop variety to get eroded.

Answer: A

78. During anaerobic respiration, what is the gross number of ATP molecules that are produced per glucose molecule?
A. Two B. Three C. Four D. Six

Explanation:
The gross amount of ATP molecules produced during anaerobic respiration from one glucose molecule is 4 ATP. However, since the cell spends 2ATP's to carry out the process, the net of 2ATP molecules are produced.

Answer: C

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79. Among the following respiration, what is the gross number of ATP molecules that are produced per glucose molecule?
A. Aerobic respiration
B. Anaerobic respiration
C. Oxidizing atmosphere
D. Photosynthesis

Explanation: The correct sequence according to the evolutionary history is:

$$B \longrightarrow D \longrightarrow C \longrightarrow A$$

Answer: B

80. Which of the following is **NOT** a learned behavior?
A. Imprinting
B. Habituation
C. Conditioning
D. Trial and error

Answer: A

81. If an actively growing potted seedling is kept horizontally, which of the following would eventually happen to the seedling?
A. The shoot would bend upward.
B. The root would bend upward.
C. The seedling would stop growing.
D. The seedling would grow horizontally.

Explanation:
When an actively growing potted seedling is kept horizontally, the shoot would bend upward because of phototropism.

Answer: A

82. Which of the following was the most possible mode of evolution by which the many species of Darwin's fiches evolved on the Galapagos islands?
A. Phyletic evolution
B. Divergent evolution
C. Convergent evolution
D. Sympatric evolution

Answer: B

83. For which of the following processes is ATP **NOT** required?
A. Diffusion of oxygen into cells.
B. Synthesis of molecules.
C. Active transport of molecules from one site to another.
D. Muscle contraction.

Explanation:
Diffusion of oxygen in the cells is a passive transport system that doesn't require energy (ATP).

Answer: A

84. Mutation may be described as:
A. Phenotypic change
B. Change in gene structure
C. Continuous variation
D. Change due to hybridization

Explanation: Mutation is a rare and sudden change in a genetic makeup of an organism.

Answer: B

85. Which of the following is the most **UNLIKELY** action that would be expected from Ethiopians who have been adequate biological literacy?
A. Engagement in activities resulting in biodiversity loss.
B. Supporting activities concerned with food security.
C. Awareness about climate change and its problems.
D. Engagement in biological resource conservation.

Explanation: A person who has adequate biological literacy is not expected to get engaged in activities that result in loss of biodiversity.

Answer: A

86. Which form of mutation is responsible for the disease known as leukemia?
A. Duplication
B. Insertion
C. Inversion
D. Deletion

Explanation:
Inversion is the orientation of parts of DNA on the chromosome. It causes genetic problems such as leukemia (due to inversion on chromosome-16), embryo miscarriage, and growth failures.

Answer: C

87. Which of the following characteristics can show the evolutionary relationships among organisms?
A. Structures having similar functions.
B. Presence of analogous structures.
C. Presence of homologous structures.
D. Structures of having different origins.

Explanation:
Homologues structures are structures that have similar physiology, but are adapted to different functions. Such kinds of structures serve as evidence for evolutionary relationships.

Answer: C

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88. Which of the following species isolating processes could lead to the evolution of a new species by the mechanism known as sympatric speciation?
A. A river that changed its course for many years.
B. A new mountain range that was created many years ago.
C. Populations of a species having different breeding seasons.
D. A land mass that separated two water bodies for many years.

Explanation: **Sympatric speciation** is the process through which new species evolve from a single ancestral species while inhabiting the same geographic region.

Answer: C

89. What is an apoenzyme?
A. A protein molecule
B. An active enzyme
C. A non – protein molecule
D. A non – protein cofactor

Explanation: **Apoenzyme** a protein component that together with a coenzyme forms an enzyme.

Answer: A

90. For which of the following can divergent evolution be taken as an alternative name?
A. Allopatric specification
B. Adaptive radiation
C. Sympatric speciation
D. Disruptive selection

Explanation: **Adaptive radiation** is a type of **divergent evolution** where a group of organisms quickly diverges into new species.

Answer: B

91. Which one of the following is the result of the similarity observed between the wings of a bird and a pterodactyl?
A. Convergent evolution
B. Directional selection
C. Stabilizing selection
D. Divergent evolution

Answer: A

92. Which of the following played an important role in the evolution of human being?
A. Emergence of wings in addition to limbs.
B. Development of even toes in the forearms.

C. Development of feathers and hollow bones.
D. Development of opposable thumb to grasp.

93. In which one of the following do we use genetically modified bacteria for purposes other than human health?
A. Production of insulin
B. Production of antibiotics
C. Production of herbicides
D. Production of vaccines

Explanation: Genetically modified bacteria are capable of producing human insulin that is used for diabetic people.

Answer: A

94. Which knowledge about cell biology did scientists discover after all the others?
A. The first understanding that living things are made of cells.
B. Knowledge about how chromosomes behave during mitosis.
C. First understanding about the double helix structure of DNA.
D. First understanding about the DNA sequence in human genome.

Answer: D

95. Which of the following statements can be considered as better definition of the biological concept of behavior?
A. The reactions of a person or an animal in response to an external or internal stimulus.
B. The responses or reactions or movements that a living plant makes in any situation.
C. A system of coordinated response by an organism to an external or internal stimulus.
D. A receptor of some kind of stimulus that an organism has which produces a response.

Answer: C

96. To which category of behavior does the human behavior that involves the strengthening of existing responses or the formation of new responses to existing stimuli that occur because of practice or repetition belong?
A. Learned behavior
B. Instinctive behavior
C. Innate behavior
D. Inborn behavior

Answer: A

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97. Which of the following is **NOT** grouped under genetically pre-programmed patterns of behavior?
A. Reflex actions in humans
B. Orientation behavior
C. Conditioned behavior
D. Instinctive behaviors
Explanation: Conditioned behaviors are behaviors that develop prior to pre-learned experience. **Answer: C**

98. Which group of bacteria reduces nitrates to nitrogen gas in the nitrogen cycle?
A. Nitrogen fixing bacteria
B. Ammonium ion forming bacteria
C. Denitrifying bacteria
D. Photosynthetic bacteria
Answer: C

99. The main factors that determine the types of terrestrial biomes in a certain geographical area are:
A. Soil and vegetation formations.
B. Soil and biological diversity.
C. Complexity in ecological succession.
D. Precipitation and temperature.
Answer: D

100. In which parts of Ethiopia are lions, elephants and giraffes found more commonly?
A. In the afroalpine and subafroalpine zones
B. In the highland areas of northern Ethiopia
C. In the grasslands of southwestern Ethiopia
D. In central and northern parts of the rift valley.
Answer: C

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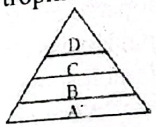
1. Cells were first seen using the microscope by:
A. Schleiden
B. schwann
C. Robert Hooke
D. Leeuwenhoek
Explanation: Robert Hooke is an English scientist who has first seen dead cork cells. He is also the fisrt perso to coin the name cell **Answer: C**

2. Of the following organic molecules, which one includes a pentose sugar?
A. Deoxyribose
B. Fructose
C. Galactose
D. Glucose
Explanation: Pentose sugar is a 5 – carbon sugar which contains deoxy ribose sugar. **Answer: A**

3. In the energy pyramid, which group of organisms has the least amount of energy?
A. Primary producers
B. Herbivores
C. Tertiary produces
D. Animals
Explanation: The energy content in a food chain decreases as trophic level increases.



A = produce (1st trophic level)
B = Primary consumer (2nd trophic level)
C = Secondary consumer (3rd trophic level)
D = Tertiary consumer (4th trophic level)
Answer: D

4. What does an enzyme do to a chemical reaction that allows it to proceed optimally?
A. It increases the amount of the substrate.
B. It lowers the required activation energy.
C. It increases the required activation energy.
D. It modifies both the p^H and the temperature.
Explanation: Enzymes make chemical reactions to proceed optimally by lowering the activation energy required. **Answer: B**

5. Which of the following tools is used by a biologist to study the internal anatomy of experimental animals?
A. Microscope
B. Hand lens
C. Petri dish
D. Dissecting kit
Explanation: A dissecting kit is used in the study of the internal structure (anatomy) of an organism **Answer: D**

6. Which of the following is **NOT** a characteristic of an enzyme?
   A. It is a globular protein
   B. It functions as a catalyst
   C. It is converted to products
   D. It lowers the energy of activation

Explanation: Enzymes have the following characteristics. These are:
- ✓ they are biological catalysts.
- ✓ they speed up the rate of the chemical reaction.
- ✓ they are globular proteins.
- ✓ They can be reused because they do not get consumed in the reaction.
- ✓ They are specific
- ✓ They lower the activation energy required for the reaction.

Answer: C

7. Who was the first person who believed to have seen live moving cells under a microscope?
   A. Robert Hooke C. Rene Dutrochet
   B. Anton van Leeuwenhoek D. Theodor Schwann

Explanation: Anton Van Leeuwenhoek is the first perso to see a living animal cells under the microscope. He has also contributed a lot in the development of microscope.

Answer: B

8. Why do micro-organisms decompose dead organic matters?
   A. To release mineral nutrients for the plants
   B. Ro clean up the ecosystem
   C. To keep the balance of atmospheric gases
   D. To drive energy for their own use

Explanation: Microorganisms decompose (break down) dead bodies of organisms (biomass) to produce energy for their own use. However the process also plays a huge in recycling of nature which cleans up the environment.

Answer: D

9. Which one of the following is the best scientific approach to come to conclusive evidence showing that a given event is caused by a certain factor?
   A. Having a strong belief
   B. Making a good guess
   C. Conducting an experiment
   D. Making systematic observation

Explanation: Experimentation is a scientific way or procedure to come up with the evidence that has caused by a certain factor. It is one step in a scientific method.

Answer: C

10. The ultimate source of genetic variation is:
    A. selection C. genetic drift
    B. migration D. mutation

Explanation: **Mutation** is a sudden change in a genetic make-up of an organism. It results in genetic variation after successive generation.

Answer: D

11. Which of the following is the first step in starting a research process?
    A. Searching sources of information to located problem.
    B. Identification of problem
    C. Survey of related literature
    D. Searching for solutions to the problem

Explanation: Scientific method is a set of procedures used to solve a problem in a scientific way. Step in scientific method are:
- i. identification of a problem.
- ii. collecting in formation
- iii. setting a hypothesis
- iv. experimentation
- v. accepting or rejecting the hypothesis
- vi. drawing conclusion on a solution retadeol to a problem.

Answer: B

12. Which of the following is lacking in prokaryotic cells?
    A. Cytosol C. Cellulose
    B. Ribosome D. DNA strand

Explanation: Prokaryotic cells are cells that have the following structures. These are, cytosol, ribosome, DNA strand, etc. However, cellulose is a component of plant cellwall and not found in prokaryotic organisms such as bacteria. Their cell wall is mainly made up of muerin.

Answer: C

13. What is the name of a major group of eukaryotic organisms that obtain their nutrition using extracellular digestion?
    A. Viruses B. Bacteria C. Algae D. Fungi

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Explanation: Fungi are mostly saprophytes and eukaryotes. They decompose the dead body of an organism this is an example of extracellular digestion. **Answer: D**

14. For which of the following purposes is Global Positioning System (GPS) used?
    A. Satellite signal
    B. Theodolite
    C. Microwave
    D. Map of an area

Explanation: Gps makes possible to record positions quickly and extremely accurately. **Answer: D**

15. Which of the following statements is **INCORRECT** with respect to an enzyme's ability to catalyze a reaction?
    A. An enzyme binds reactants such that they are positioned correctly and can attain their transition-state configurations.
    B. An enzyme provides a reaction surface and a hydrophilic environment for the reaction to take place.
    C. An enzyme allows the reaction to go through a more stable transition state than would normally be the case.
    D. An enzyme can weaken bonds in reactants through the binding process. **Answer: B**

16. The amount of organic matter present at a given time per unit area is called:
    A. carbon sequestration
    B. productivity
    C. carbon foot print
    D. standing crop

Explanation: The Total biomass of an ecosystem or any of its components at a given time is called standing crop. **Answer: D**

17. In which of the following is there a **MISMATCH** between the type of innate behavior and the resulting action/response?
    A. Orientational movement – Weaving a web in spiders
    B. Instinctive behavior – Imprinting in birds
    C. Reflex action – Blinking of the eyes in humans
    D. Instinctive behavior – Nest – building in birds

Explanation:
- Weaving web in spiders, imprinting and nest building are examples of instinctive behavior. **Answer: A**

18. Tow which genus of primates are the Neanderthal humans classified?
    A. The genus *Ardipithecus*

    B. The genus *Homo*
    C. The genus *Australopithecus*
    D. The genus *Zinjanthropus*

Explanation: Neanderthal human belong to a genus homo. Their full scientific name is *Homo Neanderthal* **Answer: B**

19. What kind of nutrition do most of the decomposers have in general?
    A. Saprobiotic nutrition
    B. Autotrophic nutrition
    C. Endosymbiont nutrition
    D. Chemosynthetic nutrition

Explanation: Chemosynthetic nutrition is decomposition of dead bodies of organisms carried out by bacterium and fungi. **Answer: D**

20. The practice of modern agriculture that reduces crop biodiversity is:
    A. Overgrazing
    B. Growing few high yielding varieties
    C. Forest clearing
    D. Growing landraces

Explanation: Growing only high yielding varieties of crops may reduce the biodiversity of an ecosystem. **Answer: B**

21. One of the following is NOT true about protein synthesis in eukaryotes.
    A. Translation takes place in the cytoplasm
    B. Transcription takes place in the nucleus
    C. mRNA is modified after transcription
    D. Both transcription and translation take place in the cytoplasm

Explanation: The process of translation takes place in the cytoplasm, while the process of transcription takes place in the nucleus. **Answer: D**

22. One of the following groups of plants carry out light dependent and light independent reactions of photosynthesis in separate cells of the leaf.
    A. Plants adapted to temperate regions

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B. Carbon – 4 plants
C. Carbon – 3 plants
D. Cacti

Explanation: C_4 plants carry of the light dependent and light independent reactions of photosynthesis in separate cells. These are:
✓ Mesophyll cells and bundle sheath cells

Answer: B

23. Through which process can inorganic ions from the soil be absorbed into the root hairs?
A. Dehydration
B. diffusion
C. Osmosis
D. Active transport

Explanation: Mineral ions are transported from the soil in to root hair cells by active transport. While, water is absorbed by osmosis.

Answer: D

24. What is the nutritional mode of those bacteria which decompose dead organic matter and thus recycle nutrients?
A. Parasitic
B. Symbiotic
C. Autotrophic
D. Saprobiotic

Explanation: Bacterias that decompose the dead organic matter and recycle nutrients are examples of **chemo autotrophs.**

Answer: C

25. The sets in which scientist usually conduct two or more sets of the same experiment at the same time is:
A. Replication
B. Experimental design
C. Experimental group
D. Control group

Answer: D

26. Which of the following properties of water makes sweat a good body coolant?
A. Its strong surface tension
B. Its low density in the solid state
C. Its high heat of vaporization
D. It strong adhesion to a surface

Explanation: Evaporation of water from the body in the form of sweat plays a role in temperature regulation by cooling the body's heat.

Answer: C

27. How many molecules of glycerol and fatty acids, respectively, are needed to form 100 molecules of triglyceride fat?
A. 50 and 50
B. 100 and 300
C. 100 and 100
D. 150 and 400

Explanation: 1 molecule of triglyceride has 3 fatty acid and 1 glycerol molecules. Thus, 100 molecules of triglyceride has 100 glycerol and 300 fatty acids.

Answer: B

28. In a metabolic pathway (A –> B –> C –>D), where step is catalyzed by a separate enzyme, which substance in the pathway serves as an allosteric regulator?
A. Substance D
B. Substance B
C. Substance C
D. substance A

Explanation: Allosteric regulators are substances that affect enzymes activity by binding to allosteric site. Thus, enzyme A is allosteric regulator. **Answer: D**

29. At which stage is most of the carbon dioxide released during aerobic respiration?
A. Substance D
B. Glycolysis
C. Electron transport
D. Chemiosmosis

Explanation: From 38 ATP's produced from a glucose molecule, 32 to 34 ATP's are produced during chemiosmosis.

Answer: D

30. Which of the following combinations is known as Neo – Darwinism?
A. Natural selection and acquired inheritance
B. Acquired inheritance and Mendelian law of inheritance
C. Natural selection and Mendelian law of inheritance
D. Natural selection and Lamarck's theory of evolution

Explanation: Neo-Darwinism incorporates knowledge of genes and gene action into the theory of natural selection.

Answer: C

31. In the process of allopatric speciation, which of the following is the mechanism that prevents populations from exchanging genes?
A. Geographic barrier
B. Polyploidy
C. Seasonal isolation
D. Temporal isolation

Explanation: Allopatric speciation is when the species become isolated by some physical feature. Such kind of isolation is called **geographic isolation.** **Answer: A**

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32. In what main way do retroviruses such as HIV differ from other types of RNA viruses?
A. They reversely copy RNA to DNA
B. They require a host cell to reproduce
C. Their genes are resistant to mutation
D. They are free living virus
Explanation: Retro virus is an RNA virus that converts its genetic information form RNA into DNA after it has infected the host.
Answer: A

33. Which of the following is NOT true about the flow of energy in the ecosystem?
A. It passes from one trophic level to the next.
B. It enters the ecosystem, in the form of light.
C. It is recycled by decomposers.
D. It leaves the ecosystem in the form of heat.
Explanation: The flow of energy in an ecosystem is always unidirectional. This is due to the reason that energy is not recycled in an ecosystem. **Answer: C**

34. A boy who had seen a snake crossing his way earlier, jumped up violently when grass touched his legs. Which behavior is it?
A. Habituation C. The royal jelly
B. Imprinting D. Mixture of iodine and sugar
Answer: D

35. What material do microbiologists use to culture bacteria in the laboratory?
A. Moist soil C. the royal jelly
B. Agar nutrient D. Mixture of iodine and sugar
Explanation: Agar is a solid nutrient medium that is made mainly from red algae. It is used to grow microorganisms the laboratory
Answer: B

36. Which of the following is usually taken as a signal for final stage of HIV infection?
A. Outbreak of some genetic diseases
B. Unifected T-helper cells present
C. Signs of the disease Kaposi's sarcoma
D. No infection by other parasitic diseases
Explanation: The final stage of HIV infection is characterized by:
✓ presence of cancers such as Kaposi's sarcoma.

✓ Infection by opportunistic diseases such as tuberculosis.
✓ Reduction in number of T-helper cells, etc.
Answer: C

37. During meiosis, which process makes the four chromatids of a homologous pair of chromosomes genetically different from one another?
A. Crossing over C. Segregation
B. Independent assortment D. Linkage
Explanation: Chromosomes produced by meiosis are genetically different from one another because of crossing over. Crossing over is an event where exchange of genetic material occurs between chromatids in a homologous pair.
Answer: A

38. In ecological context, which of the following is a population?
A. All living organisms in a habitat
B. All the producers in a habitat
C. All individuals of same species in a habitat
D. All the consumers in a habitat
Explanation: Population belong to organisms of the same species living together in a particular habitat. **Answer: C**

39. Through which of the following processes do all living organisms contribute to carbon cycle?
A. Photosynthesis C. Decomposition
B. Combustion D. Cellular respiration
Explanation: During cellular respiration all organism release carbon dioxide gas into the atmosphere which plays part in carbon cycle. **Answer: D**

40. Which of the following organelles settles out last when homogenized cells are centrifuged?
A. Chloroplast C. Ribosome
B. Nucleus D. Mitochondria
Explanation: Smaller organelles settle last when centrifuged. Thus, ribosomes are smallest organelles in cell.
Answer: C

41. Current human concerns such as famine, pestilence, and environmental degradation would be solved through the use of:
A. aquaculture C. biological principles
B. irrigation D. terracing

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Explanation: Biological principles that have solutions to resolve current human concerns such as famine, pestilence and environmental degradation. **Answer: C**

42. Form which group of micro-organisms was the first antibiotics produced?
    A. Fungal organism
    B. Gram-positive bacteria
    C. Gram- negative bacteria
    D. Viral agents

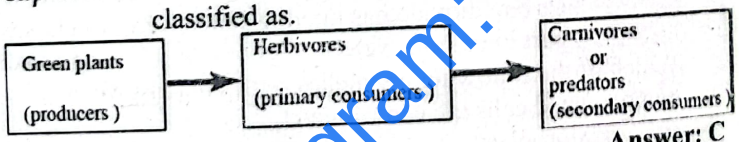
Explanation: The first model antibiotic is produced from a fugal species known as *Penicillium notatum* by sir Alexander Fleming. **Answer: A**

43. In the process of photosynthesis light is necessary to:
    A. split carbon dioxide
    B. produce ATP and a reducing substance
    C. release energy
    D. combine CO_2 and H_2O

Explanation: During the process of Photosynthesis light used to split water molecules in to hydrogen and oxygen and also, used as a chemical energy found in a glucose molecule. **Answer: C**

44. What is the other name for the primary consumers of the ecosystem?
    A. Green plants
    B. Carnivores
    C. Herbivores
    D. Predators

Explanation: In an ecosystem the organisms motioned above are classified as.



Answer: C

45. How are organic molecules distinguished from inorganic molecules? Organic molecules always contain:
    A. carbon and hydrogen.
    B. carbon and oxygen.
    C. oxygen and nitrogen
    D. oxygen and hydrogen.

Explanation: Organic molecules necessarily contain hydrogen and carbon atom. While inorganic molecules many have one of them or none. **Answer: A**

46. In which industrial enzyme technology is lactase practically applied to produce lactose-free milk for use by lactose intolerant individuals?
    A. Chocolate manufacturing industry
    B. Dairy product manufacturing industry
    C. Human hormone manufacturing industry
    D. Cereal food manufacturing industry

Explanation: Enzymes such as renin, lipase, lactase, bio chymoscysticin are applied in diary industry. **Answer: B**

47. Which kind of enzyme inhibitors bind to enzymes only weakly and the bond that holds them breaks easily releasing the inhibitor to allow the enzyme to become active again?
    A. Irreversible inhibitors
    B. Inhibitors that bind strongly to enzymes
    C. Reversible inhibitors
    D. Inhibitors that permanently alter enzyme structure

Explanation: Reversible inhibitors bind to enzymes weekly in which the bond holding enzyme molecules breaks easily **Answer: C**

48. Of the following models describing the nature of the cell membrane, which one was proposed before all the others?
    A. Proposed before all the others?
    B. Fluid mosaic model
    C. The unit membrane model
    D. The sandwich model

Explanation:
- In 1925 E.C garter and G. Grendel suggested phospholipid bilayer structure of plasma membrane.
- In 1935 Davison and Danielli suggested sandwich model.
- In 1959 J.David Robertson proposed unit membrane model.
- In 1972 J.J singer and G.L. Nicholson proposed the fluid mosaic model **Answer: A**

49. In which organ of the human body are more bacteria found?
    A. adequate distribution of condoms
    B. adequate distribution of antiviral drugs
    C. awareness creation
    D. breaking the transmission pathway

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Explanation: Bacterias may found at different parts of the body. Most of these are found in large intestine. **Answer: B**

50. The best way to control HIV/AIDS is:
A. adequate distribution of condoms
B. adequate distribution of antiviral drugs
C. awareness creation
D. breaking the transmission pathway

Explanation: Transmission of HIV can be controlled in different ways. However, the most effective way is breaking or stoping the transmission pathways. **Answer: D**

51. What is the possible effect of the weak hydrogen bond among water molecules?
A. It helps water to boil at low temperature
B. Plays a significant role in the interaction of a water molecule with hydrophobic surface.
C. It results in the crystal shape of ice
D. It speeds up photosynthesis **Answer: A**

52. Of the following activities, which one is considered to have an indirect negative effect on biodiversity?
A. Charcoal making and illegal logging
B. Human and livestock population growth
C. Deforestation for cultivation and urbanization
D. Monoculture farming with improved crop varieties

Explanation: All of the points listed have direct effect on biodiversity. However, human livestock population growth has an indirect effect on biodiversity. **Answer: B**

53. If two heterozygous tall pea plants are crossed, how many of the offspring would be phenotypically tall plants?
A. 0% B. 25% C. 75% D. 50%

Explanation: If two heterozygous pea plants are crossed the probility of having a dominat pheow type is 75% and that of recessive one is 25%. **Answer: C**

54. Cell organelles that contain hydrolytic enzymes are called:
A. Peroxisomes C. Mesosomes
B. Ribosomes D. Lysosomes

Explanation: Lysosomes are cell organelles that contains hydrolytic enzymes. **Answer: D**

55. Which part of the cell makes necessary changes, packages and secretes proteins?
A. Mitochondira C. Golgi apparatus
B. Endoplasmic reticulum D. cell wall

Explanation: Golgi apparatus is a cellular organelle that collects, packages and concentrate cellular products such as proteins and carbohydrates . **Answer: C**

56. Which of the following takes place under normal conditions, as electrons flow down the electron transport chain of the mitochondria?
A. The electrons lose free energy.
B. NADH and $FADH_2$ are oxidized.
C. The pH of the matrix increases.
D. An electrochemical gradient is formed.

Explanation: During electron transport chain electron lose free energy as they pass from one carrier to the next. **Answer: A**

57. In which of the following ways does learned behavior differ from innate behavior?
A. Learned behavior is adaptive in nature.
B. Learned behavior is modified by new experiences.
C. Learned behavior is genetically determined.
D. Learned behavior is functional at the first attempt.

Explanation:

Innate behavior	Learned behavior
→ Genetically determined → no modification → Adaptive	→ Changes or developed through experience → modified by new experience → Learned

Answer: B

58. Which of the following features doe analogous structure mostly share?
A. Functional similarity
B. Developmental similarity
C. Ancestral similarity
D. Phylogenetic similarity

Explanation:
✓ **Analogous structure** – are structures that are physiological different but have similar function.

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✓ **Homologous structures** – are structures having physiologically similarity but adapted to carry out different functions. **Answer: A**

59. How can biotechnology be applied to reduce the use of inorganic fertilizers in agriculture?
   A. By using the golden rice and producing organic fertilizers from beta carotenes.
   B. By using thermostable enzymes of genetically engineered bacteria used in washing powders
   C. By using enzymes produced by genetically engineered bacteria used in the food industry.
   D. By selecting and propagating varieties with high growth rates and high productivity. **Answer: A**

60. Why is it that longer races by athletes must be run slower that shorter races?
   A. To give time for lactic acid fermentation in muscle cells
   B. To avoid hunger as food is quickly converted to energy
   C. To adapt to the weather conditions at the racing place
   D. To allow aerobic respiration to produce the required ATP

   Explanation: In longer races athletes run slower to comfortably in hale sufficient amount of oxygen needed to carry out aerobic respiration. **Answer: D**

61. The evolution of a species which is based upon the total adaptation changes could be preserved by:
   A. natural selection C. isolation
   B. speciation D. human conservation

   Explanation: According to Darwin's theory of natural selection: those member of species which are best adapted to their environment will survive and reproduce in greater numbers than others less well adapted. **Answer: A**

62. One goat is heterozygous, long hair (Rr), and its mate carries homozygous short (rr). What is the probability of the offspring having short hair?
   A. 50% B. 100% C. 75% D. 25%

   Explanation:
   Long hair (Rr) × short hair (rr)
   Rr × rr

	R	r
r	Rr	rr
r	Rr	rr

- 2Rr: 2rr ⟶ genotypic ratio
- 50% long hair: 50% short hair ⟶ phenotypicration

Answer: A

63. Which one of the following represents the population growth curve of a species with limited resources?
   A. Lognormal C. Logistic
   B. Exponental D. Competitive exclusion

   Explanation: Competition between organisms of two different species on the same available resource in the niche makes one to be successful and the other to extinct. This condition is called **competitive exclusion**. **Answer: D**

64. Habituation and imprinting illustrate that behaviors are made up of:
   A. both innate and learned components.
   B. reversible stimulus – response learning.
   C. unchanging responses to present stimuli.
   D. simple responses to complex stimuli. **Answer: A**

65. Which of the following is not an industrial use of bacteria?
   A. Food processing
   B. Genetic engineering
   C. Waste water treatment
   D. Recycling of minerals through ecosystems

   Explanation: All the options listed; food processing, genetic engineering and waste of water treatment involve industrial use of bacteria. While, recycling on minerals through ecosystems in natural process. **Answer: D**

66. Carbon cycle is important in nature because:
   A. the amount of energy before and after a reaction must be the same.
   B. it consumes oxygen and nitrogen in the atmosphere.
   C. it continuously reuses and recycles carbon in the ecosystem.
   D. it is used in energy capture and recapture.

   Explanation: Carbon cycle continuously recycles and uses the carbon in an ecosystem. **Answer: C**

67. Which of the following is essential in order for a virus to replicate?
   A. The genome must be released into the cytoplasm.

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B. The capsid must enter the host cell cytoplasm.
C. The host cell must be undergoing mitosis.
D. The host cell must lack cell membrane.

Explanation: For virus to ruplicate inside the host cell, the most essential state is the process by which the virus ingects its genome (DNA) in the host cells cytoplasm. **Answer: A**

68. Which of the following **NOT** in situ conservation of Biodiversity?
A. Sanctuary
B. Botanical garden
C. Biosphere reserves
D. National Parks

Explanation:
Insitu concervation is conservation of ecosystem & natural habitats in their natural surroundings. These include national Parks, Wild life conservation, biosphere reserves, etc

Answer: B

69. The type of enzyme used in recombinant DNA technology to split a specific sugar phosphate bond in each strand of a DNA double helix is called:
A. esterase
B. lipase
C. restriction enzyme
D. ligase

Answer: C

70. Lamarck's theory of evolution is known as:
A. natural selection
B. inheritance of acquired characters
C. survival of the fittest
D. mutation theory

Explanation: Lamarck's theory evolution is based on inheritance of acquired characteristics. However it is inherited characteristic that determine an organisms trait. **Answer: B**

71. Among the theories about the origin of life, which one better agrees with the concept of change in organisms and their genetic composition over generations being caused by meiosis, hybridization, natural selection or mutation?
A. Cosmozoan theory
B. Special creationism theory
C. Spontaneous generation theory
D. Biochemical origin theory

Answer: D

72. Among the following crops, which one uses a non – C_4 photosynthetic pathway?
A. Maize
B. Pineapple
C. Sorghum
D. Sugarcane

Explanation: C_4 plants include grasses, maizes and sorghum.

Answer: B

73. Where is the convergence point for the metabolism of the building blocks of carbohydrates, lipids and proteins?
A. Cytoplasm
B. Calvin cycle
C. Krebs cycle
D. Electron transport

Answer: A

74. How quickly a population would increase or decrease in size is probably least influence by the presence of:
A. disproportionate human races
B. adequate supply of oxygen
C. severe inter-specific completion
D. carnivores and herbivores

Explanation:
The proportionality in human race can cannot influence a decrease or increase in human population size.

Answer: A

75. Based on differences in DNA, which of the following animals is closer to human beings?
A. Domestic pig
B. Zebra fish
C. Frog
D. Domestic chicken

Explanation:
According their closeness to human on the basis of differences in their DNA, domestic pig is relatively close to human beings.

Answer: B

76. To which group of sugars can the molecule Dihydroxy acetone be classified?
A. Hexose sugars
B. Triose sugar
C. Pentose sugars
D. Oligosugar

Explanation: Dihydroxy acetone is a 3- carbon monosaccharide molecule. $9C_3H_6O_3$) **Answer: B**

77. Choose the one in which the human digestive enzyme is incorrectly matched with the substrate that is normally catalyzes.
A. Lipase – fat
B. Pepsin – protein
C. Pancreatic amylase – starch
D. Trypsin – starch

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Explanation: The correct match of human digestive enzymes and nutrients given is as follows:

Enzyme	Nutrient
Lipase	Fat
Pepsin	Protein
Panacreatic amylase	Starch
Trypsin	Protein

Answer: D

78. The organelle of the cell which is concerned with the synthesis of lipids in addition to being associated with carbohydrate metabolism and detoxification is:
A. smooth endoplasmic reticulum
B. golgi body/ apparatus
C. thylakoid membrane
D. rough endoplasmic reticulum
Explanation: Smooth endoplasmic reticulum is concerned with lipid synthesis, carbohydrate metabolism and detoxification. **Answer: A**

79. Which of the following responsible for rapid evolution in RNA viruses?
A. High stability of its RNA.
B. Reverse transcription of it RNA.
C. Accurate replication of its RNA.
D. High mutation rate of its RNA.
Explanation: During

80. What major consequence does primary ecological succession involve normally?
A. Number of species present decrease
B. Abiotic factors become less favourable
C. Complexity of food webs increase
D. Habitats become harsher and deteriorate
Explanation: During primary succession pioneer species change the abiotic condition and allow other species to colonize. Step by step the number of species present increases and complexity of food web increase. **Answer: C**

81. Which of the following two elements are found in all organic molecules?
A. Carbon and oxygen

B. Carbon and hydrogen
C. Carbon and nitrogen
D. Carbon and phosphours
Explanation: The two elements that are necessarily found in all organic molecules are carbon and hydrogen. **Answer: B**

82. Which one of the following is the primary contribution of Darwin to biological theory?
A. New alleles arise through mutation.
B. Evolution is the change in gene frequencies over time.
C. An important mechanism of biological evolution is natural selection.
D. Genes are the units of inheritance.
Explanation: Charles Darwin is known as the father of evolution due to his detailed explanation of biological evolution on his theory of natural selection. **Answer: C**

83. A country with a high youth population but a low old age population and low infant population shows:
A. a population with wide base and narrow apex
B. an inverted pyramid
C. a pyramid assuming a rectangular shape
D. a population pyramid with bulge in the middle
Answer: D

84. Significant changes in pH that can affect an enzyme molecule include:
A. building strong ionic bonds leading to stabilization of enzymes.
B. changing the color and palatability of food items.
C. altering the changes on some substrates making it difficult to bind to enzyme.
D. increasing the boiling temperature of the substrate.
Answer: C

85. Which of the following is the most accepted theory about the origin of life on earth?
A. Theory of chemical evolution
B. Theory of spontaneous evolution
C. The cosmic theory
D. Theory of special creation **Answer: A**

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86. What is the possible cause of isolating mechanism for the formation of allopatric species?
A. Geographical isolation
B. Temporal isolation
C. Seasonal isolation
D. Behavior isolation
Explanation: Allopatric speciation is a process in which different populations of species develop due to physical/ geographical isolation.
Answer: A

87. Which of the following statements about instinctive actions is NOT true?
A. are innate.
B. require learning.
C. are adaptive
D. have fixed pattern.
Explanation: Instructive behaviors are:
✓ pre-programmed and characterized by complex sequence of actions.
✓ Common to all members of a species
✓ Show fixed action pattern
✓ adaptive
✓ are innate.
Answer: B

88. What does a low value for Simpson's index of diversity normally indicate to the researcher?
A. An area dominated by one or just a few species
B. An area with high number of endemic species
C. A biodiversity hotspot with abundant species
D. An area with plants but no species of animals
Explanation: High Simpson's index value show high diversity of species and low Simpson's index values shows less diversity (area dominated few number of species).
Answer: A

89. Retroviruses are able to reverse transcribe RNS to RNA in order to:
A. control the formation of viral protein and RNA that can be assembled into new virus particles.
B. command the host cell in the production of genomes similar to that of the hosts.
C. replicate without any interruption.
D. control the formation of viral protein and DNA that can be assembled into new viral particles.

Explanation: Retroviruses release their RNA into the cell they infected and an enzyme reverse transcriptase converts viral RNA in to DNA. The helps them to control the formation of more viral protein and RNA that can be assembled in to new virus particles
Answer: A

90. Which of the following nucleotide sequence contains four pyrimidine bases?
A. Both B and C
B. UAGCGGUAA
C. GCUAGACAA
D. GATCAATGC
Explanation: Among the nitrogenous that make up a nucleotide.
✓ **thymine, cytosine** and **uracil** are pyrimidire bases
✓ **Adenine** and **guanine** are purine bases
Answer: D

91. A small girl who had observed her mother making a telephone call, one day made a call on her own. Which learning behavior does this exemplify?
A. Insight learning
B. Latent learning
C. Trial error
D. Sensitization
Explanation:
Latent learning is a behavior that the brain acquires at certain time, without reinforcement. It is also a form of hiden learning. Thus, the girl learned the phone call form her mother without any sort of reinforcement and it is a form of hidden learning.
Answer: B

92. Suppose the age pyramid of a population has a broad base a narrow top, which of the following is NOT true about the population?
A. It is fast growing
B. It is dominated by young people
C. It has a high birth rate
D. It has high life expectancy
Explanation: If the pyramid of a population is broad at the base and narrower at the top, it shows that:
✓ there are more children that older people.
✓ The population is fast growing.
✓ There is high birth rate.
Answer: D

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93. Which of the following terms specifically refers to an ecological succession that starts in an aquatic environment?
A. Hydrosere
B. Xerosere
C. Primary succession
D. Secondary

Explanation: There are two types of succession. These are:
- ✓ **Primary succession** – occurs on a bare land where there is no any form life. It occurs on roacks or open water which is newly formed .
- ✓ **Secondary succession** – occurs on areas were there were living things earlier.

∴ If the succession has occurred in a newly earlier water body, it can be taken as a **primary succession**.

Answer: C

94. In which of the following ways does mutation contribute most to evolution?
A. It preserves adaptive genes.
B. It creates new alleles.
C. It forms a new gene pool
D. It helps genes to recombine

Explanation: Mutations produce new genes. The number of organisms with the new gene increases if they are favoured by natural selection. Thus, mutations also contribute to evolution.

Answer: A

95. If two molecules of a six carbon sugar are condensed to form a disaccharides.
A. $C_{12}H_{22}O_{11}$
B. $C_6H_{12}O_6$
C. $C_{11}H_{12}O_{11}$
D. $C_{12}H_{24}O_{12}$

Explanation: A disaccharide is a sugar that contain two sugar units. They are formed by the condensation reaction that occurs between two mono saccharides

$C_6H_{12}O_6 + C_6H_{12}O_6 \longrightarrow C_{12}H_{22}O_{11} + H_2O$ **Answer: A**

96. Which of the following is NOT a requirement for a molecule to directly diffuse across the cell membrane?
A. Small size
B. Non- polarity
C. Lipid solubility
D. Large size and polarity

Explanation: Large sized and polar substances cannot easily pass across a membrane by diffusion. They required an active transport system which requires energy. **Answer: D**

97. Which of the following molecules an provide the cell the least amount of energy per molecule?
A. Glucose
B. ATP
C. Reduced FAD
D. Reduced NAD

Explanation: The amount of ATP molecules produced per molecules is indicates as:

Molecule	Amount of energy
Glucose	36/38 ATP at aerobic respiration
Reduced FAD	2 ATP's
Reduced NAD	3 ATP's

Answer: C

98. During photosynthesis, which of the following is the first step in the transduction of light energy to chemical energy?
A. Photolysis of water
B. Carbon fixation
C. ATP formation
D. Glucose formation

Explanation: The correct step during photosynthesis is:
i. Photolysis of water
ii. carbon fixation
iii. glucose formation
iv. ATP Formation

Answer: A

99. What is the reason that Stanley Miller excluded free oxygen from the mixture of the gases that he used for his simulation experiment?
A. Early organic molecules did not contain oxygen.
B. Primitive atmosphere was reducing.
C. Oxygen cannot react with the rest of the gases.
D. The primitive atmosphere was full of oxygen.

Explanation: Oparin and Stanely Miller's experiment was based on the assumption that; the primitive atmosphere was reducing atmosphere with no free oxygen. **Answer: B**

100. Which of the following organic molecules is least informative about the evolutionary relationship between species?
A. RNA
B. Proteins
C. DNA
D. Carbohydrates

Explanation: RNA and DNA are proteins and are highly information about evolutionary relationship between species. **Answer: D**

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SECTION 5 – ENGLISH

SECTION ONE Word order (1-5)

Directions the following words are not in the proper words are not in the proper order. When put in the right order, they make correct English sentences. From the given alternatives. Choose the one that is correct and blacken the letter of your choice on the separate answer sheet provided.

1. World I the sail to around want
   - A. To sail around the world I want
   - B. I want around the world to sail
   - C. I want to sail around the world
   - D. Around the world I want to sail

   Answer: C

2. Get to at tea six up we o'clock drink
   - A. To drink at six o'clock to drink tea
   - B. We get up at six o'clock to drink tea
   - C. We drink up tea to get at six o'clock
   - D. We get up to drink tea at six o'clock

   Answer: D

3. She in do the what you evening know does
   - A. What she does do you know in the evenings?
   - B. In the evenings do you know what she does?
   - C. Do you in the evenings know what she does?
   - D. Do you know what she does in the evenings?

   Answer: D

4. Looking have we over you all been for
   - A. We have been looking for you all over
   - B. We all have been looking over for you
   - C. You have been looking for we all over
   - D. All over you have been looking for me

   Answer: A

5. Him you like son like we and you seeing laugh
   - A. You like your son and we like seeing laugh him
   - B. You laugh like seeing your son and we like him
   - C. We like him laugh and your son like seeing him
   - D. We like seeing him laugh like you and your son

   Answer: C

SECTION TWO: Paragraph Cherence (6-12)

Directions: when put in the correct order. Sentences A-E in each of the questions from6-12mach up a complete paragraph. From the alternatives A-D given in each case, choose the letter that contains the correct order

of the sentences and blacken the corresponding letter on the separate answer sheet provided.

6.
   - a. But unfortunately it got so furious
   - b. Unlike we thought, it was a wild cat
   - c. I was playing hide and seek with friends.
   - d. We were happy thinking we could take it home
   - e. Then my cousin and I discovered an abandoned cat
   - A. e a b d c
   - B. e d a b c
   - C. c e a b d
   - D. c e d a b

   Answer: D

7.
   - a. I dir, in our society, has along history
   - b. The assistance includes material financial and labor
   - c. Members contribute certain amount of money every month
   - d. Some idirs, nowadays, are involved in even more useful activities
   - e. Then they get assistance when the member or one of his family dies
   - A. a d c e b
   - B. a c e b d
   - C. d a c e b
   - D. a c d c b

   Answer: A

8.
   - a. It saves time and reduces potential expense
   - b. It also makes the communication easy and very quick
   - c. Effective communication is vital to any business any where in the world
   - d. For instance, mobile phones help to reduce travel and associated expenses
   - e. Information technology provides the tools for such communication.
   - A. d e a b c
   - B. c e a d b
   - C. c d a b e
   - D. c a b e d

   Answer: B

9.
   - a. It has only one room
   - b. Konjit's house is a hut
   - c. the room is divided in to three sections
   - d. The three sections are bedroom, kitchen and pantry
   - e. A bed and a big wooden box are in the bedroom section
   - A. e b a d c
   - B. c a d b e
   - C. b a c d e
   - D. d e c b a

   Answer: C

10.
   - a. I'm not very patient with people
   - b. I'm also good at mending things
   - c. I don't really want to work with lots of people
   - d. I help my neighbors when something is broken

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e. I'm not good at languages but I'm good at science.

A. a c d b e C. d b a e c
B. e b d a c D. b d e a e **Answer: A**

11. a. One of m favorite things to do is wash my car
b. People say they can tell about a person by their shoes
c. Here is the thoughtful process I follow to watch my car
d. A clean car says I'm proud and I feel good about my self
e. I say that you can tell about a person by how clean his car

A. b e d a c C. e b c d a
B. c a b d e D. a d b e c **Answer: B**

12. a. The first thing I do is work out
b. My morning ritual is the same every day
c. After that I put on my make up and leave home.
d. when I have school, I have to do a lot of things
e. When I finish working out, I take a shower, fix my hir and get dressed

A. a d c e b C. e a b d c
B. d e b c a D. b d a e c **Answer: D**

SECTION THREE: COMPREHENSION (13-26)

Directions: read the following assuagecarefully and answer the questions according to the information therein.

1. When I was in junior high school, my parents gave me a typewriter. my friends Amy Neff and Carol Stein often walked with me to my house after school. we closed my bedroom and spent the afternoon giggling-or rolling on the carpet, howling with lunatic laughter - as we tapped out silly limericks and satirical stories on my typewriter. once we even wrote a play.

2. Our after- school typing was not homework. we rambled wherever our teenage imaginations took us; we explored words wildly for the pure fun of it, with no exams or report cards in mind we practiced literacy risk takings and problem solving. honed our language skis and learning perseverance, all with no fear of failure - because we playing few children today have time for this king of after school play. there is too much homework parents' concerns about overload have become a familiar story as schools all over the country pile on homework in a mis-aimed response to criticism that they were teaching too little some elementary teachers routinely assign time consuming work in each of three or four academic subjects every night.

3. Amy Neff, one of the friends who helped me enjoy my typewriter in junior high, learned at a school meeting in may that in fifth grade her -year-old daughter would have an hour and a half to two hours of homework seven days a week, plus special projects. "The idea is that, under avalanche of work children should learn to organize their time efficiently." she wrote me in an e-mail message. " Nobody has mentioned creativity: ideas and content of what is learned. just doing LOTES,FAST!"

4. The parents and grandparents of today's children had time" playing around" with stamp collections or model airplanes, writing a fairytale, selling lemonade or indulging a fascination with space trav. what many people do not realize is that these kinds of interests and hobbies are just as important as academic learning.

5. Many successful adults have a childhood history of freewheel play. Louis ignarro, a Nobel prize winning Pharmacologist, had six successively complex chemistry sets and played with model rockets. when lettycottinpogerbin, president of the Authors Guild, an association of published writers, was 9 she spent afternoons writing and drawing for a monthly magazine she printed on a primitive press called hectograph she charged friends and famil 3 cents a copy and them mailed each copy with a 3-cent stamp. she recalls now. " I never understood why the magazine did not make a profit" But that is just the point when kids play. they are free to experiment and learn from experiences, without worrying about how well they are performing 6 though homework has a clear benefit in high school, there is no research showing that any amount of it advances the education of elementary school kids They may be stuffing a great deal of information into their heads, but after an hour or two, children lose any eagemess or joy they have in learning. That is important because research has shown that when children enjoy it, their learning is deeper, richer, and longer-lasting.

7. Parents do have some recourse. They can meet with principalsand teachers, organize other parents, and protest, parents in several communities have prevailed upon school districts to enact homework limits. from their children, at least, there may still be time to indulge in the ancient and powerful medium for learning known as play.

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Source: Destination: an Integrated Approach to Writing

13. The tone of the author can best be described as:
    A. dismissive
    B. argumentative
    C. emphatic
    D. compassionate

Clues:

- The TONE of a speech or piece of writing is its style and the opinions or ideasexpressed in it.
- Dismissive (adj) If you are dismissive of someone or something you sow or say that you think they are not important or novalue.
  - Eg. Mr jones was dismissive of the report
- Argumentative (adj) someone who is argumentative always ready to disagree or start quarrelling with other people
  - Eg. great chess players gave a reputation for being both accentric and argumentative.
- emphatic (adj) An emphatic response statement is one made in a forceful way, because the speaker feels very strongly about what they are saying.
  - Eg His response was immediate and emphatic
- Compassionate (adj) If you describe someone or something as compassionate, you mean that they feel o show pity sympathy, and understandingfo people who are suffering.
  - Eg My father was a deeply compassionate man

So tone the author uses through the passage iscompassionate.

14. Author narrates her childhood experience to:
    A. show that she dan good friends to play with
    B. describe how she spent hr after school hours.
    C. show how, through play, they were learning
    D ten us that she used a type writer to do her home work .

Answer: C

15. Which of the following word did the author use in paragraph 2 to refer to the development of their determined attitude through after-school play?
    A. Literacy
    B. Perseverance
    C. Imagination
    D. Risk-taking

As stated in paragraph 2, ... honed our language skills and learning perseverance all with no fear of failure because we are playing " perseverance, here is, the quality of continuing with something even thought it is difficult. Eg. she persevered in her idea despite obvious objection raised by friends. **Answer: B**

16. In paragraph 2, the author maintains that children learn best when:
    A. they are given a lot of homework.
    B. they have time for play after school.
    C. schools don't give them homework.
    D. they do homework with a type writer. **Answer: B**

17. As used in paragraph 2, which one can be the best definition of honed?
    A. Knew
    B. learned
    C. improved
    D. acquired

- If you hone something for example a skill, technique, idea , or product you carefully develop it over a long period o time so that it is exactly right for you purpose
- If you hone balde, weapon, or tool you sharpen it so, contextually, it has the meaning improved, in the passage

Answer: C

18. As implied in paragraph 2, what is the criticism directed to elmntary teacher?
    A. They are teaching too little
    B. They are not giving enough home work.
    C. They teach children a familiar story
    D. They do not assign interesting work to children.

Answer: A

19. One major issue the author mentioned about the parents and grandparents of today's children is that they:
    A. had no fear of failure at school
    B. were not given homework to do after school
    C. had time for out of school interest and hobbies.
    D. did not realize that after school play enhanced learning.

As mentioned in the paragraph 4" ... had time" playing around" and described; writing a fairy tale, selling lemonade or ... (hobbies). **Answer: C**

20. The idea is ... (paragraph 3) what does the word idea refer to?
    A. special project
    B. too much home work
    C. purpose of homework

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D. Amy's purpose for the e-mail

Answer: B

21. Elementary teachers responded to the criticism directed at them by:
   A. giving special project
   B. telling kids familiar stories
   C. giving too much homework
   D. teaching kids to organize their time effectively.

Answer: C

22. The author looks at the issue homework and after school play from:
   A. teachers' and parents' perspectives
   B. herown and her friends' perspectives.
   C. school principals' and teachers' perspectives '
   D. her friends' teachers' and parents prspectives.

Answer: D (pointed out in pa. 2,3,4)

23. who set the author an e-mail message?
   A. Amy Neff
   B. Louis Ignarro
   C. Carol Stein
   D. LettyCottinpogrbin

Answer: A

24. Which one of the following is true about homework
   A. It may advance learning in high school.
   B. It stuffs kid's head with useful information.
   C. It enhances kids' imagination and creativity
   D. It helps elementary school kids organize their time.

Answer: B

25. According to paragraph 7. which o is the recours parents are reported to have
   A. enacting homework limits.
   B. meeting with principals and teachers
   C. protesting against children's overload
   D. allowing their children to learn through play

Answer: D

26. Which one of the following is among the major issue raised by the author?
   A. Exams and report cards are a threat to children's learning
   B. too much play ad too little homework advance kids' learning
   C. all successful adults have a childhood history of free weeling play

D. through ply, children experiment and learn fromtheir experience.

Answer: D

SECTION FOUR: Vocabulary (27-57)

A. Completion (27-46)

Directions: questions 27-46 are incomplete sentences. there are four alternative words/phrases, A-D given below each question. choose the one that best completes each sentence and blacken the letter of you choice on the separate answer sheet.

27. Laptop computers are --------. They allow work to be done in man different locations
   A. ptable
   B. durable
   C. portable
   D. sizable

- **potable (adj)**-clean an saf water for drinking
- **durable (adj)** - something which strong and last a long time without beaking or be coming weaker.
- **portable (adj)** - is something such as television, radio or computer which can be easily carried or moved.
- **sizable (adj)** - something which is fairly large

Answer: C

28. Many people have a/an ------ about flying and won't travel by plane
   A. hygteria
   B. nostalgia
   C. phobia
   D. accidence

- **hysteria** - uncontrolled excitement anger or panic. A person who is suffering from hysteria is in state of violent and distrurbed emotion as aresult of shack (medical0
- **nostalgia** -is an affectionate feeling you have for the past, especially for a particularly happy time
- **Example:** He might be influenced nostalgia for the surroundings of his happy youth
- **Phobia** - is a very strong irrational fear or hatred of something
- **Example:** the man had a phobia about flying
- **Accident:** something unpleasant happens to people (car accident).

Answer: C

29. Children are naturally -------- about the world around them
   A. inquisitor
   B. requisite
   C. inquisitive
   D. notorious

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- **inquisitor (N)** - someone who is asking someone else aseries of questions esp. in arathar hostile way.
- **requisite (adj)** -necessary on account of circumstances or rules
- **inquisitive (adj)** - a person like finding out about things specially secret things
- (naturally children are inquisitive means they are very eager to feel their environment.
- Notorious (adj) - to be notorious means to be well - known for something bad.

Answer: C

30. Several students are -------- at the prospect of joining university next year.
A. incited B. relaxed C. surprisedD. excited

- Clues:

- A particular prospect is something that someone expect or know is going to happen someone's prospects are their chances of being successful, especially in their career. So, the prospects of several students of joining university next year is manifested by their being excited to have that opportunity or chance.

Answer: D

31. The -------- placed a sticker on each piece of cake she chacked for equality
A. Spectator C. inspector
B. supervisor D. director

- **Spectator (N)** - is someone who watches a game
- **Supervisor (N) -** is someone who supervises activities, or people especially workers or students.
- **inspector (N) -** Is an officer in the police who is higher in rank than a sergeant
- **Director (N) -** is someone who makes important decisions about how leads, manages a particular organization.
- Example (School's film's, business' etc director)

Answer: B

32. Linda ------- her request for a vacation to her oss
A. emitted C. submitted
B. permitted D. omitted

- **emitted** - If something emits heat light, gas, or smell it produces it and sends out by means of physical or chemical process. Eg. The amount of carbon dioxideemitted
- **permitted** - allowed something/someone to do something (to be allowed)
- **submitted** - formally sending (proposal, report, request) to someone so that he/she consider or decide about it.
- **omitted** - not included something in an activity or piece of work deliberately or accidentally. **So (C) is the correct completion**

33. The hotel ------ brought out luggag to our rooms
A. reporter C. supporter
B. porter D. importer

- A - someone who writes news articles or who broadcasts news reports Eg. TV reporter atrainee sports reporter etc.
- B - person whose job is to carry things for example people's luggage at a railway station or in hotel
- C - people who support someone or something eg. Apolitical leader or sports tam
- D- acountry firm, or prson that buys goods from another country fo use in their country.

Accordingly, the Answer is B

34. Lensa who is always worried about hr weight, eats like a/an ----
A. bird C. pig
B. ant D. elephant

- Clues:

- The idea of the sentence is bout meta'aphoric language. which is an imaginative way of describing something by referring to something else which is the same in a particular way for example, if you want to say that someone is very shy and frightened for things, you might say that they are a mouse
- the other expression of language is simile which is an expression describing a person or thing as being similar to someone or something else (with words : like, as, etc)

Examples

- she runs like a deer
- he is as whit as a sheet

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